

Sleep Study - Amy Bohnert - 3/22/2025

```
#Description of the models is here:
#https://docs.google.com/document/d/1wPB0gAGN5HowoabCOF4LSJFSpuju6Igh/edit

#Understanding geoids
#https://www.census.gov/programs-surveys/geography/guidance/geo-identifiers.html

library(foreign)
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.0      v stringr    1.5.1
v ggplot2    3.5.1      v tibble     3.2.1
v lubridate  1.9.3      v tidyr      1.3.1
v purrr      1.0.2

-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(lattice)
library(lcsm)
```

This is lcsm 0.3.2
Please report any issues or ideas at:
<https://github.com/milanwiedemann/lcsm/issues>

```
library(ggplot2)
library(tidyr)
library(dplyr)
```

```
library(stringr)
library(lmerTest)
```

Loading required package: lme4
Loading required package: Matrix

Attaching package: 'Matrix'

The following objects are masked from 'package:tidyr':

expand, pack, unpack

Attaching package: 'lmerTest'

The following object is masked from 'package:lme4':

lmer

The following object is masked from 'package:stats':

step

```
library(sjstats)
library(MuMIn)
```

#ID 2561 was on two rows. It seems very clear that this should only have been one row.

```
dat <- read.csv("../bohnert_sleep/CL_OTR_COI_Data.csv")
```

```
#dat2 <- read.csv("../bohnert_sleep/CL_OTR v1 Youth Baseline COI3 Actigraphy Demo.csv")
```

#EDA

#Observation ID 2561 had two rows in the data set. I removed one of these rows.

#ID 2561 is a problem

```
#dat %>% group_by(ID) %>% summarize(n = n()) %>% arrange(-n)
```

```
dat %>% filter(ID == 2561) %>% select(RecordedDate, ID, age.v1, gender.v1)
```

	RecordedDate	ID	age.v1	gender.v1
1	1/30/2020 12:50	2561	NA	NA
2	<NA>	2561	19	2

```

#Remove ID's that are NA
dat <- dat %>% filter(!is.na(ID))

#ID 2561 has two rows
#Fixing this here:
dat[dat$ID == 2561 &
      is.na(dat$age.v1), c("age.v1", "age_dichotomous", "gender.v1")] <-
  dat[dat$ID == 2561 & !is.na(dat$age.v1), c("age.v1", "age_dichotomous", "gender.v1")]
dat <- dat[dat$ID != 2561 |
           (dat$ID == 2561 & !is.na(dat$RecordedDate)), ]

#Merge this on to get community area in chicago
#File url is here: https://data.cityofchicago.org/Facilities-Geographic-Boundaries/Boundaries
commu_area <- read.csv("../bohnert_sleep/CensusTractsTIGER2010_20250303_small.csv")

#dat_neigh <- as.data.frame(read.spss("../bohnert_sleep/CL_OTR v1 Youth and Parent Baseline,
#write.csv(dat_neigh %>% select(famid, ID, p.mn.cohesion.mean.v1, p.mn.safety.mean.v1, p.mn.

dat_neigh <- read.csv("../bohnert_sleep/CL_OTR_COI_Data_neighborhood.csv")
dat_neigh <- dat_neigh[!duplicated(dat_neigh$ID),]

```

```

dat <- dat %>% left_join(commu_area, by = c("geoid20" = "GEOID10"))

dat <- dat %>% mutate(COMMAREA = ifelse(is.na(COMMAREA), 0, COMMAREA))

table(dat$COMMAREA)

```

0	1	2	3	7	8	13	15	18	19	21	22	23	24	25	26	27	28	29	32
100	7	4	2	3	3	1	1	1	1	1	2	6	3	14	4	6	5	8	1
35	36	37	38	39	40	41	42	43	44	45	46	48	49	50	51	53	54	56	61
5	1	1	6	4	5	3	8	16	11	3	9	6	15	3	6	12	2	1	2
63	65	66	67	68	69	70	71	73	75										
1	2	20	16	10	13	6	29	8	6										

```
sum(is.na(dat$COMMAREA))
```

```
[1] 0
```

Data Cleaning and EDA

```
dat <- dat %>% left_join(dat_neigh, by = "ID")

#Convert gender 1/2 to male/female
dat <- dat %>% mutate(
  gender.v1 = case_when(gender.v1 == 1 ~ "Male",
    gender.v1 == 2 ~ "Female")
)
```

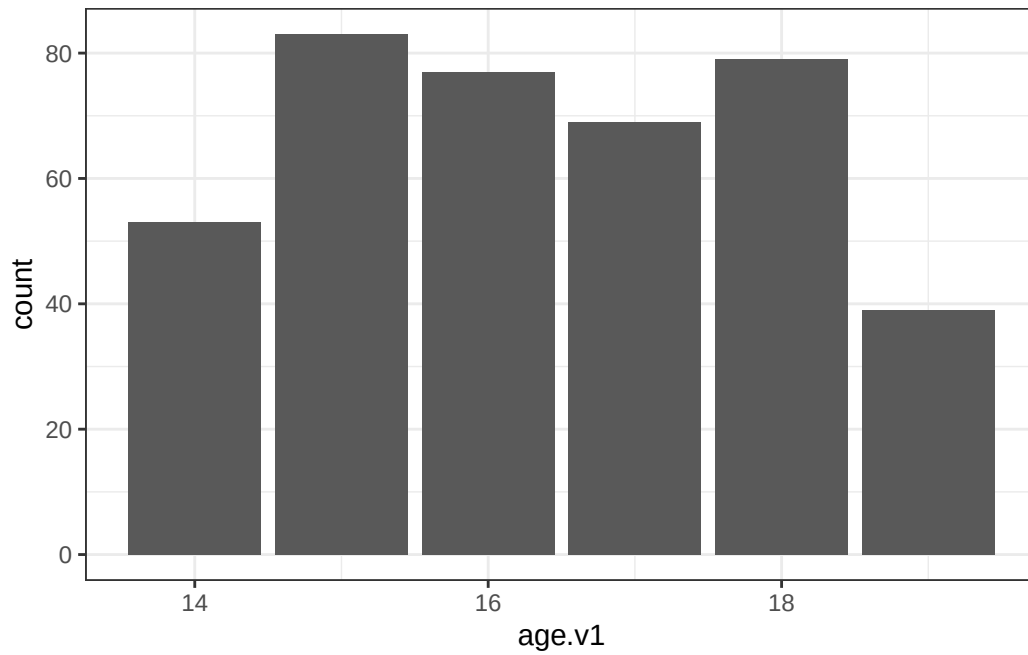
Summary stats

```
dat %>% summarize(
  mean_age = mean(age.v1, na.rm = T),
  pct_female = mean(gender.v1 == "Female", na.rm = T),
  mean_income = mean(income.v1, na.rm = T)
)
```

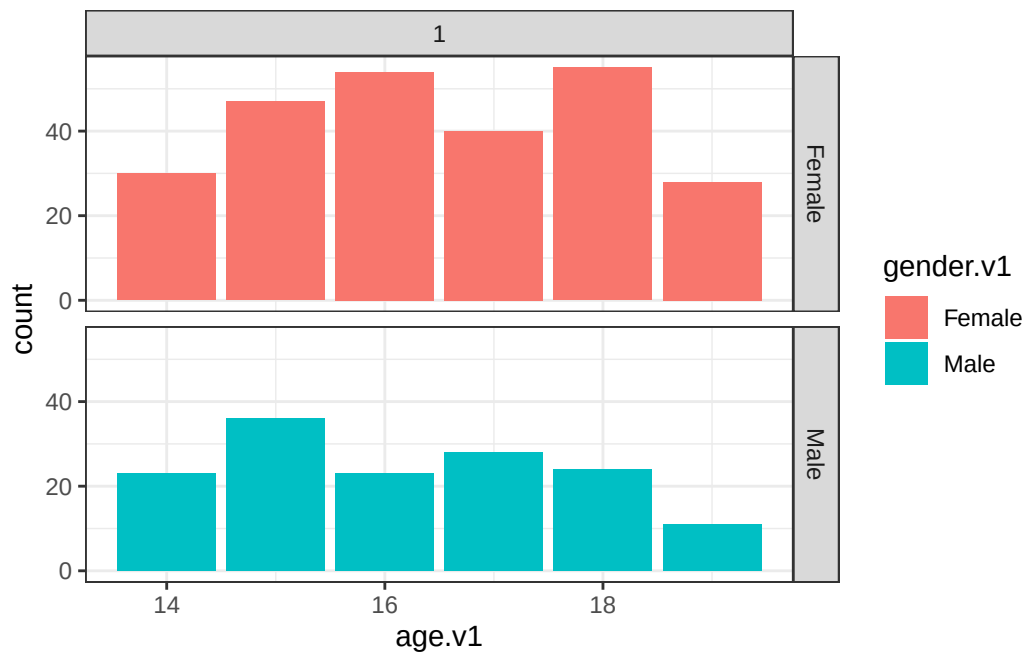
```
      mean_age pct_female mean_income
1  16.3875    0.6365915    40774.34
```

```
#ggplots for age and income
ggplot(aes(x = age.v1), data = dat) + geom_bar() + theme_bw()
```

Warning: Removed 3 rows containing non-finite outside the scale range (`stat\_count()`).



```
ggplot(aes(x = age.v1, fill = gender.v1), data = dat %>% filter(gender.v1 %in% c("Male", "Female")))
```



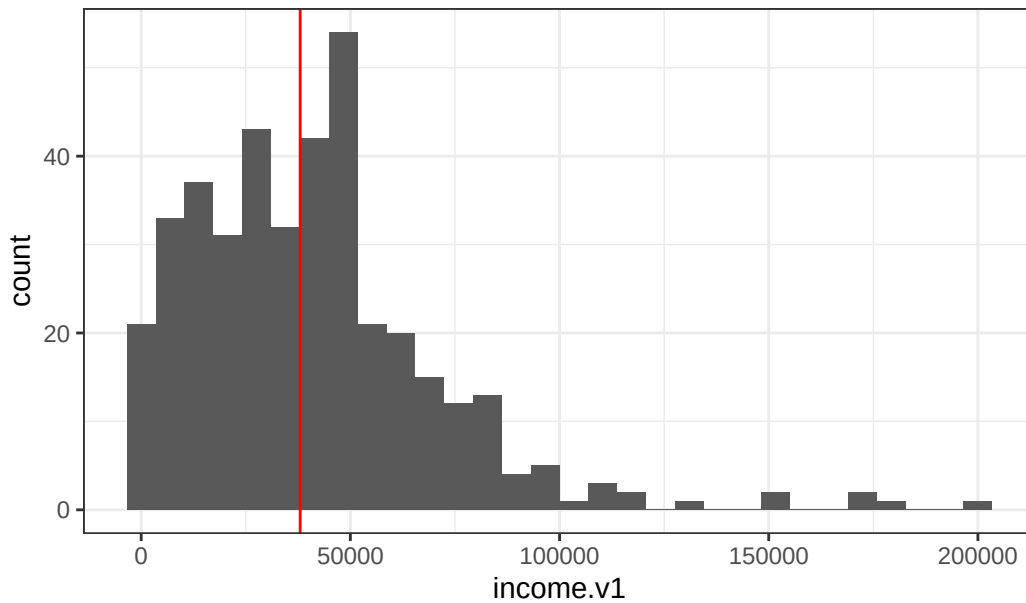
```
ggplot(aes(x = income.v1), data = dat) + geom_histogram() + theme_bw() + geom_vline(aes(xint = income.v1))
```

Warning: Use of `dat\$income.v1` is discouraged.
i Use `income.v1` instead.

`stat\_bin()` using `bins = 30`. Pick better value with `binwidth`.

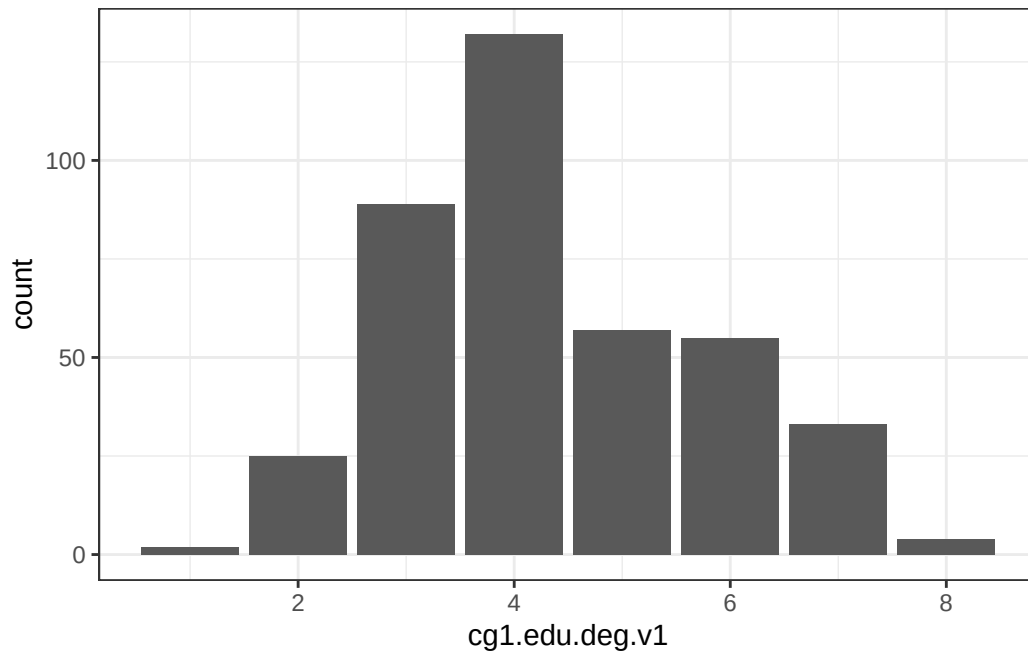
Warning: Removed 7 rows containing non-finite outside the scale range
(`stat\_bin()`).

Red line = Median



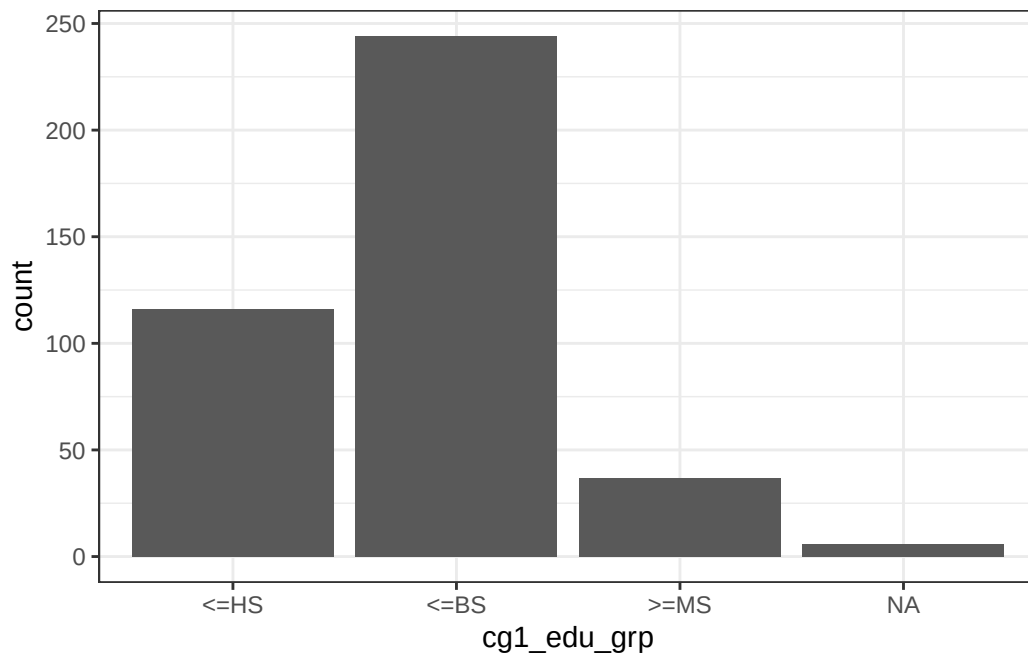
```
ggplot(aes(x = cg1.edu.deg.v1), data = dat) + geom_bar() + theme_bw()
```

Warning: Removed 6 rows containing non-finite outside the scale range
(`stat\_count()`).

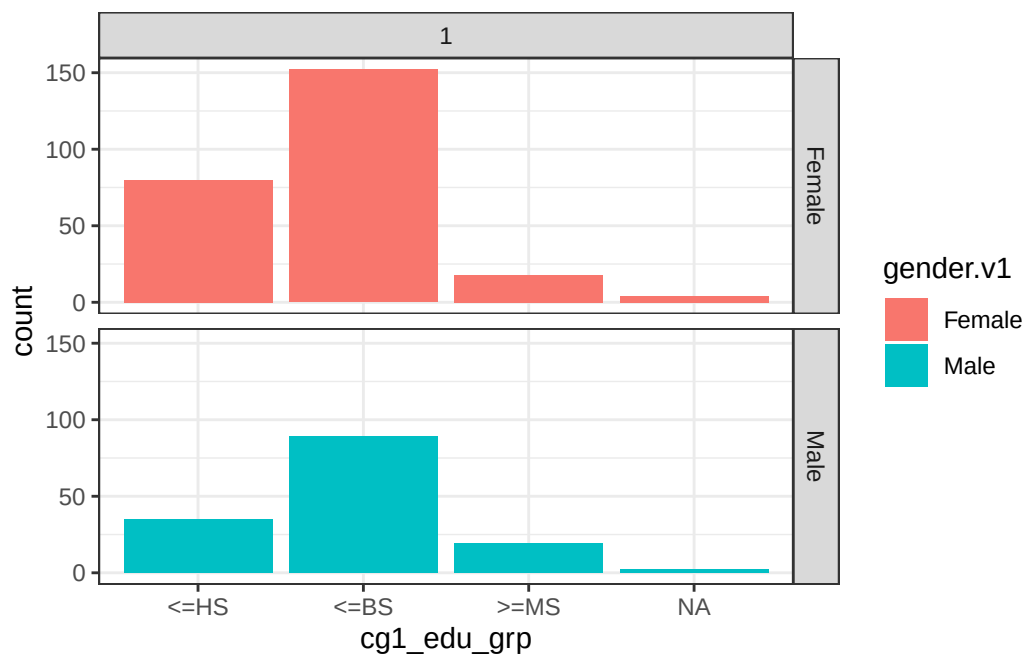


```
dat$cg1.edu.deg.v1 <- as.numeric(dat$cg1.edu.deg.v1)
dat <- dat %>% mutate(cg1_edu_grp = factor(
  case_when(
    cg1.edu.deg.v1 %in% c(1, 2, 3) ~ "<=HS",
    cg1.edu.deg.v1 %in% c(4, 5, 6) ~ "<=BS",
    cg1.edu.deg.v1 %in% c(7, 8) ~ ">=MS"
  )
))

dat <- dat %>% mutate(cg1_edu_grp = fct_relevel(cg1_edu_grp, c("<=HS", "<=BS", ">=MS")))
ggplot(aes(x = cg1_edu_grp), data = dat) + geom_bar() + theme_bw()
```



```
ggplot(aes(x = cg1_edu_grp, fill = gender.v1), data = dat %>% filter(gender.v1 %in% c("Male"
```



```
#need at least 5 days of actigraph data
table(dat$act_patotaldays)
```

```
0  1  2  3  4  5  6  7
8  2  4 10 18 15 55 254
```

```
#254+55+15 = 324
```

```
#I get 324
```

```
dat %>% filter(act_patotaldays >= 5) %>% nrow()
```

```
[1] 324
```

```
#Reproducing Table 1 stuff
```

```
dat %>% filter(act_patotaldays >= 5) %>% group_by(gender.v1) %>% summarize(age = mean(age.v1))
```

```
# A tibble: 3 x 3
```

```
  gender.v1  age      n
  <chr>      <dbl> <int>
1 Female    16.5    213
2 Male      16.2    110
3 <NA>      17       1
```

```
dat %>% group_by(gender.v1) %>% summarize(
  age = mean(age.v1),
  n = n(),
  sleep = mean(act_totsleep.min_avg, na.rm = T),
  inbed = mean(act_totinbed.min_avg, na.rm = T),
  onset = mean(act_onset.time.min_d0, na.rm = T),
  eff = mean(act_efficiency_avg, na.rm = T)
)
```

```
# A tibble: 3 x 7
```

```
  gender.v1  age      n sleep inbed onset  eff
  <chr>      <dbl> <int> <dbl> <dbl> <dbl> <dbl>
1 Female    16.5    254  381.  448. 1464.  85.5
2 Male      16.2    145  357.  427. 1518.  83.8
3 <NA>      NA       4  412.  504. 1441   82.4
```

```

dat <- dat %>% filter(act_patotaldays >= 5)

#Average inbed time is a problem because times are circular
datinbed <- dat %>%
  select(ID, gender.v1, act_inbed.time_d0:act_inbed.time_d8) %>%
  pivot_longer(
    cols = starts_with("act_inbed.time_d"),
    names_prefix = "act_inbed.time_d",
    names_to = c("day"),
    values_to = "act_inbed_time"
  ) %>% mutate(act_inbed_time_numeric = as.numeric(do.call(rbind, strsplit(act_inbed_time, "
    inbed.time_angle = 2*pi*act_inbed_time_numeric/24,
    inbed.time_angle_x = cos(inbed.time_angle) ,
    inbed.time_angle_y = sin(inbed.time_angle),
    )

datweekend_yn <- dat %>%
  select(ID, act_weekday_yn_d0:act_weekday_yn_d8) %>%
  pivot_longer(
    cols = starts_with("act_weekday_yn_d"),
    names_to = c("day"),
    values_to = "act_weekday_yn",
    names_prefix = "act_weekday_yn_d",
    )

datinbed <- datinbed %>% left_join(datweekend_yn, by = c("ID","day"))

datinbedmean <- datinbed %>% group_by(gender.v1) %>% summarize(mean_x = mean(inbed.time_angle_x),
  mean_y = mean(inbed.time_angle_y, na.rm = T),n = n()) %>% mutate(mean_angle = atan(mean_y/mean_x))

datinbedmean

```

A tibble: 3 x 8

	gender.v1	mean_x	mean_y	n	mean_angle	mean_time	mean_x_uc	mean_y_uc
	<chr>	<dbl>	<dbl>	<int>	<dbl>	<dbl>	<dbl>	<dbl>
1	Female	0.858	0.115	1917	0.133	0.507	0.991	0.132
2	Male	0.749	0.232	990	0.300	1.15	0.955	0.296
3	<NA>	0.976	0.119	9	0.122	0.465	0.993	0.121

```

ggplot(
  aes(
    x = inbed.time_angle_x,

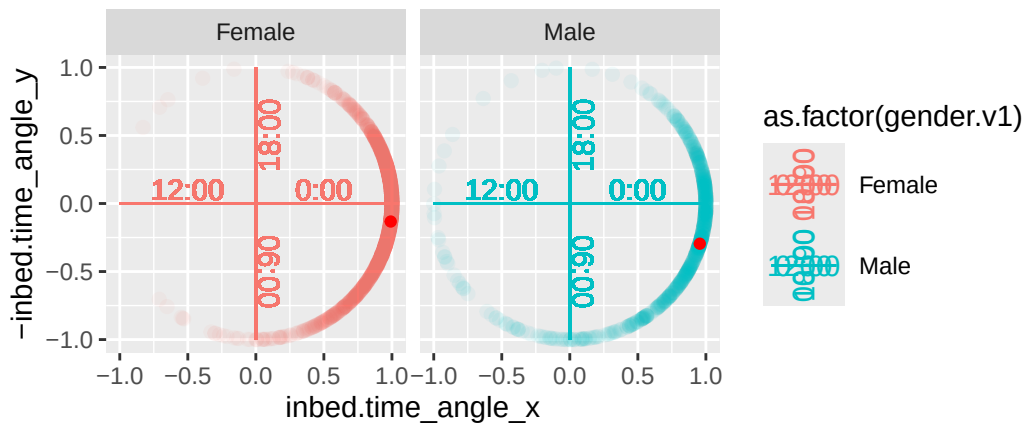
```

```

    y = -inbed.time_angle_y,
    color = as.factor(gender.v1)
  ),
  data = datinbed %>% filter(gender.v1 %in% c("Male", "Female"))
) + geom_point(alpha = 0.05, size = 2) + coord_fixed() + facet_grid( ~ gender.v1) + geom_point(
  aes(x = mean_x_uc, y = -mean_y_uc),
  color = "red",
  data = datinbedmean %>% filter(gender.v1 %in% c("Male", "Female"))
) + geom_segment(x = 0,
  y = 0,
  xend = 1,
  yend = 0) + geom_text(x = 0.5, y = 0.1, label = "0:00") +
  geom_segment(x = 0,
  y = 0,
  xend = -1,
  yend = 0) + geom_text(x = -0.5, y = 0.1, label = "12:00") +
  geom_segment(x = 0,
  y = 0,
  xend = 0,
  yend = -1) + geom_text(x = 0.1, y = -0.5, label = "06:00", angle = -90) +
  geom_segment(x = 0,
  y = 0,
  xend = 0,
  yend = 1) + geom_text(x = 0.1, y = 0.5, label = "18:00", angle = 90)

```

Warning: Removed 569 rows containing missing values or values outside the scale range (`geom\_point()`).



```

#Pivot longer
#DVs (sleep variables with models run for each sleep DV: 8 total):
# -act_totsleep.min
# -act.outbed.time
# -act.onset.time
# -act_efficiency
# -act_awakening.min
#Weekly level:
#Variability in:
#   -act_totsleep.min_sd
#   -act.outbed.time.min_sd
#   -act.onset.time_min.sd
##test
#Pivoting longer example:
# test <- dat %>%
#   select(ID, gender.v1,
#           act_inbed.time_d0:act_inbed.time_d8,
#           act_weekday_yn_d0:act_weekday_yn_d8,
#           act_onset.time_d0: act_onset.time_d8) %>%
#   pivot_longer(
#     cols = -c(ID,gender.v1),
#     cols_vary = "slowest",
#     names_to = c(".value","day"),
#     names_pattern = "(.*)_d(.*)"
#   )
#####
#Total Sleep
#####z_ED_nat; z_HE_nat; z_SE_nat
datlong_resp <- dat %>%
  filter(act_patotaldays >= 5 & gender.v1 %in% c("Male","Female")) %>%
  select(
    ID,
    RecordedDate,
    gender.v1,
    age.v1,
    income.v1,
    cg1_edu_grp,
    z_COI_nat,
    z_ED_nat,
    z_HE_nat,
    z_SE_nat,
    z_ED_EC_nat,

```



```

z_ED_EL_nat,
z_ED_ER_nat,
z_ED_SP_nat,
z_HE_EP_nat,
z_HE_HE_nat,
z_HE_HR_nat,
z_HE_SE_nat,
z_SE_EI_nat,
z_SE_EO_nat,
z_SE_ER_nat,
z_SE_HQ_nat,
z_SE_SR_nat,
z_SE_WL_nat,
p.mn.cohesion.mean.v1,
p.mn.safety.mean.v1,
p.mn.violence.mean.v1,
COMMAREA,
act_totsleep.min_d0:act_totsleep.min_d8,
act_outbed.time_d0:act_outbed.time_d8,
act_onset.time_d0: act_onset.time_d8,
act_efficiency_d0: act_efficiency_d8,
act_weekday_yn_d0:act_weekday_yn_d8,
act_awakening.min_d0: act_awakening.min_d8,
) %>%
pivot_longer(
  cols = -c(ID,
             RecordedDate,
             gender.v1,
             age.v1,
             income.v1,
             cg1_edu_grp,
             z_COI_nat,
             z_ED_nat,
             z_HE_nat,
             z_SE_nat,
             z_ED_EC_nat,
             z_ED_EL_nat,
             z_ED_ER_nat,
             z_ED_SP_nat,
             z_HE_EP_nat,
             z_HE_HE_nat,
             z_HE_HR_nat,

```

```

    z_HE_SE_nat,
    z_SE_EI_nat,
    z_SE_EO_nat,
    z_SE_ER_nat,
    z_SE_HQ_nat,
    z_SE_SR_nat,
    z_SE_WL_nat,
    p.mn.cohesion.mean.v1,
    p.mn.safety.mean.v1,
    p.mn.violence.mean.v1,
    COMMAREA),
    cols_vary = "slowest",
    names_to = c(".value", "day"),
    names_pattern = "(.*)_d(.*)"
  )

datlong_weekday_yn <- dat %>%
  filter(act_patotaldays >=5 & gender.v1 %in% c("Male", "Female")) %>%
  select(ID, gender.v1, age.v1, income.v1, cg1.edu.deg.v1, act_weekday_yn_d0:act_weekday_yn_d8,
  pivot_longer(cols = starts_with("act_weekday_yn_d"), names_prefix = "act_weekday_yn_d", name

#Weekend indicator should be Fri/Sat instead of a weekend (Sat/Sun) indicator
datlong <- datlong_resp %>% left_join(datlong_weekday_yn %>% select(ID, day, weekday_yn), by =
datlong <- datlong %>% group_by(ID) %>% mutate(I_frisat = 1 - lead(weekday_yn))
datlong <- datlong %>% group_by(ID) %>% mutate(I_frisat = ifelse(is.na(I_frisat), lag(I_frisat
#datlong$Date <- do.call(rbind, strsplit(datlong$RecordedDate, " ")[,1]
#datlong$Date <- mdy(datlong$Date)
#datlong$dayofweek <- wday(datlong$Date)

#Fix time formatting
#I cut off the day at 3pm. We could choose a different cut off
datlong <- datlong %>% ungroup() %>% mutate(
  act_outbed.time_numeric = as.numeric(do.call(rbind, strsplit(datlong$act_outbed.time, ":"))[,1],
  act_onset.time_numeric = as.numeric(do.call(rbind, strsplit(datlong$act_onset.time, ":"))[,1])
  mutate(act_onset.time_numeric = ifelse(act_onset.time_numeric >= 17, act_onset.time_numeric

datlong <- datlong %>% mutate(income.v1.10000 = income.v1 / 10000)

datlong %>% group_by(I_frisat, gender.v1) %>%
  summarize(onset_mean = mean(act_onset.time_numeric, na.rm = T),

```

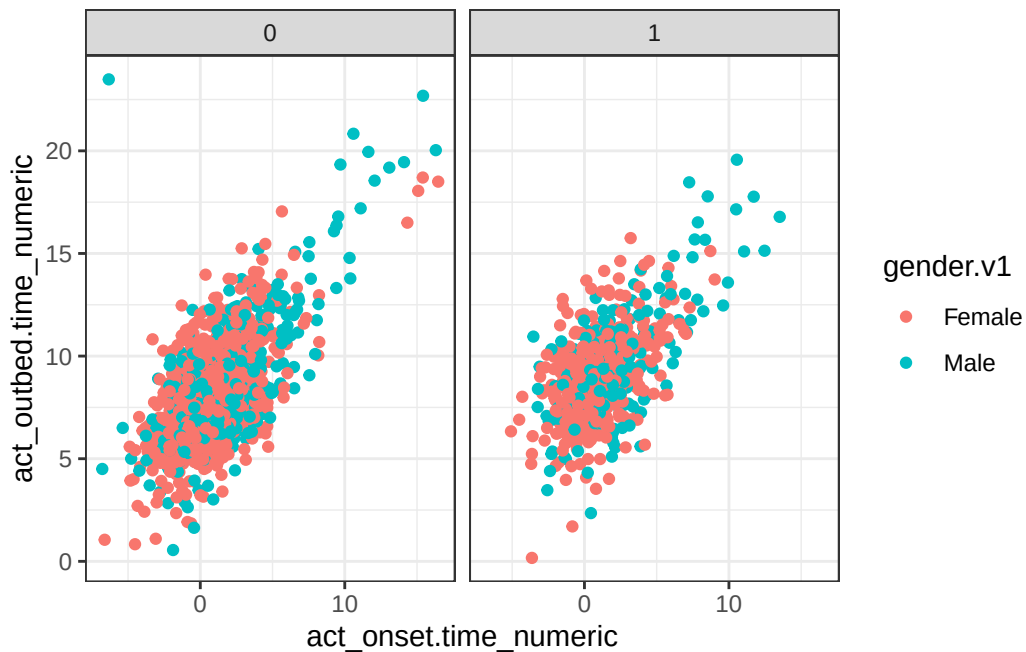
```
outbed_mean = mean(act_outbed.time_numeric, na.rm = T))
```

`summarise()` has grouped output by 'I_frisat'. You can override using the `.groups` argument.

```
# A tibble: 4 x 4
# Groups:   I_frisat [2]
  I_frisat gender.v1 onset_mean outbed_mean
    <dbl> <chr>         <dbl>     <dbl>
1      0 Female      0.487      7.78
2      0 Male       1.28      8.15
3      1 Female     0.981      8.93
4      1 Male       2.06      9.65
```

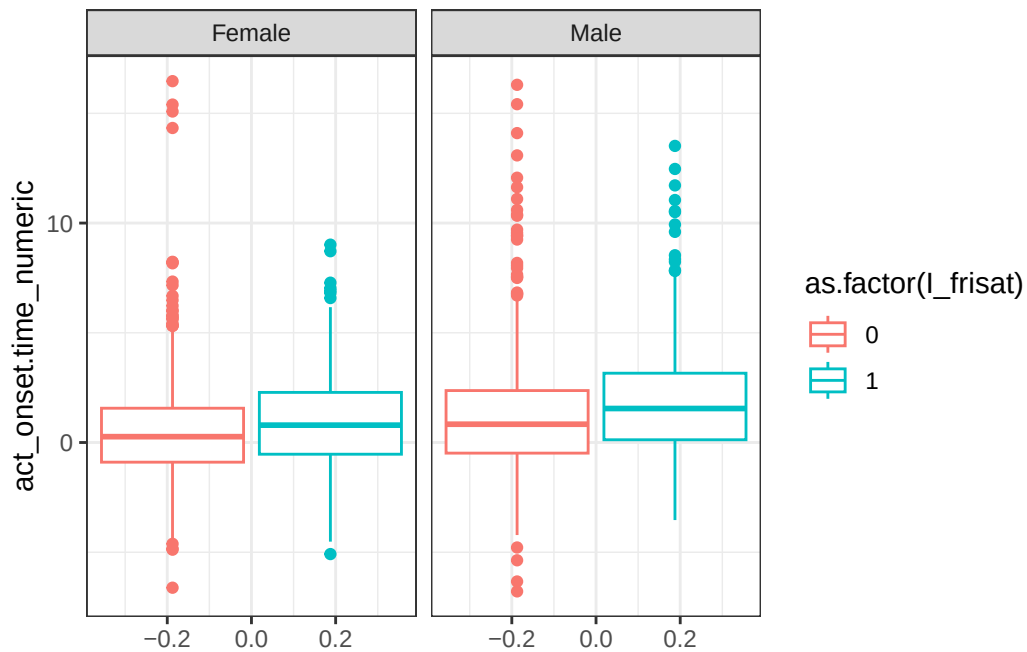
```
ggplot(aes(x = act_onset.time_numeric,
           y = act_outbed.time_numeric,
           color = gender.v1), data = datlong) + geom_point() + theme_bw() + facet_grid(~I_frisat)
```

Warning: Removed 569 rows containing missing values or values outside the scale range (`geom\_point()`).



```
ggplot(aes(y = act_onset.time_numeric,
           color = as.factor(I_frisat)), data = datlong) + geom_boxplot() + theme_bw() + fac
```

Warning: Removed 569 rows containing non-finite outside the scale range
(`stat\_boxplot()`).



```
ggplot(aes(y = act_outbed.time_numeric,
           color = as.factor(I_frisat)), data = datlong) + geom_boxplot() + theme_bw() + fac
```

Warning: Removed 569 rows containing non-finite outside the scale range
(`stat\_boxplot()`).

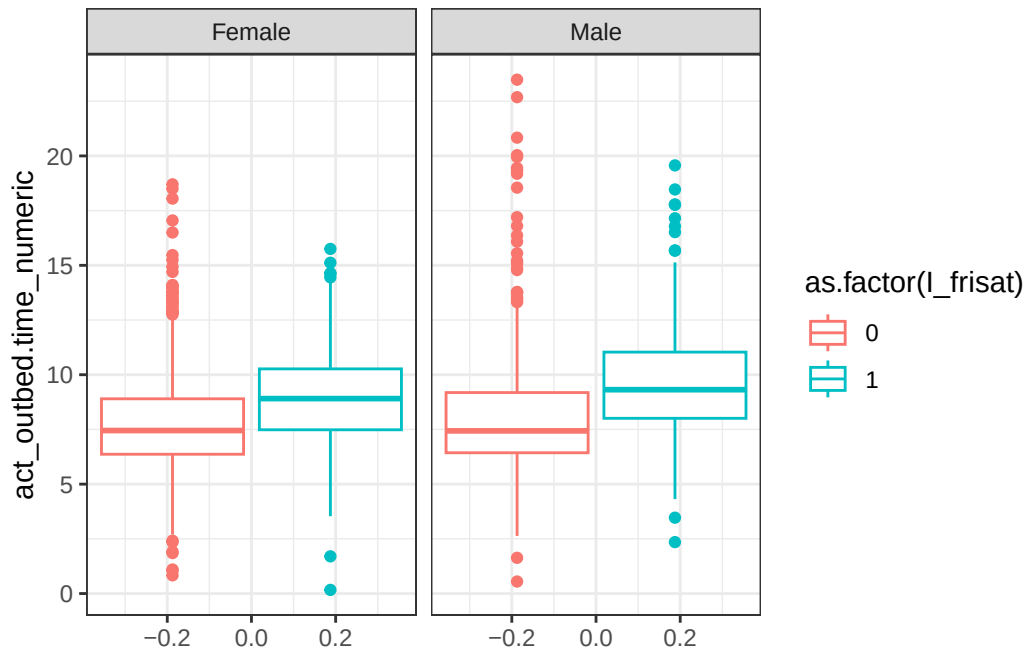


Table 1 stuff

```
library(gt)
#I lose an observation because it was missing gender. So I have 323.
datlong %>% group_by(ID) %>% summarize(n = sum(!is.na(act_efficiency))) %>% nrow()
```

```
[1] 323
```

```
#
rbind(datlong %>% filter(!duplicated(ID)) %>% group_by(gender.v1) %>% summarize(n = n(),
  age = paste0(round(mean(age.v1),2)," (",round(sd(age.v1),2),")"),
  income = paste0(round(mean(income.v1, na.rm = TRUE),2)," (",round(sd(income.v1, na.rm = TRUE),2),")"),
  cg1_edu_grp_leHS = sum(cg1_edu_grp == "<=HS", na.rm = TRUE)/n(),
  cg1_edu_grp_leBS = sum(cg1_edu_grp == "<=BS", na.rm = TRUE)/n(),
  cg1_edu_grp_geBS = sum(cg1_edu_grp == ">=MS", na.rm = TRUE)/n()
),

datlong %>% filter(!duplicated(ID)) %>% summarize(gender.v1 = "All",n = n(),
  age = paste0(round(mean(age.v1),2)," (",round(sd(age.v1),2),")"),
  income = paste0(round(mean(income.v1, na.rm = TRUE),2)," (",round(sd(income.v1, na.rm = TRUE),2),")"),
  cg1_edu_grp_leHS = sum(cg1_edu_grp == "<=HS", na.rm = TRUE)/n(),
  cg1_edu_grp_leBS = sum(cg1_edu_grp == "<=BS", na.rm = TRUE)/n(),
  cg1_edu_grp_geBS = sum(cg1_edu_grp == ">=MS", na.rm = TRUE)/n()
)
```

```

cg1_edu_grp_leHS = sum(cg1_edu_grp == "<=HS", na.rm = TRUE)/n(),
cg1_edu_grp_leBS = sum(cg1_edu_grp == "<=BS", na.rm = TRUE)/n(),
cg1_edu_grp_geBS = sum(cg1_edu_grp == ">=MS", na.rm = TRUE)/n()
)) %>% write.csv("./table1_covaraite.csv", row.names = FALSE)

rbind(datlong %>% filter(!duplicated(ID)) %>% group_by(gender.v1) %>%
  summarize(n= n(), across(c("z_COI_nat", "z_ED_nat", "z_HE_nat", "z_SE_nat",
    "z_ED_EC_nat", "z_ED_EL_nat", "z_ED_ER_nat", "z_ED_SP_nat",
    "z_HE_EP_nat", "z_HE_HE_nat", "z_HE_HR_nat", "z_HE_SE_nat",
    "z_SE_EI_nat", "z_SE_EO_nat", "z_SE_ER_nat", "z_SE_HQ_nat",
    "z_SE_SR_nat", "z_SE_WL_nat", "p.mn.cohesion.mean.v1",
    "p.mn.safety.mean.v1", "p.mn.violence.mean.v1"), function(x){paste0(round

datlong %>% filter(!duplicated(ID)) %>%
  summarize(gender.v1 = "All",
    n= n(), across(c("z_COI_nat", "z_ED_nat", "z_HE_nat", "z_SE_nat",
    "z_ED_EC_nat", "z_ED_EL_nat", "z_ED_ER_nat", "z_ED_SP_nat",
    "z_HE_EP_nat", "z_HE_HE_nat", "z_HE_HR_nat", "z_HE_SE_nat",
    "z_SE_EI_nat", "z_SE_EO_nat", "z_SE_ER_nat", "z_SE_HQ_nat",
    "z_SE_SR_nat", "z_SE_WL_nat", "p.mn.cohesion.mean.v1",
    "p.mn.safety.mean.v1", "p.mn.violence.mean.v1"), function(x){paste0(round

#To do:
#Redo this below but without stratifying by I_frisat
#Need overall averages without stratifying my male and female.
#Make an ICC table for all the 20 models.

#Now summarize variables by gender and I_frisat
rbind(datlong %>% group_by(gender.v1, I_frisat, ID) %>% summarize(
  n = n(),
  act_totsleep.min = mean(act_totsleep.min, na.rm = TRUE),
  act_efficiency = mean(act_efficiency, na.rm = TRUE),
  act_awakening.min = mean(act_awakening.min, na.rm = TRUE),
  act_outbed.time_numeric = mean(act_outbed.time_numeric, na.rm = TRUE),
  act_onset.time_numeric = mean(act_onset.time_numeric, na.rm = TRUE)
) %>% group_by(gender.v1, I_frisat) %>%
  summarize(
    n = sum(n),
    act_totsleep.min = paste0(round(mean(

```

```

    act_totsleep.min, na.rm = TRUE
  ), 2), " (", round(sd(
    act_totsleep.min, na.rm = TRUE
  ), 2), ")"),
  act_efficiency = paste0(round(mean(
    act_efficiency, na.rm = TRUE
  ), 2), " (", round(sd(
    act_efficiency, na.rm = TRUE
  ), 2), ")"),
  act_awakening.min = paste0(round(mean(
    act_awakening.min, na.rm = TRUE
  ), 2), " (", round(sd(
    act_awakening.min, na.rm = TRUE
  ), 2), ")"),
  act_onset.time_numeric = paste0(
    mean(act_onset.time_numeric, na.rm = TRUE) %/% 1,
    ":",
    sprintf("%02d", (round(60 * (
      mean(act_onset.time_numeric, na.rm = TRUE) %/% 1
    )))),
    " (",
    sd(act_onset.time_numeric, na.rm = TRUE) %/% 1,
    ":",
    sprintf("%02d", round(60 * (
      sd(act_onset.time_numeric, na.rm = TRUE) %/% 1
    )))),
    ")"
  ),
  act_outbed.time_numeric = paste0(
    mean(act_outbed.time_numeric, na.rm = TRUE) %/% 1,
    ":",
    sprintf("%02d", (round(60 * (
      mean(act_outbed.time_numeric, na.rm = TRUE) %/% 1
    )))),
    " (",
    sd(act_outbed.time_numeric, na.rm = TRUE) %/% 1,
    ":",
    sprintf("%02d", round(60 * (
      sd(act_outbed.time_numeric, na.rm = TRUE) %/% 1
    )))),
    ")"
  )
) ,

```

```

datlong %>% group_by(gender.v1, ID) %>% summarize(
  n = n(),
  act_totsleep.min = mean(act_totsleep.min, na.rm = TRUE),
  act_efficiency = mean(act_efficiency, na.rm = TRUE),
  act_awakening.min = mean(act_awakening.min, na.rm = TRUE),
  act_outbed.time_numeric = mean(act_outbed.time_numeric, na.rm = TRUE),
  act_onset.time_numeric = mean(act_onset.time_numeric, na.rm = TRUE)
) %>% group_by(gender.v1) %>%
summarize(I_frisat = 2,
  n = sum(n),
  act_totsleep.min = paste0(round(mean(
    act_totsleep.min, na.rm = TRUE
  ), 2), " (", round(sd(
    act_totsleep.min, na.rm = TRUE
  ), 2), ")"),
  act_efficiency = paste0(round(mean(
    act_efficiency, na.rm = TRUE
  ), 2), " (", round(sd(
    act_efficiency, na.rm = TRUE
  ), 2), ")"),
  act_awakening.min = paste0(round(mean(
    act_awakening.min, na.rm = TRUE
  ), 2), " (", round(sd(
    act_awakening.min, na.rm = TRUE
  ), 2), ")"),
  act_onset.time_numeric = paste0(
    mean(act_onset.time_numeric, na.rm = TRUE) %/% 1,
    ":",
    sprintf("%02d", (round(60 * (
      mean(act_onset.time_numeric, na.rm = TRUE) %/% 1
    )))),
    " (",
    sd(act_onset.time_numeric, na.rm = TRUE) %/% 1,
    ":",
    sprintf("%02d", round(60 * (
      sd(act_onset.time_numeric, na.rm = TRUE) %/% 1
    )))),
    ")",
  ),
  act_outbed.time_numeric = paste0(
    mean(act_outbed.time_numeric, na.rm = TRUE) %/% 1,
    ":",

```



```

    sprintf("%02d", (round(60 * (
      mean(act_outbed.time_numeric, na.rm = TRUE) %% 1
    )))),
    " (",
    sd(act_outbed.time_numeric, na.rm = TRUE) %% 1,
    ":",
    sprintf("%02d", round(60 * (
      sd(act_outbed.time_numeric, na.rm = TRUE) %% 1
    ))),
    ")"
  )
) ,

datlong %>% group_by(ID) %>% summarize(
  n = n(),
  act_totsleep.min = mean(act_totsleep.min, na.rm = TRUE),
  act_efficiency = mean(act_efficiency, na.rm = TRUE),
  act_awakening.min = mean(act_awakening.min, na.rm = TRUE),
  act_outbed.time_numeric = mean(act_outbed.time_numeric, na.rm = TRUE),
  act_onset.time_numeric = mean(act_onset.time_numeric, na.rm = TRUE)
) %>%
summarize(gender.v1 = "All", I_frisat = 2,
  n = sum(n),
  act_totsleep.min = paste0(round(mean(
    act_totsleep.min, na.rm = TRUE
  ), 2), " (", round(sd(
    act_totsleep.min, na.rm = TRUE
  ), 2), ")"),
  act_efficiency = paste0(round(mean(
    act_efficiency, na.rm = TRUE
  ), 2), " (", round(sd(
    act_efficiency, na.rm = TRUE
  ), 2), ")"),
  act_awakening.min = paste0(round(mean(
    act_awakening.min, na.rm = TRUE
  ), 2), " (", round(sd(
    act_awakening.min, na.rm = TRUE
  ), 2), ")"),
  act_onset.time_numeric = paste0(
    mean(act_onset.time_numeric, na.rm = TRUE) %% 1,
    ":",
    sprintf("%02d", (round(60 * (
      mean(act_onset.time_numeric, na.rm = TRUE) %% 1

```

```

))),
" (",
sd(act_onset.time_numeric, na.rm = TRUE) %/% 1,
":",
sprintf("%02d", round(60 * (
  sd(act_onset.time_numeric, na.rm = TRUE) %/% 1
))),
")"
),
act_outbed.time_numeric = paste0(
  mean(act_outbed.time_numeric, na.rm = TRUE) %/% 1,
  ":",
  sprintf("%02d", (round(60 * (
    mean(act_outbed.time_numeric, na.rm = TRUE) %/% 1
  )))),
" (",
sd(act_outbed.time_numeric, na.rm = TRUE) %/% 1,
":",
sprintf("%02d", round(60 * (
  sd(act_outbed.time_numeric, na.rm = TRUE) %/% 1
))),
")"
)) ) %>% write.csv("./table1_individual_level_vars.csv", row.names = FALSE)

```

`summarise()` has grouped output by 'gender.v1', 'I_frisat'. You can override using the `.groups` argument.

`summarise()` has grouped output by 'gender.v1'. You can override using the `.groups` argument.

`summarise()` has grouped output by 'gender.v1'. You can override using the `.groups` argument.

```

#act_onset.time_numeric
#act_totsleep.min
#act_efficiency
#act_awakening.min
#act.outbed.time
#act.onset.time
#gender.v1 + age.v1 + income.v1.10000 + cg1_edu_grp + I_frisat

```

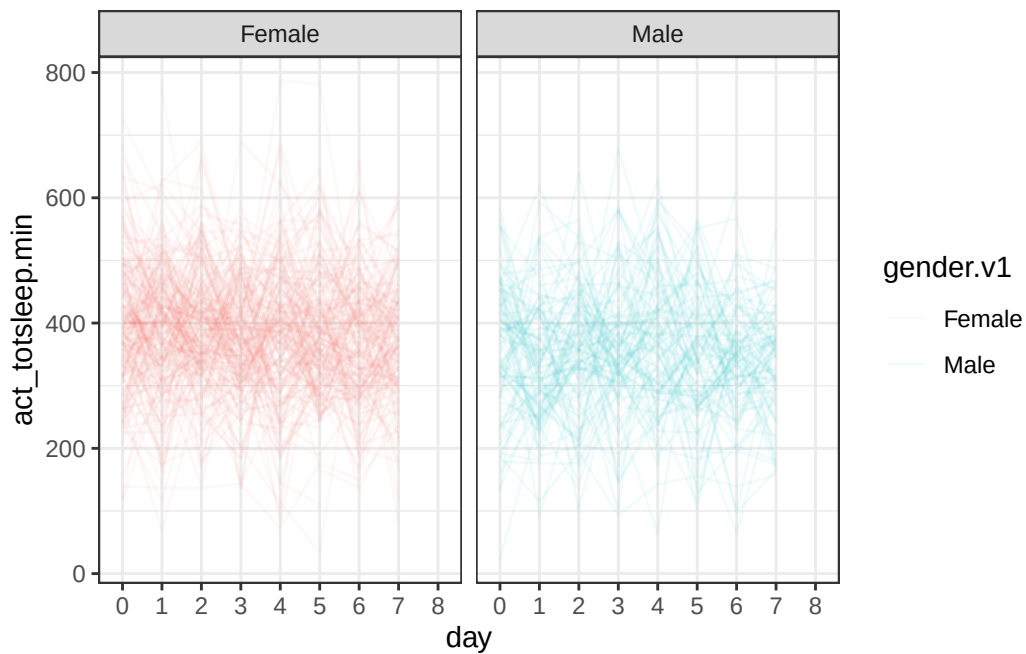
```
iccvec <- c()
```

Models

Total Sleep

```
ggplot(aes(x = day, y = act_totsleep.min, group = ID, color = gender.v1), data = datlong %>%
```

Warning: Removed 452 rows containing missing values or values outside the scale range (``geom_line()``).



```
library(lme4)
#####
#act_totalsleep.min
#####
#####
#Model 1: COI Total
#####
mod1_totsleep <- lmer(act_totsleep.min ~ gender.v1 + age.v1 + income.v1.10000 + cg1_edu_grp
summary(mod1_totsleep)
```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [

```

lmerModLmerTest]
Formula:
act_totsleep.min ~ gender.v1 + age.v1 + income.v1.10000 + cg1_edu_grp +
  I_frisat + z_COI_nat + (1 | ID)
Data: datlong

```

REML criterion at convergence: 27143

Scaled residuals:

Min	1Q	Median	3Q	Max
-3.3318	-0.5998	-0.0113	0.5645	4.2233

Random effects:

Groups	Name	Variance	Std.Dev.
ID	(Intercept)	2254	47.48
Residual		8045	89.69

Number of obs: 2269, groups: ID, 314

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	475.969	36.519	304.777	13.033	< 2e-16 ***
gender.v1Male	-32.434	7.064	302.846	-4.591	6.46e-06 ***
age.v1	-5.907	2.185	304.584	-2.703	0.00725 **
income.v1.10000	0.262	1.150	299.955	0.228	0.81987
cg1_edu_grp<=BS	-7.102	7.697	308.575	-0.923	0.35687
cg1_edu_grp>=MS	-3.796	12.857	307.151	-0.295	0.76799
I_frisat	35.286	4.240	1983.848	8.321	< 2e-16 ***
z_COI_nat	4.962	4.755	308.486	1.044	0.29743

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

	(Intr)	gnd.1M	age.v1	i.1.10	c1__<=	c1__>=	I_frst
gender.v1M1	-0.180						
age.v1	-0.975	0.130					
inc.1.10000	0.001	0.000	-0.111				
cg1_d_g<=BS	-0.084	-0.081	-0.044	-0.145			
cg1_d_g>=MS	-0.001	-0.153	-0.061	-0.254	0.468		
I_frisat	-0.019	0.002	-0.013	0.009	-0.003	-0.011	
z_COI_nat	0.248	-0.039	-0.168	-0.110	-0.059	-0.117	0.010

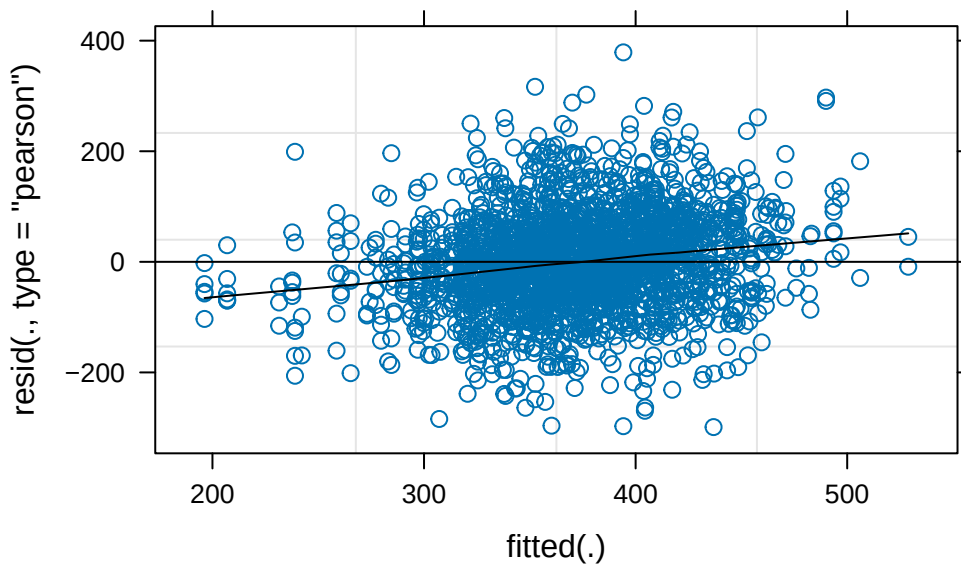
```
ci <- confint(mod1_totsleep)
```

Computing profile confidence intervals ...

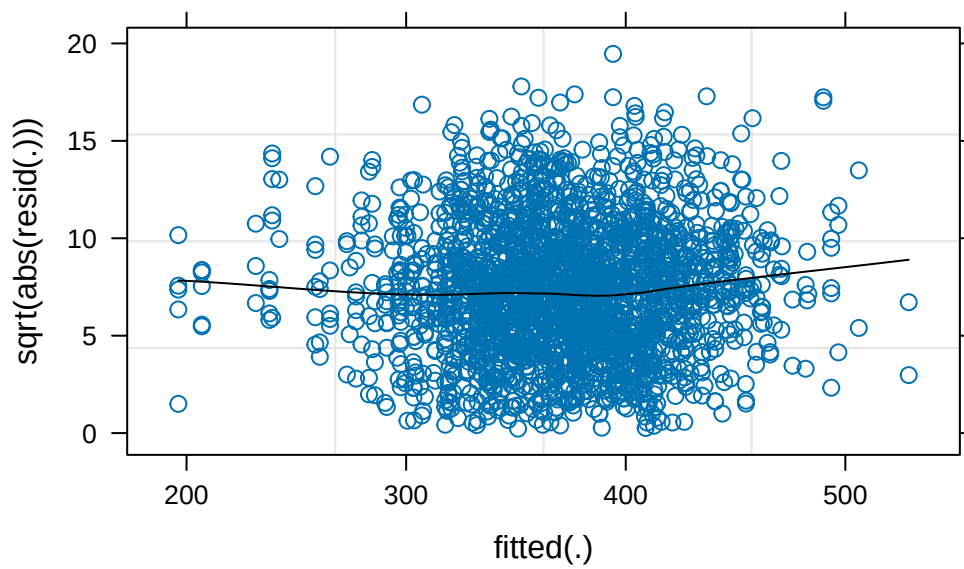
```
print(ci)
```

	2.5 %	97.5 %
.sig01	41.282439	52.543796
.sigma	86.931393	92.553988
(Intercept)	405.041365	547.029266
gender.v1Male	-46.165238	-18.703203
age.v1	-10.160143	-1.663527
income.v1.10000	-1.972485	2.497145
cg1_edu_grp<=BS	-22.058534	7.866421
cg1_edu_grp>=MS	-28.781692	21.201858
I_frisat	26.974400	43.599522
z_COI_nat	-4.282910	14.201264

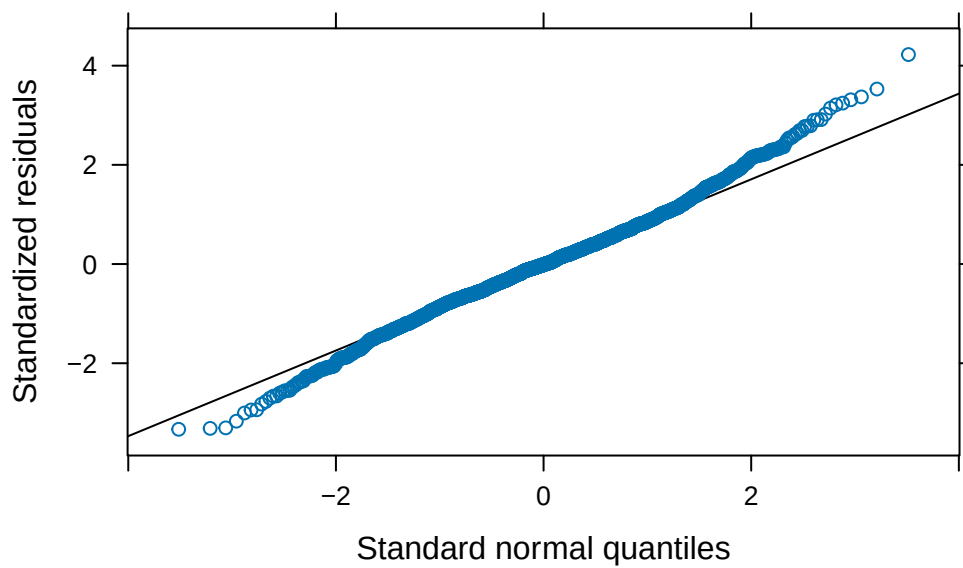
```
plot(mod1_totsleep, type=c("p","smooth"), col.line=1)
```



```
plot(mod1_totsleep,  
      sqrt(abs(resid(.)))~fitted(.),  
      type=c("p","smooth"), col.line=1)
```



```
lattice::qqmath(mod1_totsleep)
```



```
#Table
#Fixed effects
mod1_totsleep_table <- data.frame(
  predictor = attr(summary(mod1_totsleep)$coef,"dimnames")[[1]],
  B = summary(mod1_totsleep)$coef["Estimate"],
  beta = effectsize::standardize_parameters(mod1_totsleep)$Std_Coefficient,
  CI = paste0("(",round(ci[-c(1,2),1],4),", ",round(ci[-c(1,2),2],4),")"),
```

```

CI_STD = paste0("(",round(effectsize::standardize_parameters(mod1_totsleep)$CI_low,4),", ",round(effectsize::standardize_parameters(mod1_totsleep)$CI_high,4),")",", ",round(pval,4))
pval = summary(mod1_totsleep)$coef[,5]
)

mod1_totsleep_table[,c("B","beta","pval")] <-
round(mod1_totsleep_table[,c("B","beta","pval")],4)
write.csv(mod1_totsleep_table,file = "../bohnert_sleep_git/mod1_totsleep_table.csv")

#Random effects
#within
sigma2 <- summary(mod1_totsleep)$sigma^2
#between
tau00 <- attr(summary(mod1_totsleep)$varcor$ID,"stddev")^2
iccvec[1] <- ICC <- tau00/(tau00 + sigma2)
N <- summary(mod1_totsleep)$devcomp$dims["q"]
obs <- summary(mod1_totsleep)$devcomp$dims["n"]
R2marg <- r.squaredGLMM(mod1_totsleep)[,"R2m"]
R2cond <- r.squaredGLMM(mod1_totsleep)[,"R2c"]
mod1_totsleep_table_re <- data.frame(sigma2, tau00, ICC, N, obs, R2marg, R2cond)
mod1_totsleep_table_re

```

	sigma2	tau00	ICC	N	obs	R2marg	R2cond
(Intercept)	8044.628	2254.018	0.2188655	314	2269	0.05047032	0.2582896

```

#####
#Model 2: COI Sub-domains
#####
mod2_totsleep <- lmer(act_totsleep.min ~ gender.v1 + age.v1 + income.v1.10000 + cg1_edu_grp +
summary(mod2_totsleep)

```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [lmerModLmerTest]

Formula:

act_totsleep.min ~ gender.v1 + age.v1 + income.v1.10000 + cg1_edu_grp +
I_frisat + z_ED_nat + z_HE_nat + z_SE_nat + (1 | ID)

Data: datlong

REML criterion at convergence: 27129.9

Scaled residuals:

Min	1Q	Median	3Q	Max
-----	----	--------	----	-----

-3.3267 -0.6009 -0.0137 0.5619 4.2382

Random effects:

Groups	Name	Variance	Std.Dev.
ID	(Intercept)	2263	47.57
Residual		8044	89.69

Number of obs: 2269, groups: ID, 314

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	472.122	36.763	303.141	12.842	< 2e-16 ***
gender.v1Male	-32.382	7.088	300.743	-4.569	7.17e-06 ***
age.v1	-5.875	2.188	302.921	-2.685	0.00765 **
income.v1.10000	0.519	1.169	299.284	0.444	0.65738
cg1_edu_grp<=BS	-6.769	7.731	307.112	-0.876	0.38198
cg1_edu_grp>=MS	-3.484	12.881	305.539	-0.270	0.78698
I_frisat	35.293	4.240	1984.001	8.323	< 2e-16 ***
z_ED_nat	9.291	8.333	310.082	1.115	0.26576
z_HE_nat	4.856	8.007	300.027	0.606	0.54465
z_SE_nat	-5.469	7.007	305.613	-0.780	0.43571

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

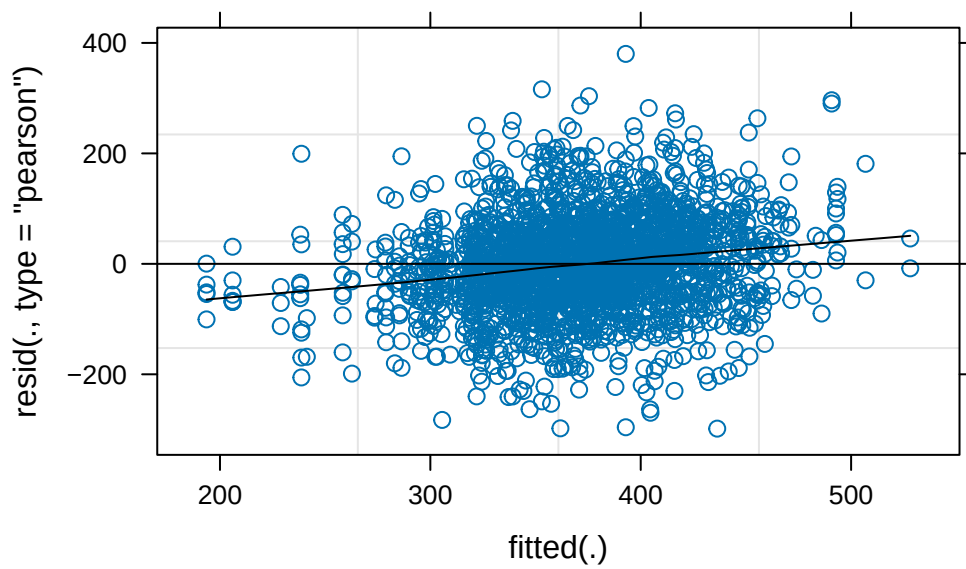
Correlation of Fixed Effects:

	(Intr)	gnd.1M	age.v1	i.1.10	c1__<=	c1__>=	I_frst	z_ED_n	z_HE_n
gender.v1M1	-0.175								
age.v1	-0.971	0.129							
inc.1.10000	-0.010	0.002	-0.109						
cg1_d_g<=BS	-0.081	-0.076	-0.044	-0.134					
cg1_d_g>=MS	0.000	-0.150	-0.061	-0.246	0.469				
I_frisat	-0.019	0.001	-0.013	0.009	-0.003	-0.011			
z_ED_nat	0.027	0.038	-0.031	0.131	0.061	0.014	0.000		
z_HE_nat	-0.072	-0.060	-0.001	0.034	-0.057	-0.033	0.006	-0.247	
z_SE_nat	0.155	-0.025	-0.067	-0.202	-0.060	-0.062	0.003	-0.702	-0.252

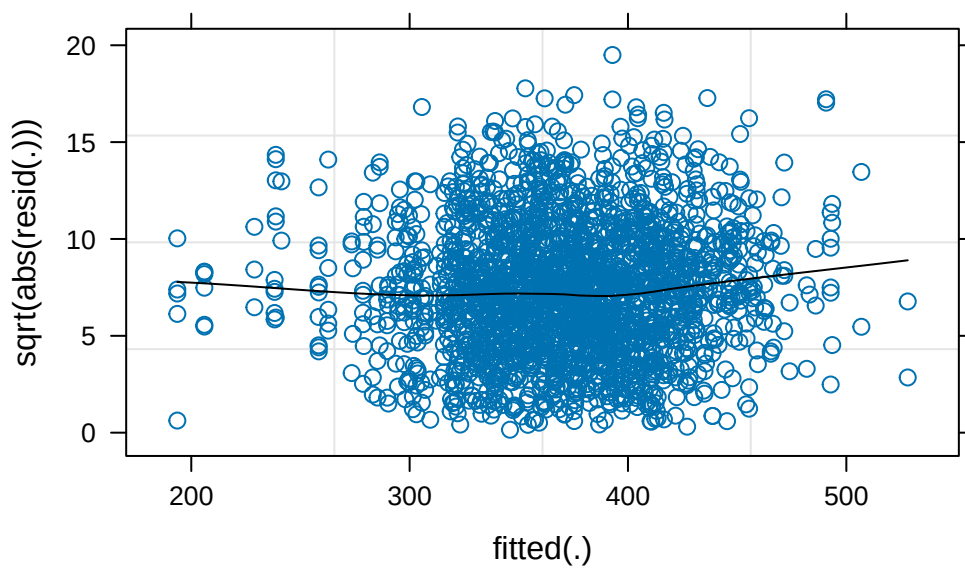
```
ci <- confint(mod2_totsleep)
```

Computing profile confidence intervals ...

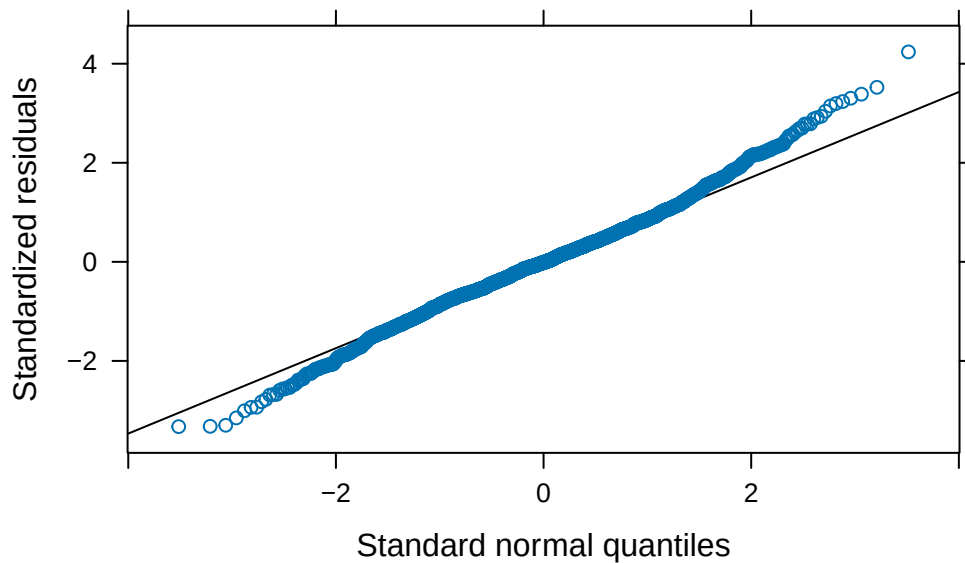
```
plot(mod2_totsleep, type=c("p","smooth"), col.line=1)
```

```
plot(mod2_totsleep,
     sqrt(abs(resid(.)))~fitted(.),
     type=c("p","smooth"), col.line=1)
```



```
lattice::qqmath(mod2_totsleep)
```



```

mod2_totsleep_table <- data.frame(
  predictor = attr(summary(mod2_totsleep)$coef,"dimnames")[[1]],
  B = summary(mod2_totsleep)$coef[,"Estimate"],
  beta = effectsize::standardize_parameters(mod2_totsleep)$Std_Coefficient,
  CI = paste0("(",round(ci[-c(1,2),1],4),", ",round(ci[-c(1,2),2],4),")"),
  CI_STD = paste0("(",round(effectsize::standardize_parameters(mod2_totsleep)$CI_low,4),", ",round(effectsize::standardize_parameters(mod2_totsleep)$CI_high,4),")"),
  pval = summary(mod2_totsleep)$coef[,5]
)

mod2_totsleep_table[,c("B","beta","pval")] <-
round(mod2_totsleep_table[,c("B","beta","pval")],4)
write.csv(mod2_totsleep_table,file = "../bohnert_sleep_git/mod2_totsleep_table.csv")

#Random effects
#within
sigma2 <- summary(mod2_totsleep)$sigma^2
#between
tau00 <- attr(summary(mod2_totsleep)$varcor$ID,"stddev")^2
iccvec[2] <- ICC <- tau00/(tau00 + sigma2)
N <- summary(mod2_totsleep)$devcomp$dims["q"]
obs <- summary(mod2_totsleep)$devcomp$dims["n"]
R2marg <- r.squaredGLMM(mod2_totsleep)[,"R2m"]
R2cond <- r.squaredGLMM(mod2_totsleep)[,"R2c"]
mod2_totsleep_table_re <- data.frame(sigma2, tau00, ICC, N, obs, R2marg, R2cond)
mod2_totsleep_table_re

```

	sigma2	tau00	ICC	N	obs	R2marg	R2cond
(Intercept)	8043.567	2262.57	0.2195362	314	2269	0.05186329	0.2600136

```
#####
#Model 3: COI Sub-domains
#####
mod3_totsleep <- lmer(act_totsleep.min ~ gender.v1 + age.v1 + income.v1.10000 + cg1_edu_grp +
  z_ED_EL_nat +
  z_ED_ER_nat +
  z_ED_SP_nat +
  z_HE_EP_nat +
  z_HE_HE_nat +
  z_HE_HR_nat +
  z_HE_SE_nat +
  z_SE_EI_nat +
  z_SE_EO_nat +
  z_SE_ER_nat +
  z_SE_HQ_nat +
  z_SE_SR_nat +
  z_SE_WL_nat + (1|ID) , data = datlong)
summary(mod3_totsleep)
```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]

Formula:

```
act_totsleep.min ~ gender.v1 + age.v1 + income.v1.10000 + cg1_edu_grp +
  I_frisat + z_ED_EC_nat + z_ED_EL_nat + z_ED_ER_nat + z_ED_SP_nat +
  z_HE_EP_nat + z_HE_HE_nat + z_HE_HR_nat + z_HE_SE_nat + z_SE_EI_nat +
  z_SE_EO_nat + z_SE_ER_nat + z_SE_HQ_nat + z_SE_SR_nat + z_SE_WL_nat +
  (1 | ID)
```

Data: datlong

REML criterion at convergence: 23607.4

Scaled residuals:

	Min	1Q	Median	3Q	Max
	-3.3259	-0.5955	-0.0199	0.5793	4.2658

Random effects:

Groups	Name	Variance	Std.Dev.
ID	(Intercept)	2063	45.42
Residual		8166	90.36

Number of obs: 1981, groups: ID, 276

Fixed effects:

	Estimate	Std. Error	df	t	value	Pr(> t)
(Intercept)	471.11204	39.90135	253.82101	11.807	< 2e-16	***
gender.v1Male	-36.49730	7.50684	251.94898	-4.862	2.05e-06	***
age.v1	-5.87411	2.31471	253.17204	-2.538	0.0118	*
income.v1.10000	0.13880	1.24567	250.40555	0.111	0.9114	
cg1_edu_grp<=BS	-12.48691	8.21592	256.20426	-1.520	0.1298	
cg1_edu_grp>=MS	-11.31765	13.91349	255.81184	-0.813	0.4167	
I_frisat	36.24695	4.57709	1729.24353	7.919	4.24e-15	***
z_ED_EC_nat	2.92514	4.06134	249.56059	0.720	0.4721	
z_ED_EL_nat	9.23692	9.19562	259.09170	1.004	0.3161	
z_ED_ER_nat	-7.68365	16.13967	255.04871	-0.476	0.6344	
z_ED_SP_nat	-0.21155	8.57179	254.32747	-0.025	0.9803	
z_HE_EP_nat	55.22773	30.74996	248.81670	1.796	0.0737	.
z_HE_HE_nat	18.17240	20.01274	255.52560	0.908	0.3647	
z_HE_HR_nat	-8.62337	8.23676	251.46520	-1.047	0.2961	
z_HE_SE_nat	8.44666	5.86308	257.08625	1.441	0.1509	
z_SE_EI_nat	0.04928	16.19976	255.04437	0.003	0.9976	
z_SE_EO_nat	-3.48177	10.49166	256.36155	-0.332	0.7403	
z_SE_ER_nat	-3.82071	10.92694	254.59181	-0.350	0.7269	
z_SE_HQ_nat	-6.11226	9.75836	251.54066	-0.626	0.5316	
z_SE_SR_nat	-8.61709	9.65132	251.74010	-0.893	0.3728	
z_SE_WL_nat	17.58839	11.41792	254.53085	1.540	0.1247	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

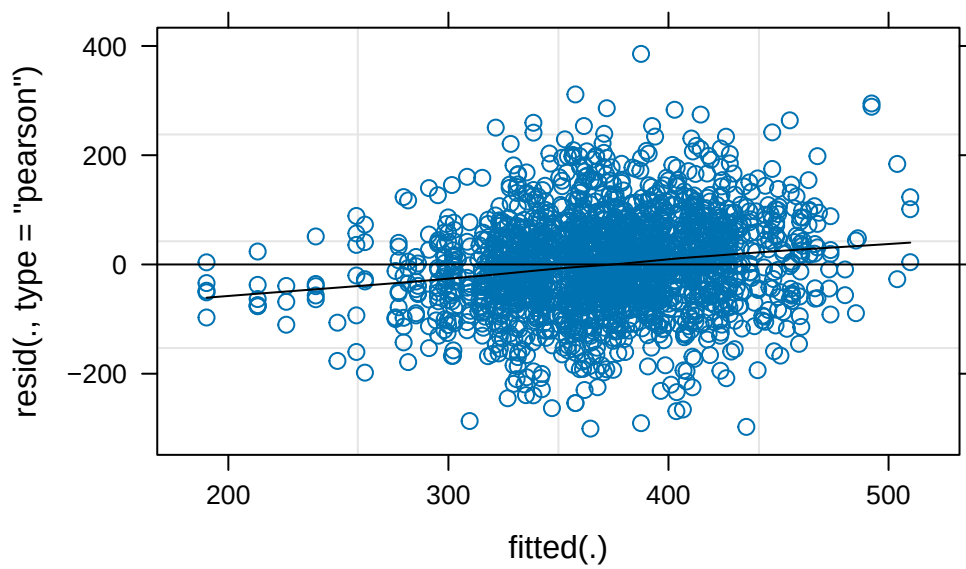
Correlation matrix not shown by default, as p = 21 > 12.

Use print(x, correlation=TRUE) or
vcov(x) if you need it

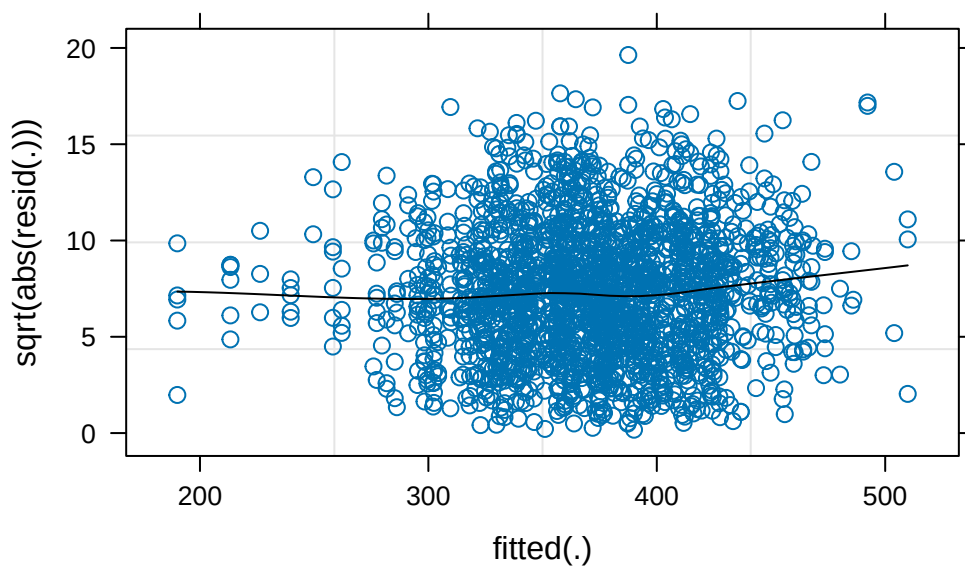
```
ci <- confint(mod3_totsleep)
```

Computing profile confidence intervals ...

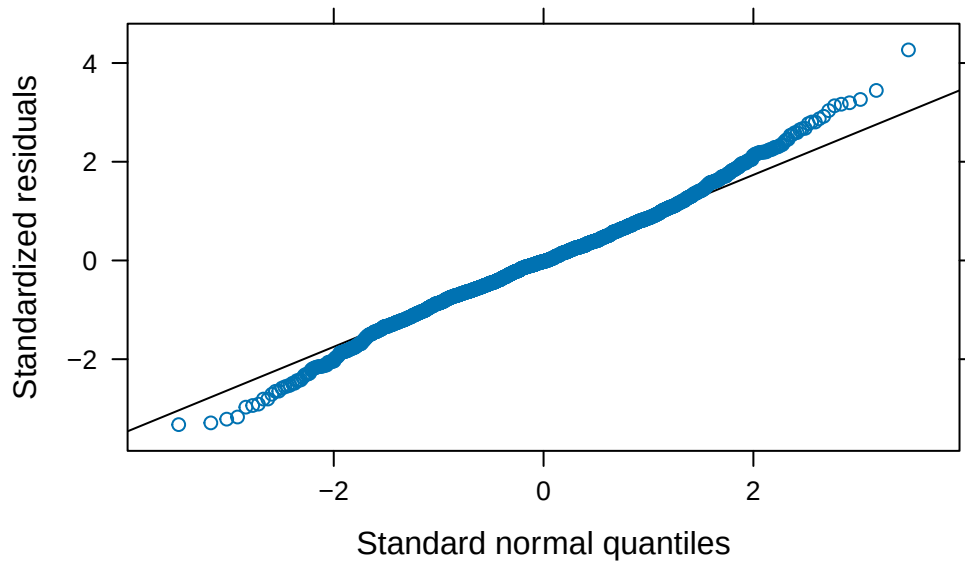
```
plot(mod3_totsleep, type=c("p","smooth"), col.line=1)
```



```
plot(mod3_totsleep,
      sqrt(abs(resid(.)))~fitted(.),
      type=c("p","smooth"), col.line=1)
```



```
lattice::qqmath(mod3_totsleep)
```



```
#Table
#Fixed Effects
mod3_totsleep_table <- data.frame(
  predictor = attr(summary(mod3_totsleep)$coef,"dimnames")[[1]],
  B = summary(mod3_totsleep)$coef["Estimate"],
  beta = effectsize::standardize_parameters(mod3_totsleep)$Std_Coefficient,
  CI = paste0("(",round(ci[-c(1,2),1],4),", ",round(ci[-c(1,2),2],4),")"),
  CI_STD = paste0("(",round(effectsize::standardize_parameters(mod3_totsleep)$CI_low,4),", ",round(effectsize::standardize_parameters(mod3_totsleep)$CI_high,4),")"),
  pval = summary(mod3_totsleep)$coef[,5]
)

mod3_totsleep_table[,c("B","beta","pval")] <-
round(mod3_totsleep_table[,c("B","beta","pval")],4)
write.csv(mod3_totsleep_table,file = "../bohnert_sleep_git/mod3_totsleep_table.csv")

#Random effects
#within
sigma2 <- summary(mod3_totsleep)$sigma^2
#between
tau00 <- attr(summary(mod3_totsleep)$varcor$ID,"stddev")^2
iccvec[3] <- ICC <- tau00/(tau00 + sigma2)
N <- summary(mod3_totsleep)$devcomp$dims["q"]
obs <- summary(mod3_totsleep)$devcomp$dims["n"]
R2marg <- r.squaredGLMM(mod3_totsleep)[,"R2m"]
R2cond <- r.squaredGLMM(mod3_totsleep)[,"R2c"]
mod3_totsleep_table_re <- data.frame(sigma2, tau00, ICC, N, obs, R2marg, R2cond)
```

```
mod3_totsleep_table_re
```

```
          sigma2    tau00      ICC    N  obs      R2marg    R2cond
(Intercept) 8165.7 2062.572 0.201654 276 1981 0.07031717 0.2577915
```

```
#####
```

```
#Model 4: Neighborhood
```

```
#####
```

```
mod4_totsleep <- lmer(act_totsleep.min ~ gender.v1 + age.v1 + income.v1.10000 + cg1_edu_grp +  
summary(mod4_totsleep)
```

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [  
lmerModLmerTest]
```

```
Formula:
```

```
act_totsleep.min ~ gender.v1 + age.v1 + income.v1.10000 + cg1_edu_grp +  
  I_frisat + +p.mn.cohesion.mean.v1 + p.mn.safety.mean.v1 +  
  p.mn.violence.mean.v1 + (1 | ID)
```

```
Data: datlong
```

```
REML criterion at convergence: 25947.8
```

```
Scaled residuals:
```

```
      Min      1Q  Median      3Q      Max  
-3.3585 -0.6002 -0.0134  0.5571  4.1889
```

```
Random effects:
```

```
Groups   Name      Variance Std.Dev.  
ID       (Intercept) 2303      47.99  
Residual                8089      89.94
```

```
Number of obs: 2169, groups: ID, 300
```

```
Fixed effects:
```

	Estimate	Std. Error	df	t value	Pr(> t)	
(Intercept)	454.9508	41.8484	292.4600	10.871	< 2e-16	***
gender.v1Male	-32.1336	7.2185	287.4302	-4.452	1.22e-05	***
age.v1	-5.0884	2.2851	290.2625	-2.227	0.0267	*
income.v1.10000	0.3635	1.1871	286.2637	0.306	0.7597	
cg1_edu_grp<=BS	-4.0916	8.1558	293.9457	-0.502	0.6163	
cg1_edu_grp>=MS	1.7474	13.4738	291.2471	0.130	0.8969	
I_frisat	34.5395	4.3496	1896.6053	7.941	3.41e-15	***
p.mn.cohesion.mean.v1	-0.5889	5.3763	290.0948	-0.110	0.9129	

```

p.mn.safety.mean.v1      -1.1548      5.0320  289.5823  -0.229   0.8186
p.mn.violence.mean.v1    3.7967      5.6250  290.0395   0.675   0.5002
---

```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

```

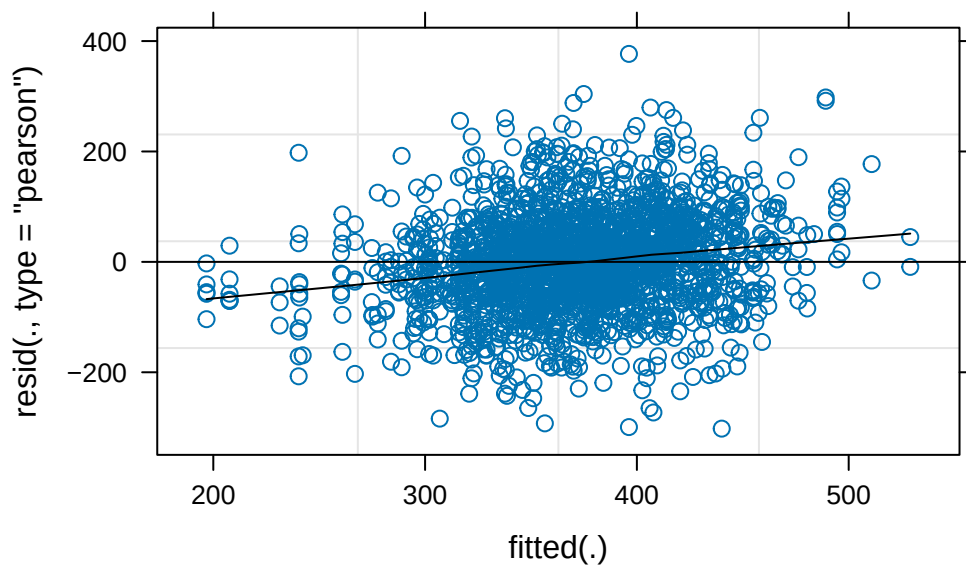
(Intr) gnd.1M age.v1 i.1.10 c1__<= c1__>= I_first p.mn.c..1
gender.v1M1 -0.101
age.v1      -0.816  0.107
inc.1.10000 0.051  0.001 -0.115
cg1_d_g<=BS -0.035 -0.063 -0.106 -0.168
cg1_d_g>=MS  0.079 -0.145 -0.098 -0.272  0.490
I_frisat    -0.019  0.002 -0.013  0.010 -0.002 -0.010
p.mn.chs..1 -0.184 -0.011 -0.034  0.056 -0.085 -0.171  0.003
p.mn.sft..1 -0.207 -0.042 -0.058 -0.162  0.097  0.081 -0.001 -0.558
p.mn.vln..1 -0.420 -0.085 -0.056 -0.022  0.049 -0.006  0.002 -0.014
p.mn.s..1
gender.v1M1
age.v1
inc.1.10000
cg1_d_g<=BS
cg1_d_g>=MS
I_frisat
p.mn.chs..1
p.mn.sft..1
p.mn.vln..1  0.576

```

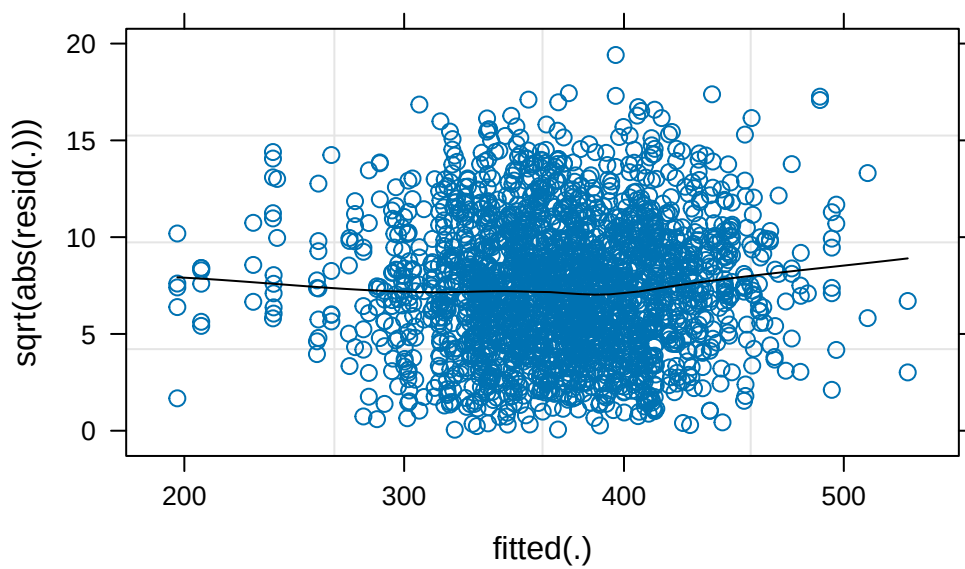
```
ci <- confint(mod4_totsleep)
```

Computing profile confidence intervals ...

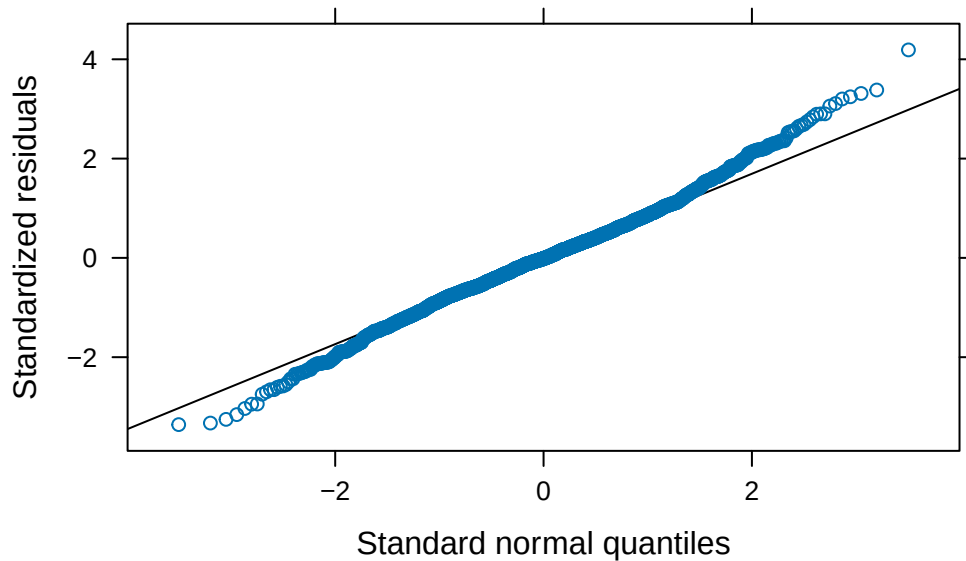
```
plot(mod4_totsleep, type=c("p","smooth"), col.line=1)
```

```
plot(mod4_totsleep,
     sqrt(abs(resid(.)))~fitted(.),
     type=c("p","smooth"), col.line=1)
```



```
lattice::qqmath(mod4_totsleep)
```



```
#Table
#Fixed Effects
mod4_totsleep_table <- data.frame(
  predictor = attr(summary(mod4_totsleep)$coef,"dimnames")[[1]],
  B = summary(mod4_totsleep)$coef["Estimate"],
  beta = effectsize::standardize_parameters(mod4_totsleep)$Std_Coefficient,
  CI = paste0("(",round(ci[-c(1,2),1],4),", ",round(ci[-c(1,2),2],4),")"),
  CI_STD = paste0("(",round(effectsize::standardize_parameters(mod4_totsleep)$CI_low,4),", ",round(effectsize::standardize_parameters(mod4_totsleep)$CI_high,4),")"),
  pval = summary(mod4_totsleep)$coef[,5]
)

mod4_totsleep_table[,c("B","beta","pval")] <-
round(mod4_totsleep_table[,c("B","beta","pval")],4)
write.csv(mod4_totsleep_table,file = "../bohnert_sleep_git/mod4_totsleep_table.csv")

#Random effects
#within
sigma2 <- summary(mod4_totsleep)$sigma^2
#between
tau00 <- attr(summary(mod4_totsleep)$varcor$ID,"stddev")^2
iccvec[4] <- ICC <- tau00/(tau00 + sigma2)
N <- summary(mod4_totsleep)$devcomp$dims["q"]
obs <- summary(mod4_totsleep)$devcomp$dims["n"]
R2marg <- r.squaredGLMM(mod4_totsleep)[,"R2m"]
R2cond <- r.squaredGLMM(mod4_totsleep)[,"R2c"]
mod4_totsleep_table_re <- data.frame(sigma2, tau00, ICC, N, obs, R2marg, R2cond)
```

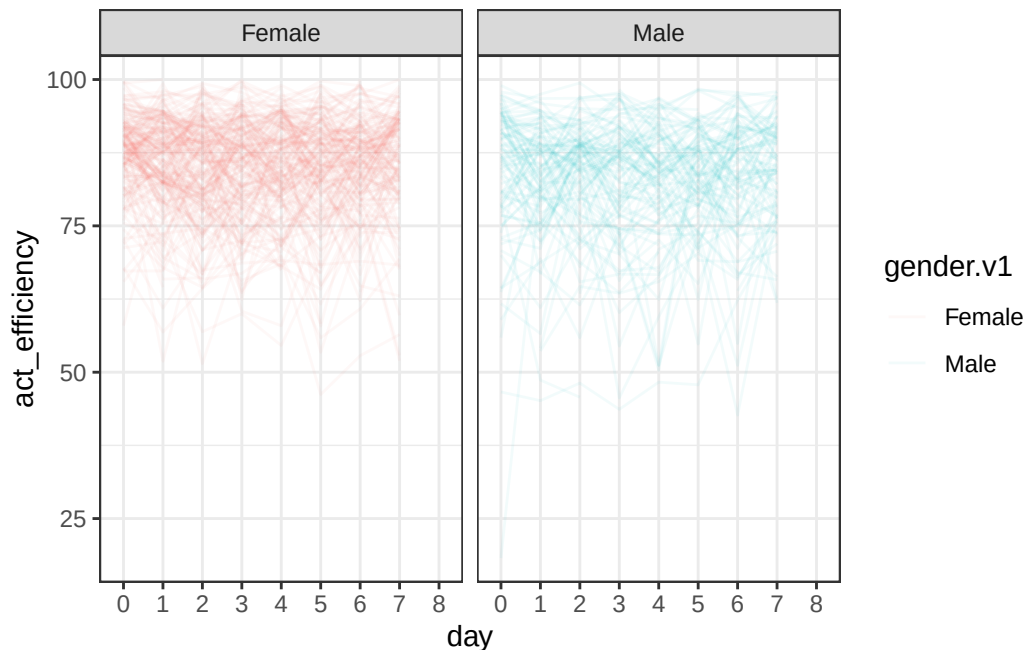
```
mod4_totsleep_table_re
```

```
          sigma2    tau00      ICC    N obs      R2marg    R2cond  
(Intercept) 8089.284 2302.851 0.2215956 300 2169 0.04877636 0.2595633
```

Sleep Efficiency

```
ggplot(aes(x = day, y = act_efficiency, group = ID, color = gender.v1), data = datlong %>% f
```

Warning: Removed 452 rows containing missing values or values outside the scale range (`geom\_line()`).



```
library(lme4)  
#####  
#act_efficiency  
#####  
#####  
#Model 1: COI Total  
#####  
mod1_eff <- lmer(act_efficiency ~ gender.v1 + age.v1 + income.v1.10000 + cg1_edu_grp + I_fri  
summary(mod1_eff)
```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [
 lmerModLmerTest]
 Formula: act_efficiency ~ gender.v1 + age.v1 + income.v1.10000 + cg1_edu_grp +
 I_frisat + z_COI_nat + (1 | ID)
 Data: datlong

REML criterion at convergence: 15683.2

Scaled residuals:

Min	1Q	Median	3Q	Max
-8.3086	-0.4821	0.1151	0.5782	2.9123

Random effects:

Groups	Name	Variance	Std.Dev.
ID	(Intercept)	39.61	6.293
	Residual	44.89	6.700

Number of obs: 2269, groups: ID, 314

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	96.41085	4.25015	306.48683	22.684	< 2e-16 ***
gender.v1Male	-2.28616	0.82316	304.99363	-2.777	0.00582 **
age.v1	-0.77169	0.25436	306.27768	-3.034	0.00262 **
income.v1.10000	0.39939	0.13421	303.25607	2.976	0.00316 **
cg1_edu_grp<=BS	-0.05268	0.89358	309.36931	-0.059	0.95303
cg1_edu_grp>=MS	-0.79144	1.49467	307.27587	-0.530	0.59684
I_frisat	0.54148	0.31751	1967.75374	1.705	0.08828 .
z_COI_nat	0.12900	0.55231	308.05857	0.234	0.81547

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

	(Intr)	gnd.1M	age.v1	i.1.10	c1__<=	c1__>=	I_frst
gender.v1M1	-0.181						
age.v1	-0.975	0.131					
inc.1.10000	0.000	-0.002	-0.110				
cg1_d_g<=BS	-0.082	-0.084	-0.045	-0.147			
cg1_d_g>=MS	0.001	-0.154	-0.061	-0.255	0.465		
I_frisat	-0.012	0.001	-0.009	0.006	-0.003	-0.008	
z_COI_nat	0.249	-0.038	-0.170	-0.111	-0.054	-0.110	0.007

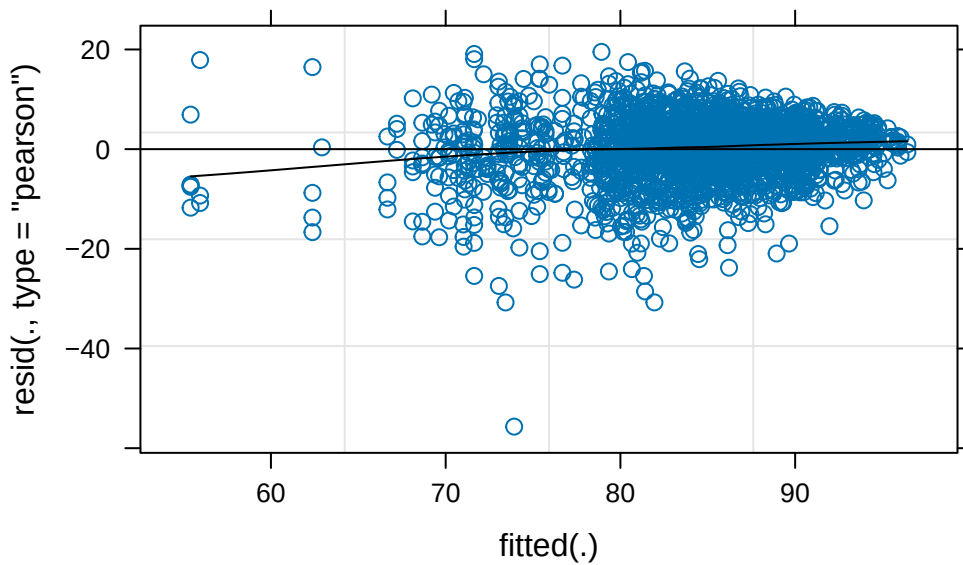
```
ci <- confint(mod1_eff)
```

Computing profile confidence intervals ...

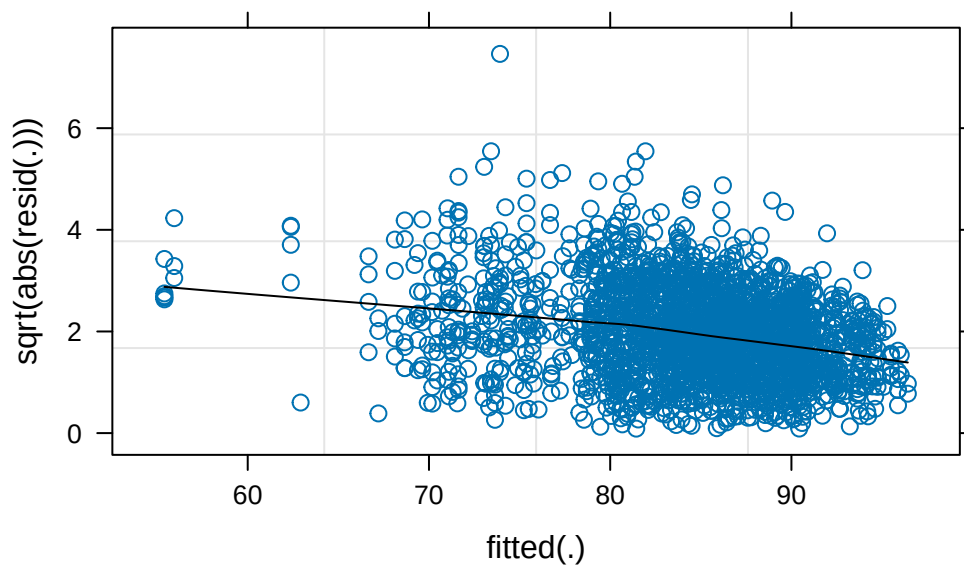
```
ci
```

	2.5 %	97.5 %
.sig01	5.67402671	6.8147459
.sigma	6.49395426	6.9141021
(Intercept)	88.14937811	104.6729987
gender.v1Male	-3.88639793	-0.6861656
age.v1	-1.26616320	-0.2772831
income.v1.10000	0.13853843	0.6602846
cg1_edu_grp<=BS	-1.78953645	1.6845914
cg1_edu_grp>=MS	-3.69669709	2.1142744
I_frisat	-0.08132475	1.1635536
z_COI_nat	-0.94453612	1.2027634

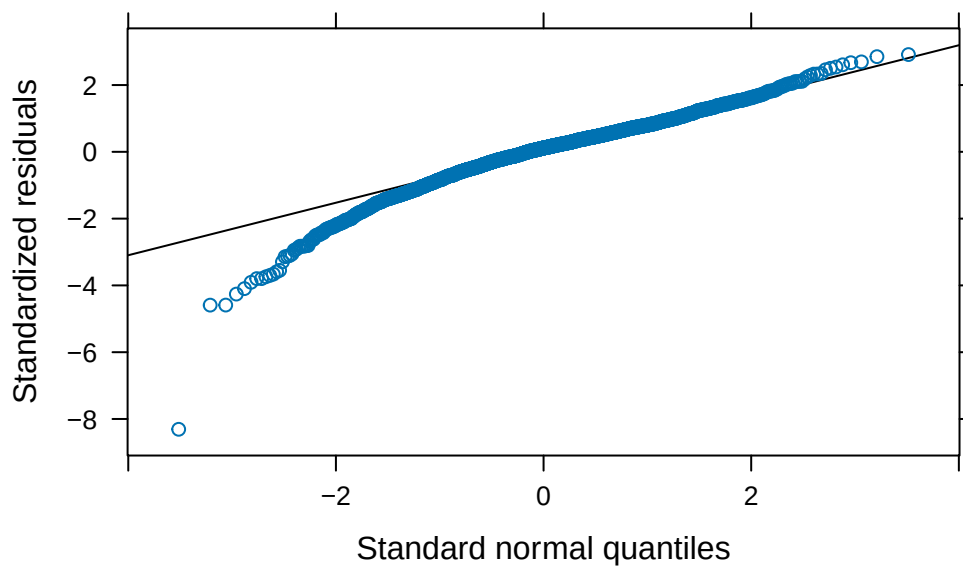
```
plot(mod1_eff, type=c("p","smooth"), col.line=1)
```



```
plot(mod1_eff,
      sqrt(abs(resid(.)))~fitted(.),
      type=c("p","smooth"), col.line=1)
```



```
lattice::qqmath(mod1_eff)
```



```
#Table
#fixed effects
mod1_eff_table <- data.frame(
  predictor = attr(summary(mod1_eff)$coef,"dimnames")[[1]],
  B = summary(mod1_eff)$coef["Estimate"],
  beta = effectsize::standardize_parameters(mod1_eff)$Std_Coefficient,
  CI = paste0("(",round(ci[-c(1,2),1],4),", ",round(ci[-c(1,2),2],4),")"),
```

```

CI_STD = paste0("(",round(effectsize::standardize_parameters(mod1_eff)$CI_low,4),", ",round(
pval = summary(mod1_eff)$coef[,5]
)

mod1_eff_table[,c("B","beta","pval")] <-
round(mod1_eff_table[,c("B","beta","pval")],4)
write.csv(mod1_eff_table,file = "../bohnert_sleep_git/mod1_eff_table.csv")

#Random effects
#within
sigma2 <- summary(mod1_eff)$sigma^2
#between
tau00 <- attr(summary(mod1_eff)$varcor$ID,"stddev")^2
iccvec[5] <- ICC <- tau00/(tau00 + sigma2)
N <- summary(mod1_eff)$devcomp$dims["q"]
obs <- summary(mod1_eff)$devcomp$dims["n"]
R2marg <- r.squaredGLMM(mod1_eff)[,"R2m"]
R2cond <- r.squaredGLMM(mod1_eff)[,"R2c"]
mod1_eff_table_re <- data.frame(sigma2, tau00, ICC, N, obs, R2marg, R2cond)
mod1_eff_table_re

```

	sigma2	tau00	ICC	N	obs	R2marg	R2cond
(Intercept)	44.8932	39.60511	0.4687089	314	2269	0.03813267	0.4889685

```

#####
#Model 2: COI Sub-domains
#####
mod2_eff <- lmer(act_efficiency ~ gender.v1 + age.v1 + income.v1.10000 + cg1_edu_grp + I_fr
summary(mod2_eff)

```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [lmerModLmerTest]

Formula: act_efficiency ~ gender.v1 + age.v1 + income.v1.10000 + cg1_edu_grp + I_frisat + z_ED_nat + z_HE_nat + z_SE_nat + (1 | ID)

Data: datlong

REML criterion at convergence: 15678.3

Scaled residuals:

	Min	1Q	Median	3Q	Max
	-8.3022	-0.4814	0.1084	0.5790	2.9094

Random effects:

Groups	Name	Variance	Std.Dev.
ID	(Intercept)	39.62	6.294
Residual		44.89	6.700

Number of obs: 2269, groups: ID, 314

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	95.74491	4.27351	304.72206	22.404	< 2e-16 ***
gender.v1Male	-2.33987	0.82518	302.86253	-2.836	0.00488 **
age.v1	-0.76516	0.25438	304.40613	-3.008	0.00285 **
income.v1.10000	0.41763	0.13623	301.92732	3.066	0.00237 **
cg1_edu_grp<=BS	-0.08532	0.89650	307.49059	-0.095	0.92424
cg1_edu_grp>=MS	-0.80566	1.49577	305.39243	-0.539	0.59054
I_frisat	0.54277	0.31751	1967.80345	1.709	0.08753 .
z_ED_nat	0.08104	0.96485	308.83441	0.084	0.93311
z_HE_nat	1.25837	0.93186	304.08137	1.350	0.17790
z_SE_nat	-0.56551	0.81339	306.03221	-0.695	0.48743

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

	(Intr)	gnd.1M	age.v1	i.1.10	c1__<=	c1__>=	I_frst	z_ED_n	z_HE_n
gender.v1M1	-0.175								
age.v1	-0.971	0.130							
inc.1.10000	-0.011	0.001	-0.108						
cg1_d_g<=BS	-0.079	-0.078	-0.045	-0.136					
cg1_d_g>=MS	0.002	-0.151	-0.062	-0.247	0.467				
I_frisat	-0.012	0.001	-0.009	0.006	-0.003	-0.008			
z_ED_nat	0.027	0.040	-0.031	0.128	0.062	0.015	0.000		
z_HE_nat	-0.072	-0.063	-0.002	0.033	-0.053	-0.030	0.004	-0.248	
z_SE_nat	0.155	-0.025	-0.067	-0.199	-0.060	-0.061	0.002	-0.701	-0.255

```
ci <- confint(mod2_eff)
```

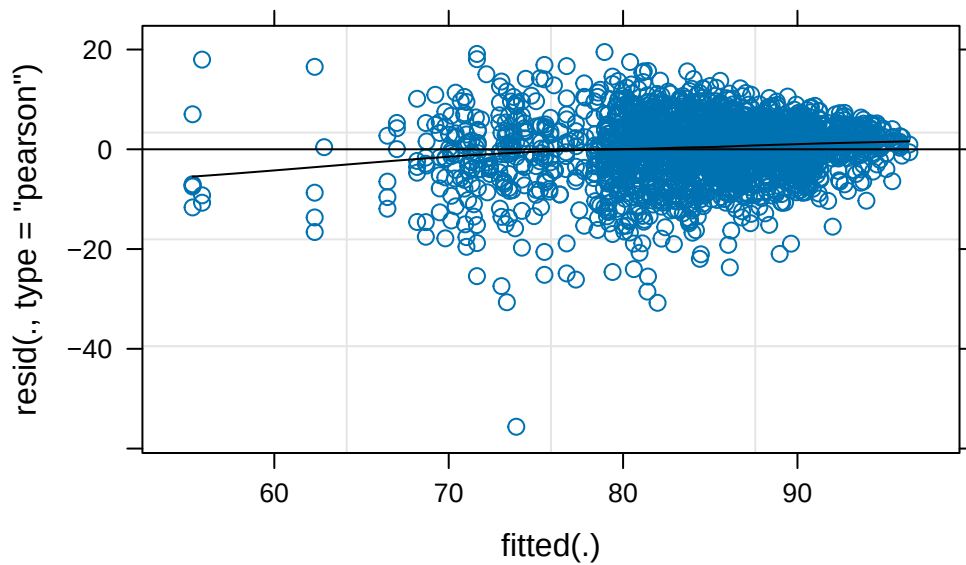
Computing profile confidence intervals ...

```
ci
```

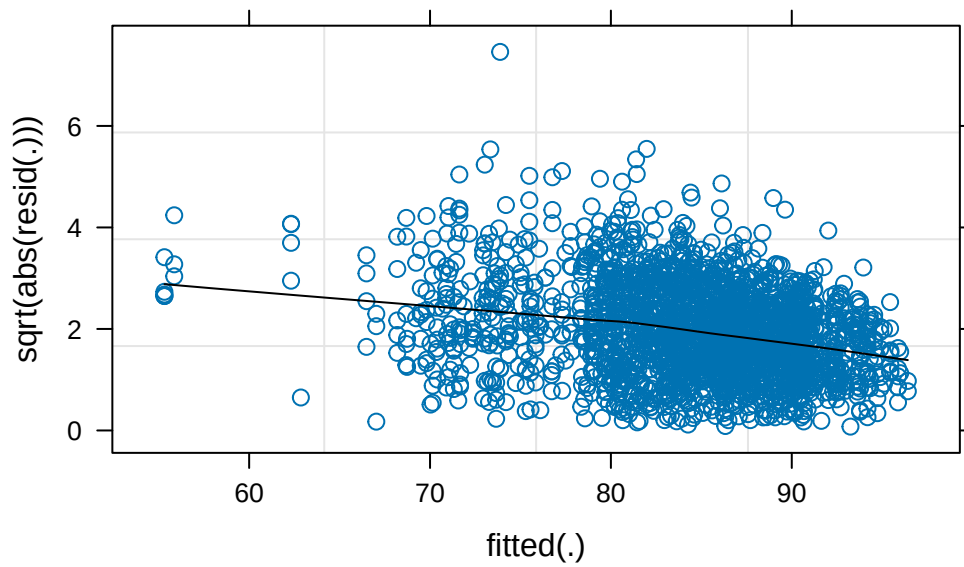
2.5 % 97.5 %

.sig01	5.65285098	6.7904122
.sigma	6.49388673	6.9140182
(Intercept)	87.46524023	104.0256198
gender.v1Male	-3.93876203	-0.7411832
age.v1	-1.25808908	-0.2723225
income.v1.10000	0.15375559	0.6816335
cg1_edu_grp<=BS	-1.82203031	1.6521353
cg1_edu_grp>=MS	-3.70346416	2.0928706
I_frisat	-0.08009454	1.1647623
z_ED_nat	-1.78728482	1.9519240
z_HE_nat	-0.54771696	3.0633544
z_SE_nat	-2.14211255	1.0099677

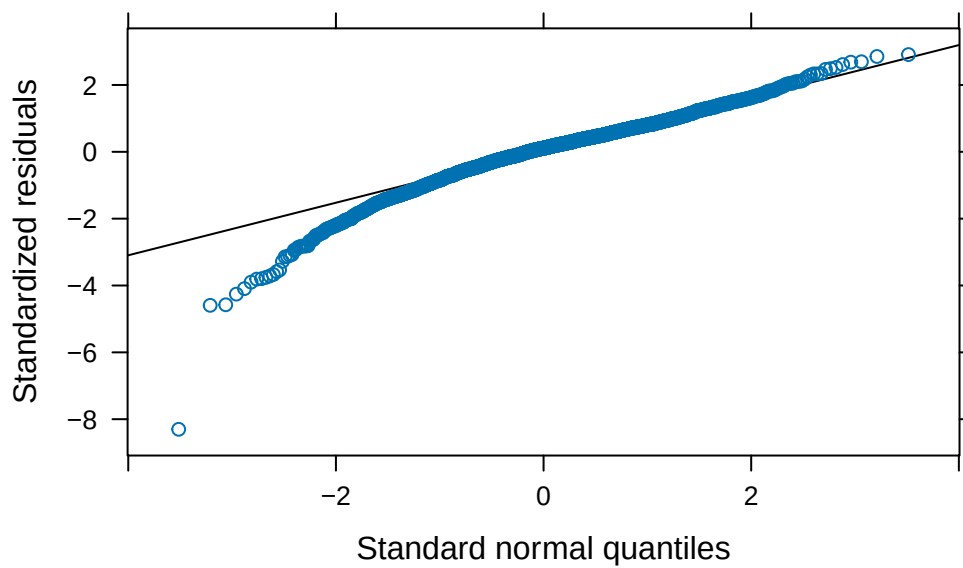
```
plot(mod2_eff, type=c("p","smooth"), col.line=1)
```



```
plot(mod2_eff,
      sqrt(abs(resid(.)))~fitted(.),
      type=c("p","smooth"), col.line=1)
```



```
lattice::qqmath(mod2_eff)
```



```
#Table
#fixed effects
mod2_eff_table <- data.frame(
  predictor = attr(summary(mod2_eff)$coef,"dimnames")[[1]],
  B = summary(mod2_eff)$coef[, "Estimate"],
  beta = effectsize::standardize_parameters(mod2_eff)$Std_Coefficient,
  CI = paste0("(",round(ci[-c(1,2),1],4),", ",round(ci[-c(1,2),2],4),")"),
```

```

CI_STD = paste0("(",round(effectsize::standardize_parameters(mod2_eff)$CI_low,4),", ",round(
pval = summary(mod2_eff)$coef[,5]
)

mod2_eff_table[,c("B","beta","pval")] <-
round(mod2_eff_table[,c("B","beta","pval")],4)
write.csv(mod2_eff_table,file = "../bohnert_sleep_git/mod2_eff_table.csv")

#Random effects
#within
sigma2 <- summary(mod2_eff)$sigma^2
#between
tau00 <- attr(summary(mod2_eff)$varcor$ID,"stddev")^2
iccvec[6] <- ICC <- tau00/(tau00 + sigma2)
N <- summary(mod2_eff)$devcomp$dims["q"]
obs <- summary(mod2_eff)$devcomp$dims["n"]
R2marg <- r.squaredGLMM(mod2_eff)[,"R2m"]
R2cond <- r.squaredGLMM(mod2_eff)[,"R2c"]
mod2_eff_table_re <- data.frame(sigma2, tau00, ICC, N, obs, R2marg, R2cond)
mod2_eff_table_re

```

	sigma2	tau00	ICC	N	obs	R2marg	R2cond
(Intercept)	44.89236	39.61737	0.4687906	314	2269	0.0414154	0.4907909

```

#####
#Model 3: COI Sub-domains
#####
mod3_eff <- lmer(act_efficiency ~ gender.v1 + age.v1 + income.v1.10000 + cg1_edu_grp +I_fris
  z_ED_EL_nat +
  z_ED_ER_nat +
  z_ED_SP_nat +
  z_HE_EP_nat +
  z_HE_HE_nat +
  z_HE_HR_nat +
  z_HE_SE_nat +
  z_SE_EI_nat +
  z_SE_EO_nat +
  z_SE_ER_nat +
  z_SE_HQ_nat +
  z_SE_SR_nat +
  z_SE_WL_nat + (1|ID) , data = datlong)
summary(mod3_eff)

```

```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: act_efficiency ~ gender.v1 + age.v1 + income.v1.10000 + cg1_edu_grp +
  I_frisat + z_ED_EC_nat + z_ED_EL_nat + z_ED_ER_nat + z_ED_SP_nat +
  z_HE_EP_nat + z_HE_HE_nat + z_HE_HR_nat + z_HE_SE_nat + z_SE_EI_nat +
  z_SE_EO_nat + z_SE_ER_nat + z_SE_HQ_nat + z_SE_SR_nat + z_SE_WL_nat +
  (1 | ID)
Data: datlong

```

REML criterion at convergence: 13691.8

Scaled residuals:

Min	1Q	Median	3Q	Max
-8.2318	-0.4821	0.1081	0.5783	2.8276

Random effects:

Groups	Name	Variance	Std.Dev.
ID	(Intercept)	38.69	6.220
	Residual	46.08	6.788

Number of obs: 1981, groups: ID, 276

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	94.9906	4.7242	256.2787	20.107	< 2e-16 ***
gender.v1Male	-2.3465	0.8902	254.6794	-2.636	0.00890 **
age.v1	-0.7766	0.2742	255.7424	-2.832	0.00499 **
income.v1.10000	0.4352	0.1479	253.7589	2.943	0.00355 **
cg1_edu_grp<=BS	-0.7900	0.9707	258.4183	-0.814	0.41648
cg1_edu_grp>=MS	-1.3202	1.6455	256.8543	-0.802	0.42311
I_frisat	0.5096	0.3446	1716.1049	1.479	0.13938
z_ED_EC_nat	0.3913	0.4821	254.4343	0.812	0.41772
z_ED_EL_nat	-1.4322	1.0849	259.0874	-1.320	0.18794
z_ED_ER_nat	0.3301	1.9087	257.3760	0.173	0.86283
z_ED_SP_nat	0.8104	1.0143	256.7667	0.799	0.42503
z_HE_EP_nat	9.3468	3.6546	252.8229	2.558	0.01113 *
z_HE_HE_nat	0.3055	2.3665	257.2450	0.129	0.89738
z_HE_HR_nat	-0.9774	0.9768	254.8380	-1.001	0.31794
z_HE_SE_nat	0.8567	0.6922	258.9907	1.238	0.21700
z_SE_EI_nat	1.7477	1.9159	257.2854	0.912	0.36250
z_SE_EO_nat	-2.3731	1.2402	257.3271	-1.913	0.05680 .
z_SE_ER_nat	-0.6598	1.2928	256.9126	-0.510	0.61022
z_SE_HQ_nat	2.7816	1.1572	254.8436	2.404	0.01694 *
z_SE_SR_nat	-1.1027	1.1443	255.0794	-0.964	0.33614

```

z_SE_WL_nat      0.1173      1.3516 256.1105    0.087 0.93090
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Correlation matrix not shown by default, as $p = 21 > 12$.
 Use `print(x, correlation=TRUE)` or
`vcov(x)` if you need it

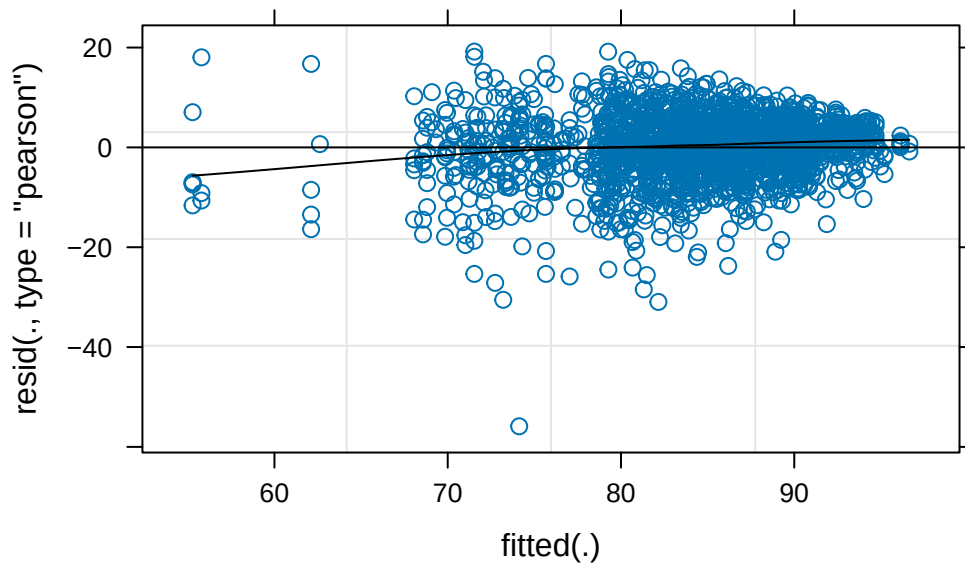
```
ci <- confint(mod3_eff)
```

Computing profile confidence intervals ...

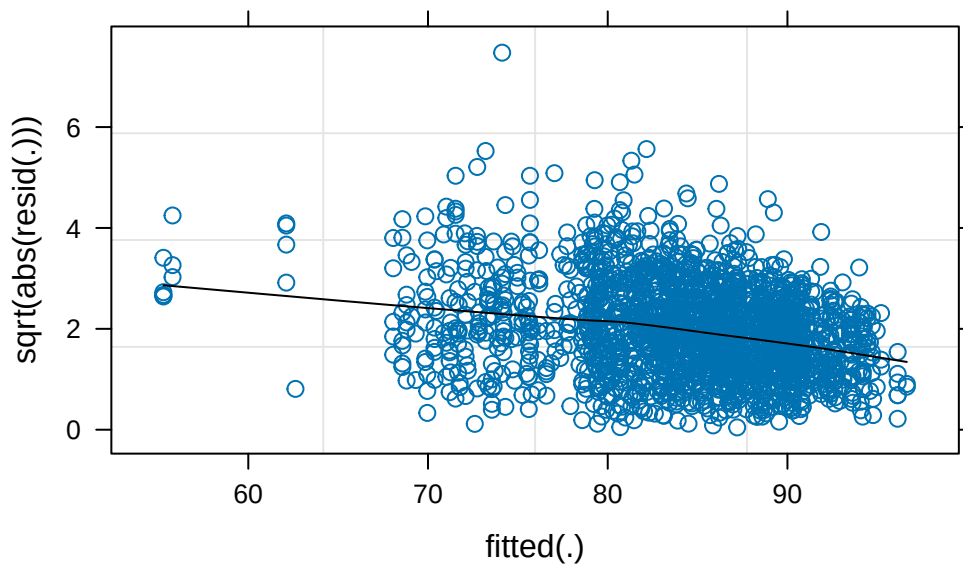
```
ci
```

	2.5 %	97.5 %
.sig01	5.3935417	6.58025171
.sigma	6.5643150	7.02001624
(Intercept)	86.0455716	103.94310596
gender.v1Male	-4.0331170	-0.66102487
age.v1	-1.2962024	-0.25744831
income.v1.10000	0.1554112	0.71556306
cg1_edu_grp<=BS	-2.6258077	1.05204159
cg1_edu_grp>=MS	-4.4356222	1.79857224
I_frisat	-0.1667496	1.18423941
z_ED_EC_nat	-0.5216992	1.30461669
z_ED_EL_nat	-3.4865276	0.62417151
z_ED_ER_nat	-3.2858759	3.94579841
z_ED_SP_nat	-1.1063508	2.73664866
z_HE_EP_nat	2.4183325	16.26093525
z_HE_HE_nat	-4.1785595	4.78751560
z_HE_HR_nat	-2.8288042	0.87145852
z_HE_SE_nat	-0.4562883	2.16673833
z_SE_EI_nat	-1.8868350	5.37206295
z_SE_EO_nat	-4.7207647	-0.02188028
z_SE_ER_nat	-3.1056988	1.79230112
z_SE_HQ_nat	0.5897884	4.97354735
z_SE_SR_nat	-3.2689072	1.06597601
z_SE_WL_nat	-2.4468656	2.67353498

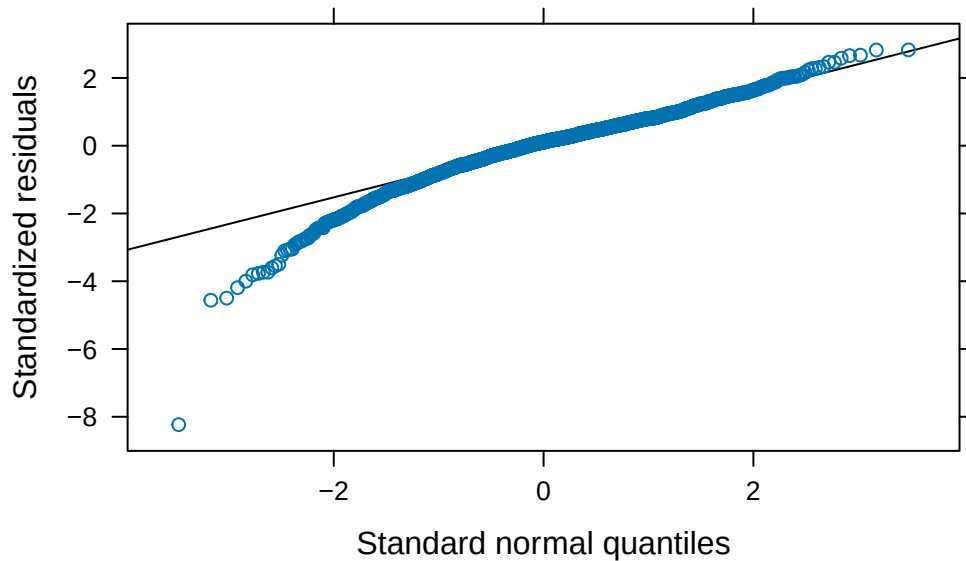
```
plot(mod3_eff, type=c("p","smooth"), col.line=1)
```



```
plot(mod3_eff,
      sqrt(abs(resid(.)))~fitted(.),
      type=c("p","smooth"), col.line=1)
```



```
lattice::qqmath(mod3_eff)
```



```
#Table
#fixed effects
mod3_eff_table <- data.frame(
  predictor = attr(summary(mod3_eff)$coef,"dimnames")[[1]],
  B = summary(mod3_eff)$coef["Estimate"],
  beta = effectsize::standardize_parameters(mod3_eff)$Std_Coefficient,
  CI = paste0("(",round(ci[-c(1,2),1],4),", ",round(ci[-c(1,2),2],4),")"),
  CI_STD = paste0("(",round(effectsize::standardize_parameters(mod3_eff)$CI_low,4),", ",round(
  pval = summary(mod3_eff)$coef[,5]
)

mod3_eff_table[,c("B","beta","pval")] <-
round(mod3_eff_table[,c("B","beta","pval")],4)
write.csv(mod3_eff_table,file = "../bohnert_sleep_git/mod3_eff_table.csv")

#Random effects
#within
sigma2 <- summary(mod3_eff)$sigma^2
#between
tau00 <- attr(summary(mod3_eff)$varcor$ID,"stddev")^2
iccvec[7] <- ICC <- tau00/(tau00 + sigma2)
N <- summary(mod3_eff)$devcomp$dims["q"]
obs <- summary(mod3_eff)$devcomp$dims["n"]
R2marg <- r.squaredGLMM(mod3_eff)[,"R2m"]
R2cond <- r.squaredGLMM(mod3_eff)[,"R2c"]
mod3_eff_table_re <- data.frame(sigma2, tau00, ICC, N, obs, R2marg, R2cond)
```

```
mod3_eff_table_re
```

```
              sigma2   tau00      ICC   N   obs      R2marg   R2cond
(Intercept) 46.07512 38.6921 0.4564512 276 1981 0.07784319 0.4987628
```

```
#####
```

```
#Model 4: Neighborhood
```

```
#####
```

```
mod4_eff <- lmer(act_efficiency ~ gender.v1 + age.v1 + income.v1.10000 + cg1_edu_grp + I_frisat + p.mn.cohesion.mean.v1 + p.mn.safety.mean.v1 + p.mn.violence.mean.v1 + (1 | ID))
summary(mod4_eff)
```

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
```

```
Formula: act_efficiency ~ gender.v1 + age.v1 + income.v1.10000 + cg1_edu_grp +
  I_frisat + p.mn.cohesion.mean.v1 + p.mn.safety.mean.v1 +
  p.mn.violence.mean.v1 + (1 | ID)
Data: datlong
```

```
REML criterion at convergence: 15006.1
```

```
Scaled residuals:
```

```
      Min       1Q   Median       3Q      Max
-8.2797 -0.4799  0.1173  0.5824  2.9054
```

```
Random effects:
```

```
Groups   Name      Variance Std.Dev.
ID        (Intercept) 40.27    6.346
Residual                45.22    6.725
```

```
Number of obs: 2169, groups: ID, 300
```

```
Fixed effects:
```

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	96.90604	4.85950	292.40710	19.942	< 2e-16 ***
gender.v1Male	-2.13335	0.84108	288.56476	-2.536	0.011725 *
age.v1	-0.89449	0.26579	290.37822	-3.365	0.000867 ***
income.v1.10000	0.38870	0.13843	287.74018	2.808	0.005326 **
cg1_edu_grp<=BS	0.27264	0.94606	293.52543	0.288	0.773407
cg1_edu_grp>=MS	-0.51575	1.56642	290.68359	-0.329	0.742201
I_frisat	0.46965	0.32594	1880.58205	1.441	0.149777
p.mn.cohesion.mean.v1	0.29822	0.62548	289.94402	0.477	0.633876
p.mn.safety.mean.v1	0.02459	0.58547	290.09535	0.042	0.966522


```
p.mn.violence.mean.v1    0.02725    0.65405  291.17596    0.042 0.966801
```

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Correlation of Fixed Effects:

```
(Intr) gnd.1M age.v1 i.1.10 c1__<= c1__>= I_frst p.mn.c...1
gender.v1M1 -0.100
age.v1      -0.816  0.109
inc.1.10000 0.050  0.000 -0.114
cg1_d_g<=BS -0.033 -0.066 -0.105 -0.169
cg1_d_g>=MS  0.080 -0.145 -0.095 -0.272  0.487
I_friscat   -0.012  0.001 -0.009  0.007 -0.002 -0.007
p.mn.chs...1 -0.182 -0.011 -0.039  0.056 -0.081 -0.169  0.001
p.mn.sft...1 -0.209 -0.045 -0.056 -0.161  0.094  0.076 -0.001 -0.557
p.mn.vln...1 -0.419 -0.090 -0.058 -0.023  0.046 -0.011  0.001 -0.011
p.mn.s...1
gender.v1M1
age.v1
inc.1.10000
cg1_d_g<=BS
cg1_d_g>=MS
I_friscat
p.mn.chs...1
p.mn.sft...1
p.mn.vln...1 0.575
```

```
ci <- confint(mod4_eff)
```

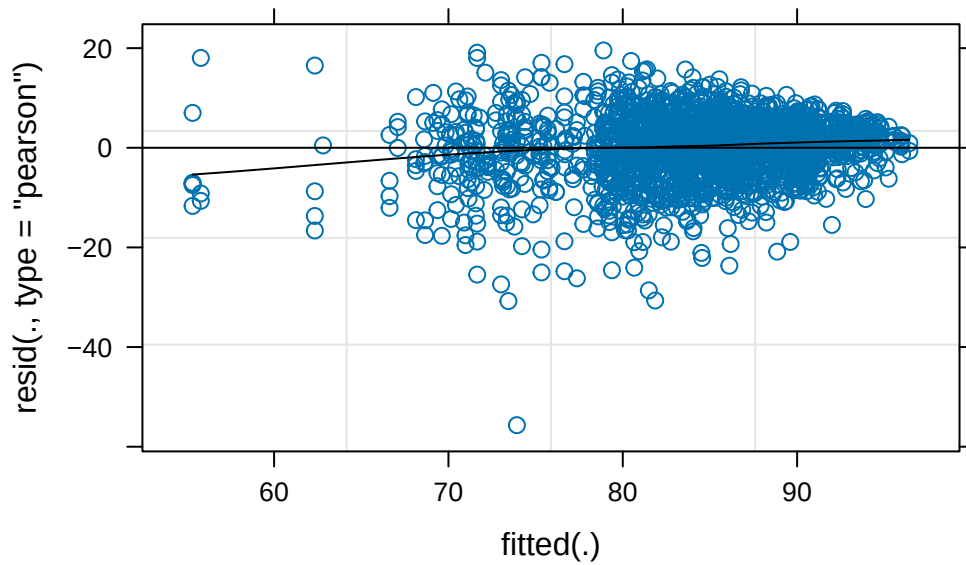
Computing profile confidence intervals ...

```
ci
```

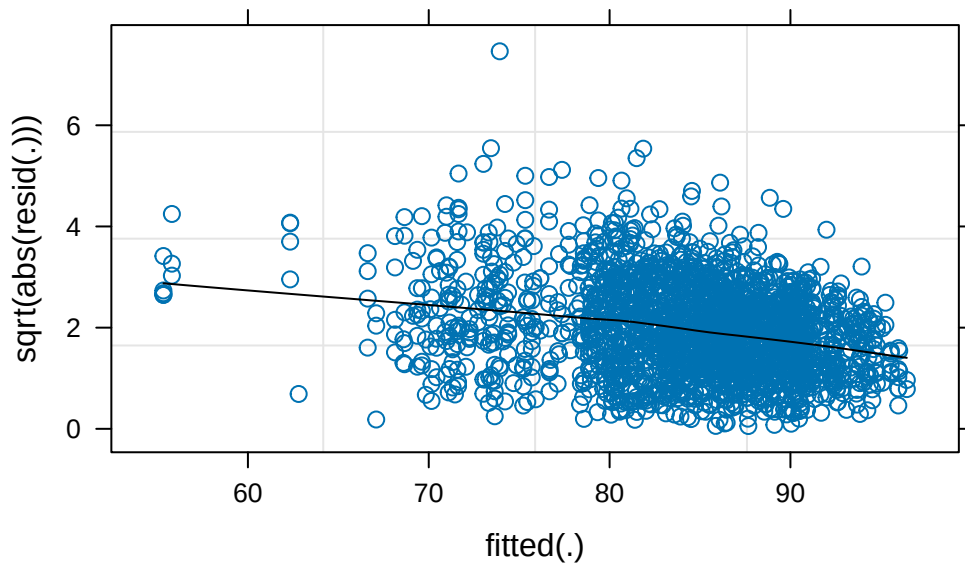
	2.5 %	97.5 %
.sig01	5.6832815	6.8554034
.sigma	6.5128886	6.9441985
(Intercept)	87.4966694	106.3175930
gender.v1Male	-3.7621352	-0.5048267
age.v1	-1.4092598	-0.3798910
income.v1.10000	0.1206963	0.6567826
cg1_edu_grp<=BS	-1.5588564	2.1053282
cg1_edu_grp>=MS	-3.5483209	2.5182997

I_frisat	-0.1697693	1.1081539
p.mn.cohesion.mean.v1	-0.9129855	1.5094180
p.mn.safety.mean.v1	-1.1091889	1.1582473
p.mn.violence.mean.v1	-1.2394413	1.2936415

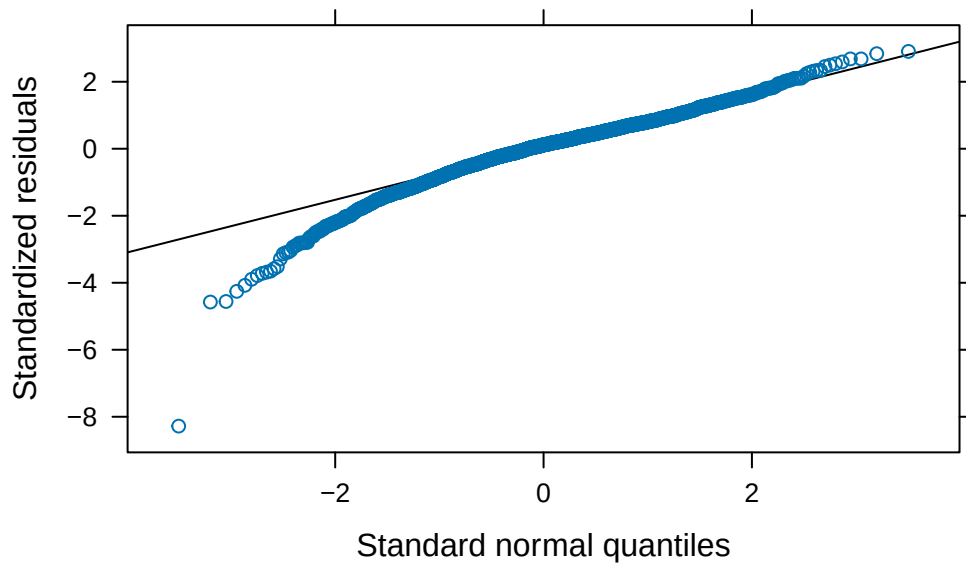
```
plot(mod4_eff, type=c("p","smooth"), col.line=1)
```



```
plot(mod4_eff,
      sqrt(abs(resid(.)))~fitted(.),
      type=c("p","smooth"), col.line=1)
```



```
lattice::qqmath(mod4_eff)
```



```
#Table
#Fixed Effects
mod4_eff_table <- data.frame(
  predictor = attr(summary(mod4_eff)$coef,"dimnames")[[1]],
  B = summary(mod4_eff)$coef[, "Estimate"],
  beta = effectsize::standardize_parameters(mod4_eff)$Std_Coefficient,
  CI = paste0("(",round(ci[-c(1,2),1],4),", ",round(ci[-c(1,2),2],4),")"),
  CI_STD = paste0("(",round(effectsize::standardize_parameters(mod4_eff)$CI_low,4),", ",round(
  pval = summary(mod4_eff)$coef[,5]
)

mod4_eff_table[,c("B","beta","pval")] <-
round(mod4_eff_table[,c("B","beta","pval")],4)
write.csv(mod4_eff_table,file = "../bohnert_sleep_git/mod4_eff_table.csv")

#Random effects
#within
sigma2 <- summary(mod4_eff)$sigma^2
#between
tau00 <- attr(summary(mod4_eff)$varcor$ID,"stddev")^2
iccvec[8] <- ICC <- tau00/(tau00 + sigma2)
N <- summary(mod4_eff)$devcomp$dims["q"]
obs <- summary(mod4_eff)$devcomp$dims["n"]
```

```
R2marg <- r.squaredGLMM(mod4_eff)[,"R2m"]
R2cond <- r.squaredGLMM(mod4_eff)[,"R2c"]
mod4_eff_table_re <- data.frame(sigma2, tau00, ICC, N, obs, R2marg, R2cond)
mod4_eff_table_re
```

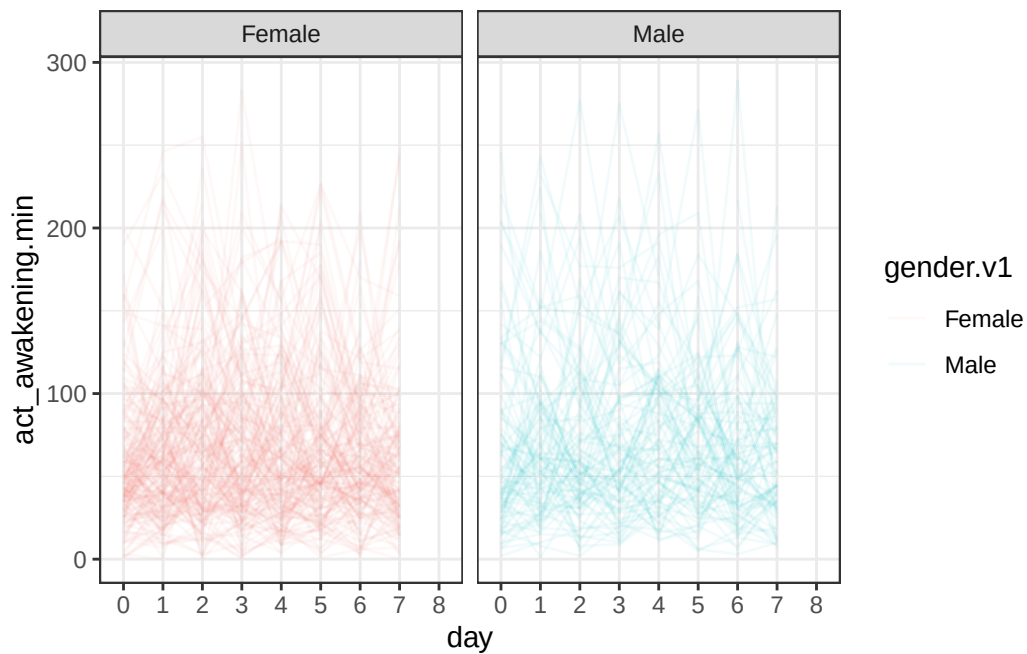
```
      sigma2    tau00      ICC    N  obs    R2marg    R2cond
(Intercept) 45.21947 40.27327 0.4710724 300 2169 0.04092477 0.4927186
```

Awakening

There are some classic violations of the constant variance assumption here. So I took a log transformation (plus 1 to prevent $\log(0)$) of the awakening variable and this alleviates much of the problem.

```
ggplot(aes(x = day, y = act_awakening.min, group = ID, color = gender.v1), data = datlong %>%
```

Warning: Removed 452 rows containing missing values or values outside the scale range (``geom_line()``).



```

library(lme4)
#####
#act_awakening.min
#####
#####
#Model 1: COI Total
#####
mod1_awake <- lmer(log(act_awakening.min + 1) ~ gender.v1 + age.v1 + income.v1.10000 + cg1_e
summary(mod1_awake)

```

```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: log(act_awakening.min + 1) ~ gender.v1 + age.v1 + income.v1.10000 +
      cg1_edu_grp + I_frisat + z_COI_nat + (1 | ID)
Data: datlong

```

REML criterion at convergence: 4792.9

Scaled residuals:

	Min	1Q	Median	3Q	Max
	-5.8695	-0.4848	0.0810	0.6092	3.2281

Random effects:

Groups	Name	Variance	Std.Dev.
ID	(Intercept)	0.2255	0.4749
	Residual	0.3801	0.6165

Number of obs: 2269, groups: ID, 314

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	3.47554	0.33143	304.54947	10.487	< 2e-16 ***
gender.v1Male	0.08448	0.06416	302.90945	1.317	0.18896
age.v1	0.03686	0.01983	304.36114	1.858	0.06407 .
income.v1.10000	-0.03179	0.01046	300.76775	-3.040	0.00257 **
cg1_edu_grp<=BS	-0.02313	0.06974	307.92429	-0.332	0.74036
cg1_edu_grp>=MS	0.01165	0.11659	305.83730	0.100	0.92047
I_frisat	0.05981	0.02919	1971.07399	2.049	0.04062 *
z_COI_nat	0.02367	0.04309	306.79813	0.549	0.58324

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

```

(Intr) gnd.1M age.v1 i.1.10 c1__<= c1__>= I_frst
gender.v1M1 -0.181
age.v1      -0.975  0.131
inc.1.10000 0.000 -0.001 -0.110
cg1_d_g<=BS -0.083 -0.083 -0.044 -0.146
cg1_d_g>=MS 0.000 -0.153 -0.061 -0.254 0.466
I_friscat   -0.014 0.001 -0.010 0.007 -0.003 -0.009
z_COI_nat    0.249 -0.038 -0.169 -0.111 -0.055 -0.112 0.008

```

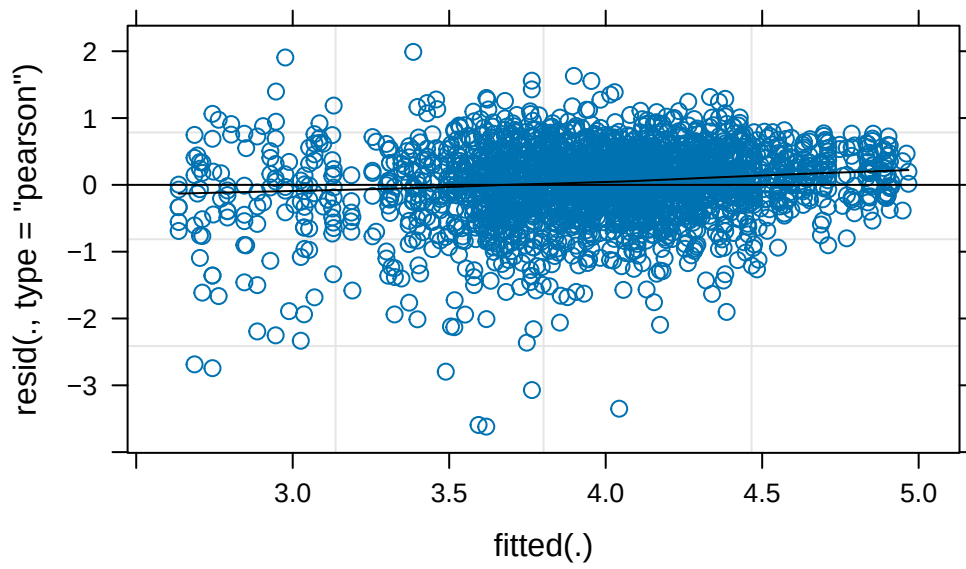
```
ci <- confint(mod1_awake)
```

Computing profile confidence intervals ...

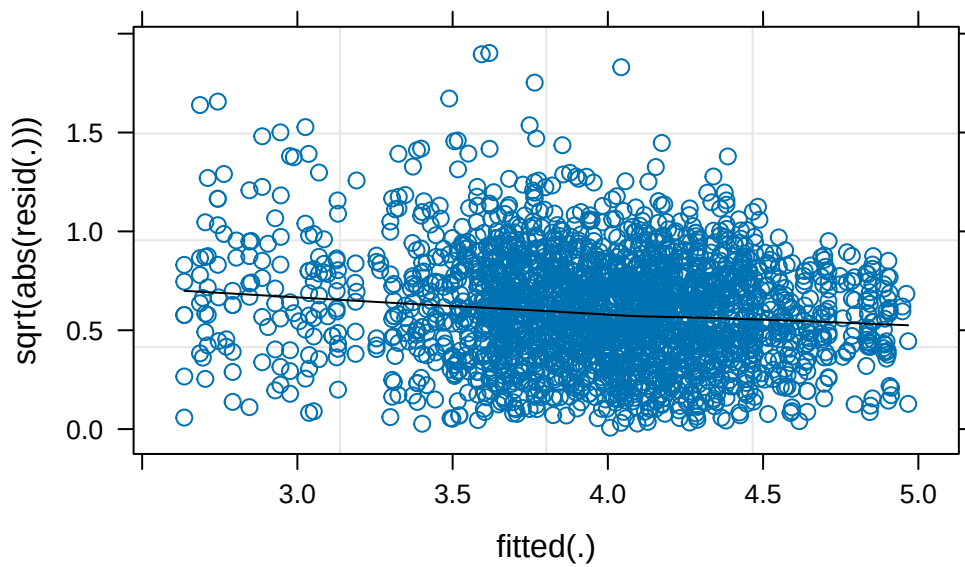
```
ci
```

	2.5 %	97.5 %
.sig01	0.424625217	0.51692378
.sigma	0.597502431	0.63617255
(Intercept)	2.831377820	4.11981846
gender.v1Male	-0.040227062	0.20920651
age.v1	-0.001695030	0.07540935
income.v1.10000	-0.052111920	-0.01146742
cg1_edu_grp<=BS	-0.158706493	0.11242008
cg1_edu_grp>=MS	-0.214983236	0.23825825
I_friscat	0.002612133	0.11707538
z_COI_nat	-0.060121131	0.10740453

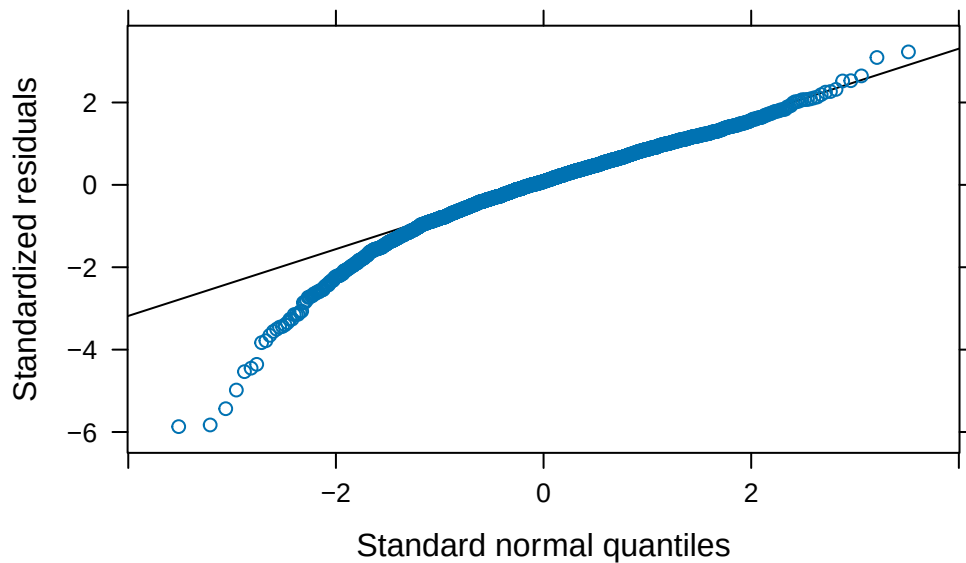
```
plot(mod1_awake, type=c("p","smooth"), col.line=1)
```



```
plot(mod1_awake,
     sqrt(abs(resid(.)))~fitted(.),
     type=c("p","smooth"), col.line=1)
```



```
lattice::qqmath(mod1_awake)
```



```
#Table
#Fixed effects
mod1_awake_table <- data.frame(
  predictor = attr(summary(mod1_awake)$coef,"dimnames")[[1]],
  B = summary(mod1_awake)$coef[, "Estimate"],
  beta = effectsize::standardize_parameters(mod1_awake)$Std_Coefficient,
  CI = paste0("(",round(ci[-c(1,2),1],4),", ",round(ci[-c(1,2),2],4),")"),
  CI_STD = paste0("(",round(effectsize::standardize_parameters(mod1_awake)$CI_low,4),", ",round(
  pval = summary(mod1_awake)$coef[,5]
)
```

Formula contains log- or sqrt-terms.

See `help("standardize")` for how such terms are standardized.

Formula contains log- or sqrt-terms.

See `help("standardize")` for how such terms are standardized.

Formula contains log- or sqrt-terms.

See `help("standardize")` for how such terms are standardized.

```
mod1_awake_table[,c("B","beta","pval")] <-
round(mod1_awake_table[,c("B","beta","pval")],4)
write.csv(mod1_awake_table,file = "../bohnert_sleep_git/mod1_awake_table.csv")

#Random effects
#within
```



```

sigma2 <- summary(mod1_awake)$sigma^2
#between
tau00 <- attr(summary(mod1_awake)$varcor$ID,"stddev")^2
iccvec[9] <- ICC <- tau00/(tau00 + sigma2)
N <- summary(mod1_awake)$devcomp$dims["q"]
obs <- summary(mod1_awake)$devcomp$dims["n"]
R2marg <- r.squaredGLMM(mod1_awake)[,"R2m"]
R2cond <- r.squaredGLMM(mod1_awake)[,"R2c"]
mod1_awake_table_re <- data.frame(sigma2, tau00, ICC, N, obs, R2marg, R2cond)
mod1_awake_table_re

```

	sigma2	tau00	ICC	N	obs	R2marg	R2cond
(Intercept)	0.3800511	0.2255243	0.3724133	314	2269	0.02079916	0.3854666

```

#####
#Model 2: COI Sub-domains
#####
mod2_awake <- lmer(log(act_awakening.min + 1) ~ gender.v1 + age.v1 + income.v1.10000 + cg1_e
summary(mod2_awake)

```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [

lmerModLmerTest]

Formula: log(act_awakening.min + 1) ~ gender.v1 + age.v1 + income.v1.10000 +

cg1_edu_grp + I_frisat + z_ED_nat + z_HE_nat + z_SE_nat + (1 | ID)

Data: datlong

REML criterion at convergence: 4798.1

Scaled residuals:

	Min	1Q	Median	3Q	Max
	-5.8819	-0.4829	0.0829	0.6075	3.2465

Random effects:

Groups	Name	Variance	Std.Dev.
ID	(Intercept)	0.2256	0.4750
	Residual	0.3800	0.6165

Number of obs: 2269, groups: ID, 314

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	3.51117	0.33326	302.90424	10.536	< 2e-16 ***

gender.v1Male	0.09039	0.06431	300.86066	1.405	0.16091
age.v1	0.03639	0.01984	302.63121	1.835	0.06754 .
income.v1.10000	-0.03137	0.01061	299.72607	-2.955	0.00337 **
cg1_edu_grp<=BS	-0.01639	0.06997	306.24137	-0.234	0.81497
cg1_edu_grp>=MS	0.01623	0.11667	304.11196	0.139	0.88943
I_frisat	0.05967	0.02919	1971.24943	2.044	0.04110 *
z_ED_nat	0.07472	0.07533	308.10825	0.992	0.32204
z_HE_nat	-0.08799	0.07265	301.66726	-1.211	0.22681
z_SE_nat	0.00421	0.06346	304.67490	0.066	0.94715

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

	(Intr)	gnd.1M	age.v1	i.1.10	c1__<=	c1__>=	I_frst	z_ED_n	z_HE_n
gender.v1M1	-0.175								
age.v1	-0.971	0.130							
inc.1.10000	-0.011	0.001	-0.108						
cg1_d_g<=BS	-0.080	-0.078	-0.045	-0.135					
cg1_d_g>=MS	0.001	-0.151	-0.062	-0.247	0.468				
I_frisat	-0.014	0.001	-0.010	0.007	-0.003	-0.009			
z_ED_nat	0.027	0.039	-0.031	0.129	0.062	0.014	0.000		
z_HE_nat	-0.072	-0.062	-0.002	0.033	-0.054	-0.031	0.005	-0.248	
z_SE_nat	0.155	-0.025	-0.067	-0.200	-0.060	-0.061	0.002	-0.701	-0.254

```
ci <- confint(mod2_awake)
```

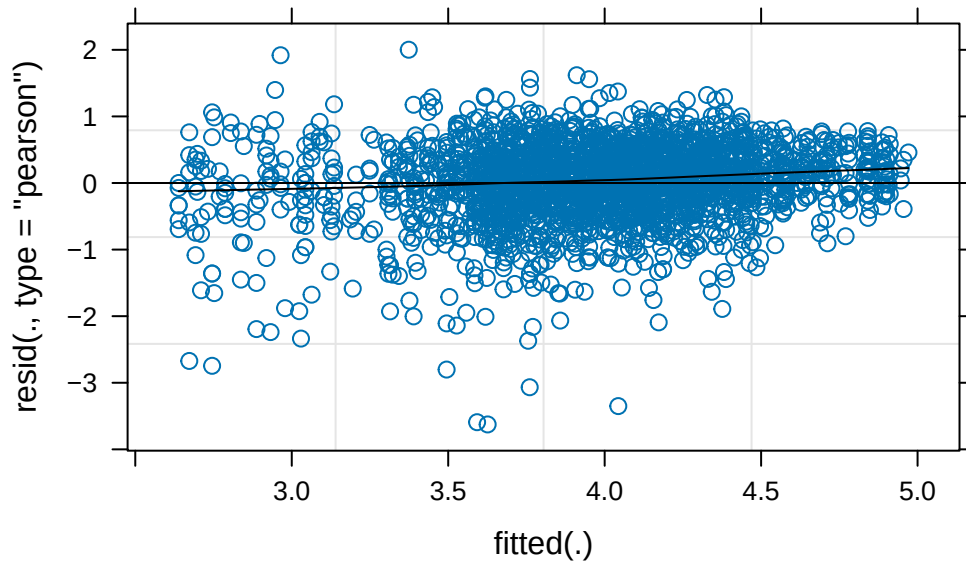
Computing profile confidence intervals ...

```
ci
```

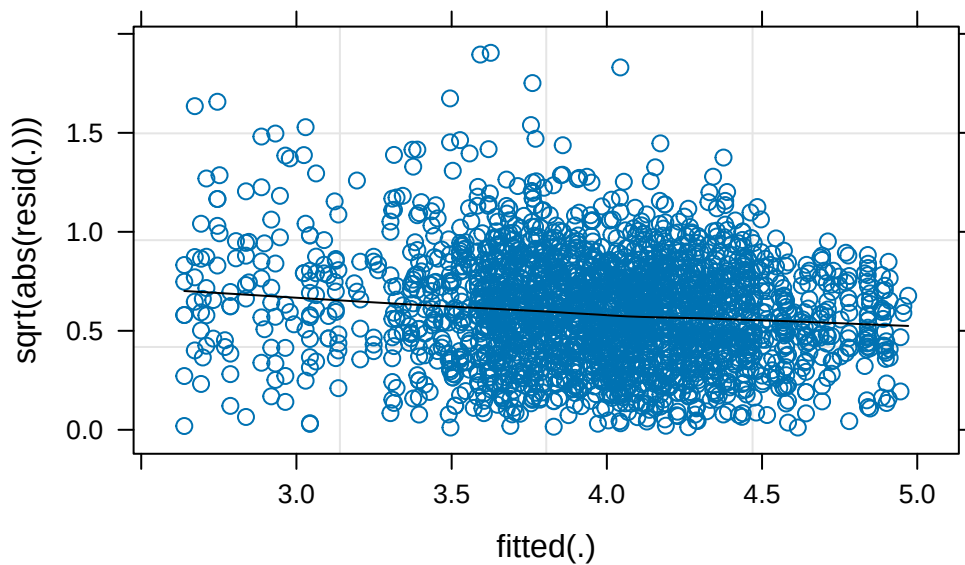
	2.5 %	97.5 %
.sig01	0.422913539	0.51495770
.sigma	0.597490122	0.63615689
(Intercept)	2.865637535	4.15696756
gender.v1Male	-0.034197265	0.21500636
age.v1	-0.002043923	0.07481699
income.v1.10000	-0.051941898	-0.01081327
cg1_edu_grp<=BS	-0.151976900	0.11915369
cg1_edu_grp>=MS	-0.209835324	0.24226196
I_frisat	0.002474889	0.11693464
z_ED_nat	-0.071379645	0.22056704

```
z_HE_nat      -0.228761221  0.05274542  
z_SE_nat      -0.118663736  0.12723301
```

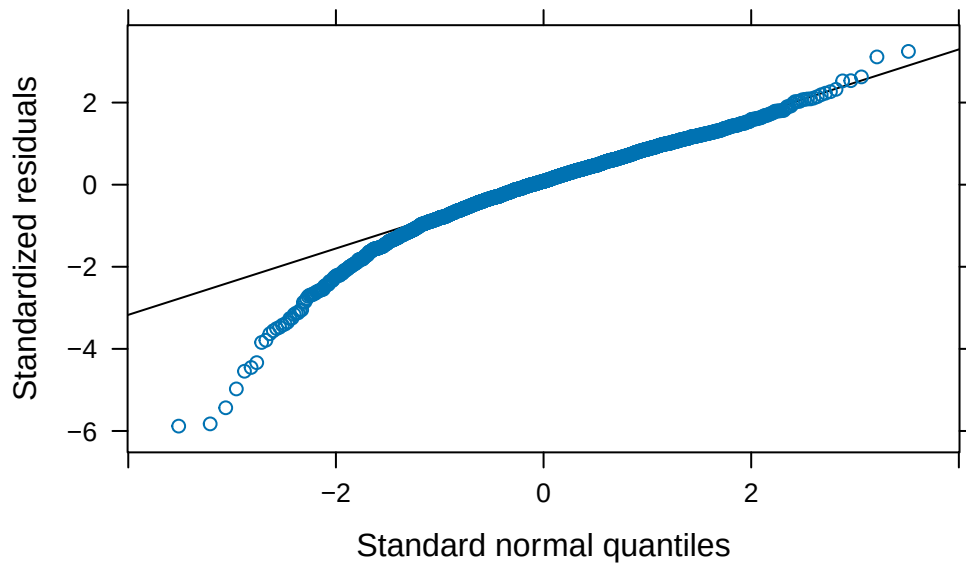
```
plot(mod2_awake, type=c("p","smooth"), col.line=1)
```



```
plot(mod2_awake,  
      sqrt(abs(resid(.)))~fitted(.),  
      type=c("p","smooth"), col.line=1)
```



```
lattice::qqmath(mod2_awake)
```



```
#Table
#Fixed effects
mod2_awake_table <- data.frame(
  predictor = attr(summary(mod2_awake)$coef,"dimnames")[[1]],
  B = summary(mod2_awake)$coef[, "Estimate"],
  beta = effectsize::standardize_parameters(mod2_awake)$Std_Coefficient,
  CI = paste0("(",round(ci[-c(1,2),1],4),", ", "round(ci[-c(1,2),2],4),")"),
  CI_STD = paste0("(",round(effectsize::standardize_parameters(mod2_awake)$CI_low,4),", ", "round(
  pval = summary(mod2_awake)$coef[,5]
)
```

Formula contains log- or sqrt-terms.

See help("standardize") for how such terms are standardized.

Formula contains log- or sqrt-terms.

See help("standardize") for how such terms are standardized.

Formula contains log- or sqrt-terms.

See help("standardize") for how such terms are standardized.

```
mod2_awake_table[,c("B","beta","pval")] <-
round(mod2_awake_table[,c("B","beta","pval")],4)
write.csv(mod2_awake_table,file = "../bohnert_sleep_git/mod2_awake_table.csv")
```

```

#Random effects
#within
sigma2 <- summary(mod2_awake)$sigma^2
#between
tau00 <- attr(summary(mod2_awake)$varcor$ID,"stddev")^2
iccvec[10] <- ICC <- tau00/(tau00 + sigma2)
N <- summary(mod2_awake)$devcomp$dims["q"]
obs <- summary(mod2_awake)$devcomp$dims["n"]
R2marg <- r.squaredGLMM(mod2_awake)[,"R2m"]
R2cond <- r.squaredGLMM(mod2_awake)[,"R2c"]
mod2_awake_table_re <- data.frame(sigma2, tau00, ICC, N, obs, R2marg, R2cond)
mod2_awake_table_re

```

	sigma2	tau00	ICC	N	obs	R2marg	R2cond
(Intercept)	0.3800343	0.2256006	0.3725027	314	2269	0.02374537	0.3874029

```

#####
#Model 3: COI Sub-domains
#####
mod3_awake <- lmer(log(act_awakening.min + 1) ~ gender.v1 + age.v1 + income.v1.10000 + cg1_e
  z_ED_EL_nat +
  z_ED_ER_nat +
  z_ED_SP_nat +
  z_HE_EP_nat +
  z_HE_HE_nat +
  z_HE_HR_nat +
  z_HE_SE_nat +
  z_SE_EI_nat +
  z_SE_EO_nat +
  z_SE_ER_nat +
  z_SE_HQ_nat +
  z_SE_SR_nat +
  z_SE_WL_nat + (1|ID) , data = datlong)
summary(mod3_awake)

```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]

Formula: log(act_awakening.min + 1) ~ gender.v1 + age.v1 + income.v1.10000 +
cg1_edu_grp + I_frisat + z_ED_EC_nat + z_ED_EL_nat + z_ED_ER_nat +
z_ED_SP_nat + z_HE_EP_nat + z_HE_HE_nat + z_HE_HR_nat + z_HE_SE_nat +
z_SE_EI_nat + z_SE_EO_nat + z_SE_ER_nat + z_SE_HQ_nat + z_SE_SR_nat +

```

      z_SE_WL_nat + (1 | ID)
Data: datlong

```

REML criterion at convergence: 4266.4

Scaled residuals:

Min	1Q	Median	3Q	Max
-5.8801	-0.4800	0.0911	0.5936	3.2037

Random effects:

Groups	Name	Variance	Std.Dev.
ID	(Intercept)	0.2198	0.4688
	Residual	0.3926	0.6266

Number of obs: 1981, groups: ID, 276

Fixed effects:

	Estimate	Std. Error	df	t	value	Pr(> t)
(Intercept)	3.585e+00	3.689e-01	2.555e+02	9.719	< 2e-16	***
gender.v1Male	7.832e-02	6.947e-02	2.538e+02	1.127	0.26064	
age.v1	3.436e-02	2.141e-02	2.549e+02	1.605	0.10965	
income.v1.10000	-3.327e-02	1.154e-02	2.526e+02	-2.884	0.00427	**
cg1_edu_grp<=BS	3.796e-02	7.585e-02	2.579e+02	0.501	0.61712	
cg1_edu_grp>=MS	4.692e-02	1.285e-01	2.565e+02	0.365	0.71533	
I_frisat	6.016e-02	3.179e-02	1.720e+03	1.892	0.05861	.
z_ED_EC_nat	-2.981e-02	3.762e-02	2.529e+02	-0.792	0.42882	
z_ED_EL_nat	1.496e-01	8.480e-02	2.592e+02	1.765	0.07878	.
z_ED_ER_nat	-3.426e-02	1.491e-01	2.568e+02	-0.230	0.81845	
z_ED_SP_nat	-2.356e-02	7.922e-02	2.561e+02	-0.297	0.76635	
z_HE_EP_nat	-6.272e-01	2.850e-01	2.515e+02	-2.201	0.02867	*
z_HE_HE_nat	2.699e-02	1.849e-01	2.568e+02	0.146	0.88404	
z_HE_HR_nat	6.182e-02	7.623e-02	2.538e+02	0.811	0.41815	
z_HE_SE_nat	-5.328e-02	5.410e-02	2.586e+02	-0.985	0.32561	
z_SE_EI_nat	-1.379e-01	1.497e-01	2.567e+02	-0.922	0.35759	
z_SE_EO_nat	1.941e-01	9.688e-02	2.571e+02	2.003	0.04619	*
z_SE_ER_nat	-2.563e-03	1.010e-01	2.563e+02	-0.025	0.97977	
z_SE_HQ_nat	-2.254e-01	9.031e-02	2.538e+02	-2.495	0.01321	*
z_SE_SR_nat	8.114e-02	8.931e-02	2.540e+02	0.909	0.36447	
z_SE_WL_nat	2.681e-02	1.055e-01	2.556e+02	0.254	0.79968	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation matrix not shown by default, as $p = 21 > 12$.
 Use `print(x, correlation=TRUE)` or
`vcov(x)` if you need it

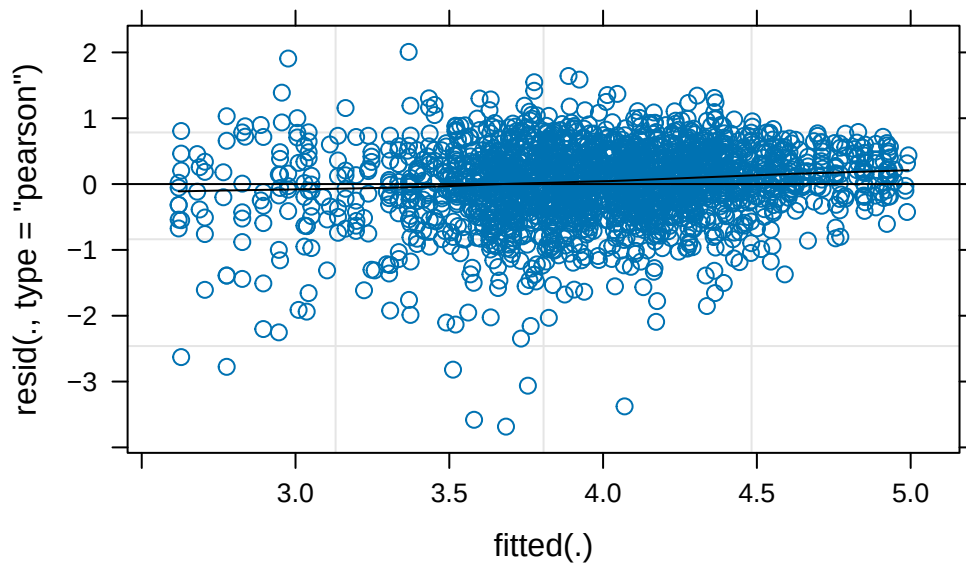
```
ci <- confint(mod3_awake)
```

Computing profile confidence intervals ...

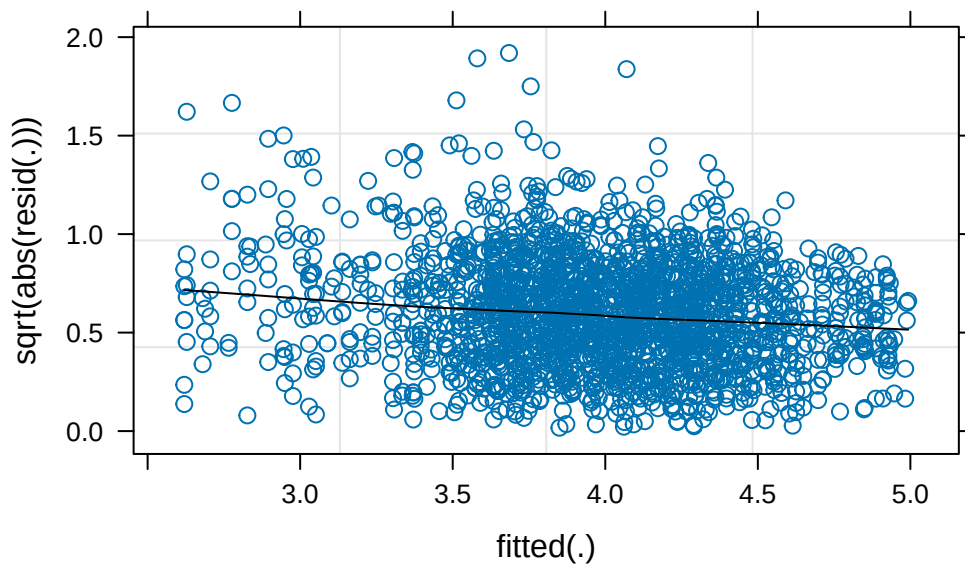
```
ci
```

	2.5 %	97.5 %
.sig01	0.401417917	0.49787202
.sigma	0.605950130	0.64801663
(Intercept)	2.886899864	4.28423928
gender.v1Male	-0.053174514	0.20995405
age.v1	-0.006201194	0.07488415
income.v1.10000	-0.055157141	-0.01146286
cg1_edu_grp<=BS	-0.105998033	0.18136785
cg1_edu_grp>=MS	-0.196671652	0.29019800
I_frisat	-0.002105801	0.12251742
z_ED_EC_nat	-0.100989944	0.04148928
z_ED_EL_nat	-0.011309054	0.30999088
z_ED_ER_nat	-0.316606284	0.24822897
z_ED_SP_nat	-0.173979477	0.12611749
z_HE_EP_nat	-1.166181874	-0.08679964
z_HE_HE_nat	-0.323005707	0.37728787
z_HE_HR_nat	-0.082575241	0.20616824
z_HE_SE_nat	-0.155861445	0.04912742
z_SE_EI_nat	-0.420926007	0.14602496
z_SE_EO_nat	0.010362114	0.37739220
z_SE_ER_nat	-0.193900878	0.18859821
z_SE_HQ_nat	-0.396463301	-0.05438378
z_SE_SR_nat	-0.088163761	0.25012849
z_SE_WL_nat	-0.172661198	0.22711486

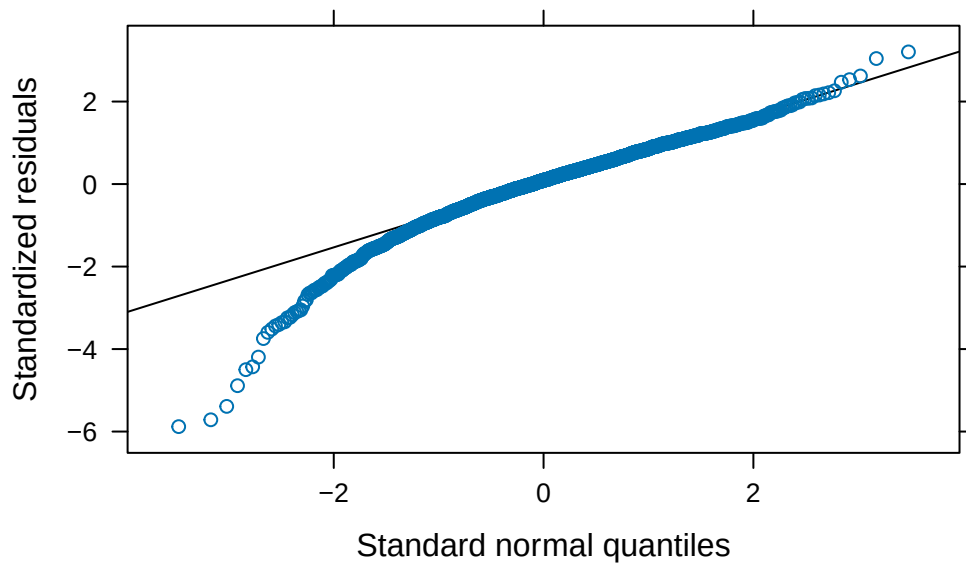
```
plot(mod3_awake, type=c("p","smooth"), col.line=1)
```



```
plot(mod3_awake,
      sqrt(abs(resid(.)))~fitted(.),
      type=c("p","smooth"), col.line=1)
```



```
lattice::qqmath(mod3_awake)
```

```
#Table
#Fixed effects
mod3_awake_table <- data.frame(
  predictor = attr(summary(mod3_awake)$coef,"dimnames")[[1]],
  B = summary(mod3_awake)$coef["Estimate"],
  beta = effectsize::standardize_parameters(mod3_awake)$Std_Coefficient,
  CI = paste0("(",round(ci[-c(1,2),1],4),", ", "(",round(ci[-c(1,2),2],4),")"),
  CI_STD = paste0("(",round(effectsize::standardize_parameters(mod3_awake)$CI_low,4),", ", "(",round(
  pval = summary(mod3_awake)$coef[,5]
)
```

Formula contains log- or sqrt-terms.

See help("standardize") for how such terms are standardized.

Formula contains log- or sqrt-terms.

See help("standardize") for how such terms are standardized.

Formula contains log- or sqrt-terms.

See help("standardize") for how such terms are standardized.

```
mod3_awake_table[,c("B","beta","pval")] <-
round(mod3_awake_table[,c("B","beta","pval")],4)
write.csv(mod3_awake_table,file = "../bohnert_sleep_git/mod3_awake_table.csv")

#Random effects
#within
sigma2 <- summary(mod3_awake)$sigma^2
```

```
#between
tau00 <- attr(summary(mod3_awake)$varcor$ID,"stddev")^2
iccvec[11] <- ICC <- tau00/(tau00 + sigma2)
N <- summary(mod3_awake)$devcomp$dims["q"]
obs <- summary(mod3_awake)$devcomp$dims["n"]
R2marg <- r.squaredGLMM(mod3_awake)[,"R2m"]
R2cond <- r.squaredGLMM(mod3_awake)[,"R2c"]
mod3_awake_table_re <- data.frame(sigma2, tau00, ICC, N, obs, R2marg, R2cond)
mod3_awake_table_re
```

	sigma2	tau00	ICC	N	obs	R2marg	R2cond
(Intercept)	0.3926031	0.2197512	0.3588628	276	1981	0.05447786	0.3937906

```
#####
```

```
#Model 4: Neighborhood
```

```
#####
```

```
mod4_awake <- lmer(act_awakening.min ~ gender.v1 + age.v1 + income.v1.10000 + cg1_edu_grp +
summary(mod4_awake)
```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]

Formula:

```
act_awakening.min ~ gender.v1 + age.v1 + income.v1.10000 + cg1_edu_grp +
  I_frisat + +p.mn.cohesion.mean.v1 + p.mn.safety.mean.v1 +
  p.mn.violence.mean.v1 + (1 | ID)
```

Data: datlong

REML criterion at convergence: 22239.6

Scaled residuals:

	Min	1Q	Median	3Q	Max
	-2.6236	-0.5949	-0.1487	0.4413	5.0960

Random effects:

Groups	Name	Variance	Std.Dev.
ID	(Intercept)	848.2	29.12
Residual		1340.1	36.61

Number of obs: 2169, groups: ID, 300

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
--	----------	------------	----	---------	----------

(Intercept)	22.0446	22.9345	290.4992	0.961	0.33725	
gender.v1Male	4.7480	3.9655	286.2659	1.197	0.23217	
age.v1	3.6091	1.2537	288.3741	2.879	0.00429	**
income.v1.10000	-2.0275	0.6525	285.3107	-3.107	0.00208	**
cg1_edu_grp<=BS	-2.5185	4.4663	291.8509	-0.564	0.57327	
cg1_edu_grp>=MS	1.5202	7.3897	288.8583	0.206	0.83716	
I_frisat	5.0295	1.7734	1882.6553	2.836	0.00461	**
p.mn.cohesion.mean.v1	-2.0477	2.9502	287.9994	-0.694	0.48818	
p.mn.safety.mean.v1	-0.2691	2.7615	288.0247	-0.097	0.92243	
p.mn.violence.mean.v1	0.5518	3.0857	289.0018	0.179	0.85822	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

	(Intr)	gnd.1M	age.v1	i.1.10	c1__<=	c1__>=	I_frst	p.mn.c..1
gender.v1M1	-0.100							
age.v1	-0.816	0.108						
inc.1.10000	0.050	0.000	-0.114					
cg1_d_g<=BS	-0.034	-0.065	-0.105	-0.169				
cg1_d_g>=MS	0.080	-0.145	-0.096	-0.272	0.487			
I_frisat	-0.014	0.002	-0.010	0.008	-0.002	-0.008		
p.mn.chs..1	-0.182	-0.011	-0.037	0.056	-0.082	-0.170	0.002	
p.mn.sft..1	-0.209	-0.044	-0.056	-0.161	0.095	0.077	-0.001	-0.557
p.mn.vln..1	-0.419	-0.088	-0.058	-0.023	0.047	-0.009	0.001	-0.012
p.mn.s..1								
gender.v1M1								
age.v1								
inc.1.10000								
cg1_d_g<=BS								
cg1_d_g>=MS								
I_frisat								
p.mn.chs..1								
p.mn.sft..1								
p.mn.vln..1								0.575

```
ci <- confint(mod4_awake)
```

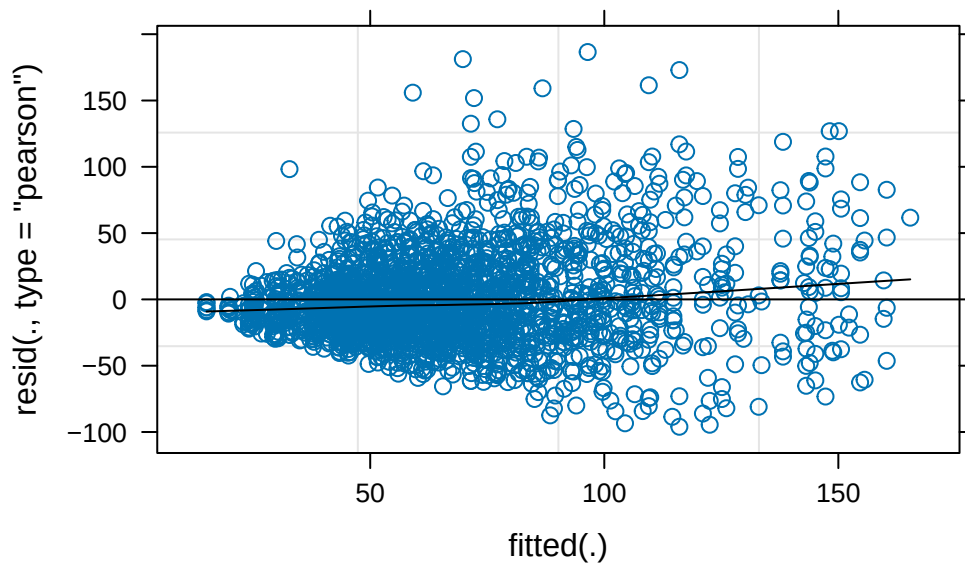
Computing profile confidence intervals ...

```
ci
```

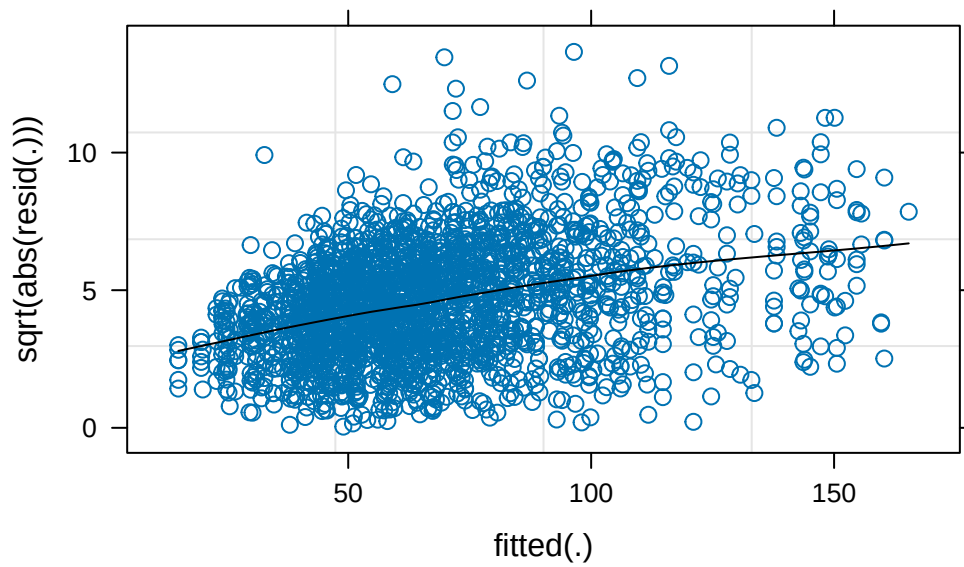
2.5 % 97.5 %

```
.sig01          25.886395 31.5957088
.sigma          35.454744 37.8037737
(Intercept)    -22.350399 66.4679160
gender.v1Male   -2.928501 12.4276502
age.v1          1.181022  6.0362282
income.v1.10000 -3.291211 -0.7644717
cg1_edu_grp<=BS -11.168821  6.1281003
cg1_edu_grp>=MS -12.791996 15.8251622
I_frisat        1.555473  8.5083436
p.mn.cohesion.mean.v1 -7.761555  3.6630453
p.mn.safety.mean.v1  -5.614898  5.0790578
p.mn.violence.mean.v1 -5.424034  6.5258167
```

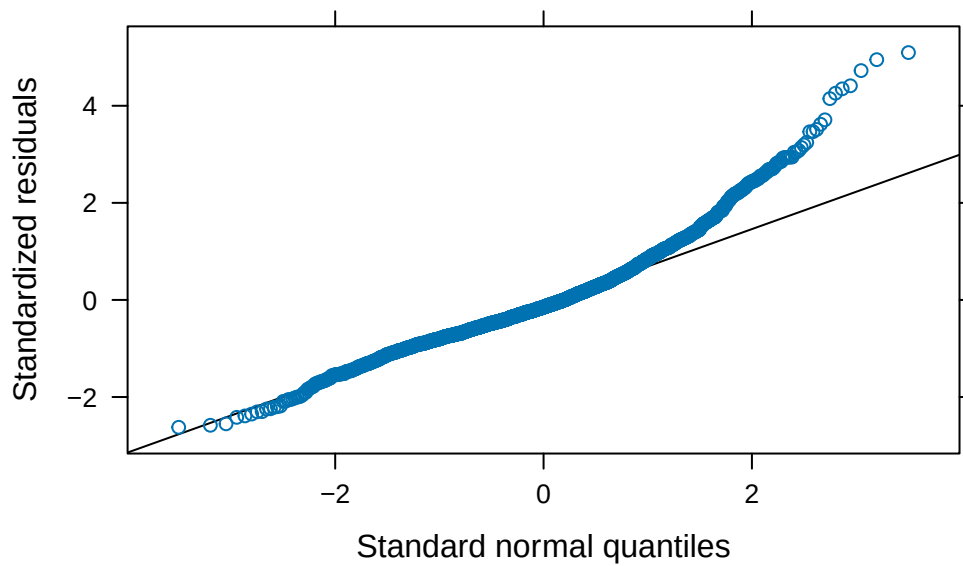
```
plot(mod4_awake, type=c("p","smooth"), col.line=1)
```



```
plot(mod4_awake,
      sqrt(abs(resid(.)))~fitted(.),
      type=c("p","smooth"), col.line=1)
```



```
lattice::qqmath(mod4_awake)
```



```
#Table
#Fixed Effects
mod4_awake_table <- data.frame(
  predictor = attr(summary(mod4_awake)$coef,"dimnames")[[1]],
  B = summary(mod4_awake)$coef["Estimate"],
  beta = effectsize::standardize_parameters(mod4_awake)$Std_Coefficient,
  CI = paste0("(",round(ci[-c(1,2),1],4),", ",round(ci[-c(1,2),2],4),")"),
```

```

CI_STD = paste0("(",round(effectsize::standardize_parameters(mod4_awake)$CI_low,4),", ",round
pval = summary(mod4_awake)$coef[,5]
)

mod4_awake_table[,c("B","beta","pval")] <-
round(mod4_awake_table[,c("B","beta","pval")],4)
write.csv(mod4_awake_table,file = "../bohnert_sleep_git/mod4_awake_table.csv")

#Random effects
#within
sigma2 <- summary(mod4_awake)$sigma^2
#between
tau00 <- attr(summary(mod4_awake)$varcor$ID,"stddev")^2
iccvec[12] <- ICC <- tau00/(tau00 + sigma2)
N <- summary(mod4_awake)$devcomp$dims["q"]
obs <- summary(mod4_awake)$devcomp$dims["n"]
R2marg <- r.squaredGLMM(mod4_awake)[,"R2m"]
R2cond <- r.squaredGLMM(mod4_awake)[,"R2c"]
mod4_awake_table_re <- data.frame(sigma2, tau00, ICC, N, obs, R2marg, R2cond)
mod4_awake_table_re

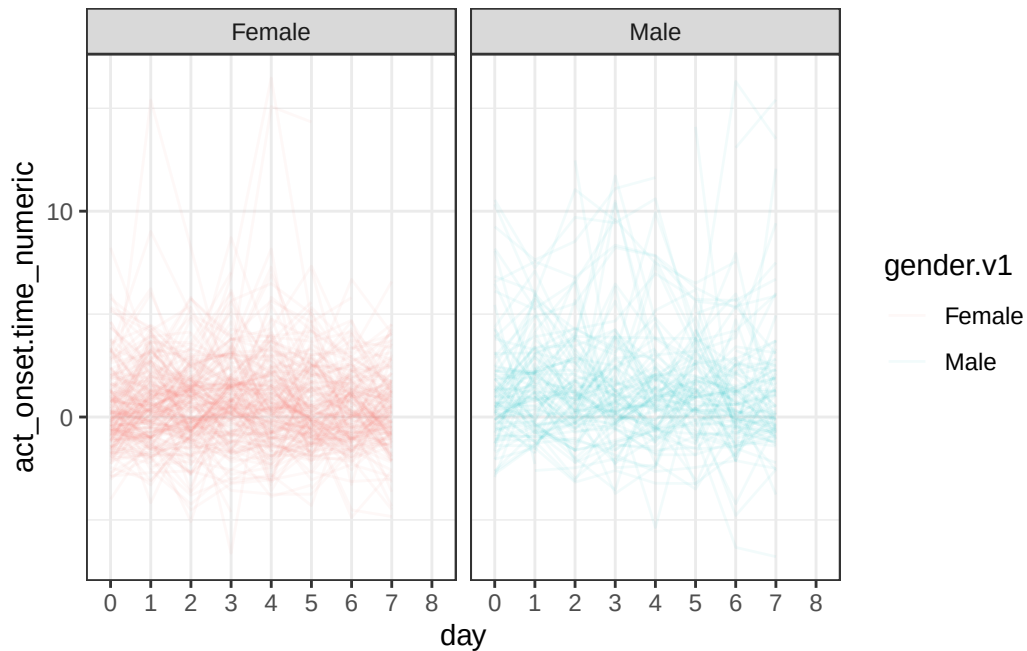
```

	sigma2	tau00	ICC	N	obs	R2marg	R2cond
(Intercept)	1340.063	848.2489	0.387627	300	2169	0.03300979	0.4078413

Sleep Onset

```
ggplot(aes(x = day, y = act_onset.time_numeric, group = ID, color = gender.v1), data = datlon)
```

Warning: Removed 452 rows containing missing values or values outside the scale range (`geom\_line()`).



```
library(lme4)
#####
#act_onset.time_numeric
#####
#####
#Model 1: COI Total
#####
mod1_onset <- lmer(act_onset.time_numeric ~ gender.v1 + age.v1 + income.v1.10000 +
  cg1_edu_grp + I_frisat + z_COI_nat + (1|ID) , data = datlong)
summary(mod1_onset)
```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: act_onset.time_numeric ~ gender.v1 + age.v1 + income.v1.10000 +
cg1_edu_grp + I_frisat + z_COI_nat + (1 | ID)
Data: datlong

REML criterion at convergence: 9589

Scaled residuals:

Min	1Q	Median	3Q	Max
-5.4674	-0.5114	-0.0455	0.4569	7.1612

Random effects:

Groups	Name	Variance	Std.Dev.
ID	(Intercept)	2.852	1.689
Residual		3.005	1.734

Number of obs: 2269, groups: ID, 314

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	-0.13421	1.13485	308.68009	-0.118	0.905937
gender.v1Male	0.76501	0.21981	307.20901	3.480	0.000573 ***
age.v1	0.03525	0.06792	308.46650	0.519	0.604165
income.v1.10000	0.01241	0.03584	305.53493	0.346	0.729481
cg1_edu_grp<=BS	0.08345	0.23856	311.47437	0.350	0.726726
cg1_edu_grp>=MS	0.21366	0.39908	309.39211	0.535	0.592779
I_frisat	0.59000	0.08216	1969.02813	7.181	9.76e-13 ***
z_COI_nat	0.15928	0.14746	310.14648	1.080	0.280938

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

	(Intr)	gnd.1M	age.v1	i.1.10	c1__<=	c1__>=	I_frst
gender.v1M1	-0.181						
age.v1	-0.975	0.132					
inc.1.10000	-0.001	-0.002	-0.110				
cg1_d_g<=BS	-0.082	-0.084	-0.045	-0.147			
cg1_d_g>=MS	0.001	-0.154	-0.061	-0.255	0.465		
I_frisat	-0.011	0.001	-0.009	0.006	-0.003	-0.007	
z_COI_nat	0.249	-0.037	-0.170	-0.111	-0.054	-0.110	0.007

```
ci <- confint(mod1_onset)
```

Computing profile confidence intervals ...

```
ci
```

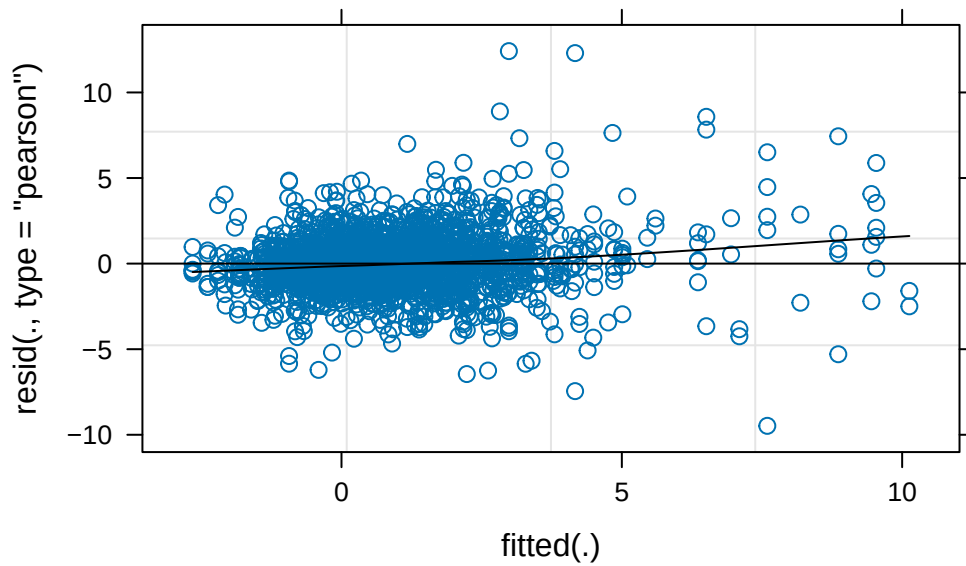
	2.5 %	97.5 %
.sig01	1.52498714	1.82693071
.sigma	1.68021649	1.78886180
(Intercept)	-2.34049698	2.07180535
gender.v1Male	0.33771473	1.19233732
age.v1	-0.09679064	0.16727266


```

income.v1.10000 -0.05725757 0.08208756
cg1_edu_grp<=BS -0.38022810 0.54733809
cg1_edu_grp>=MS -0.56200662 0.98964459
I_frisat        0.42890837 0.75102966
z_COI_nat       -0.12749096 0.44586448

```

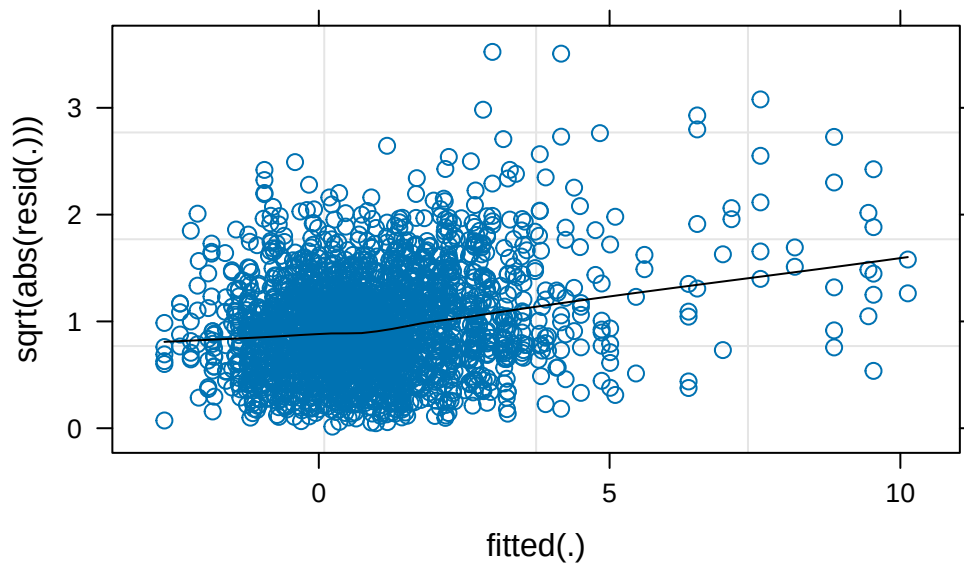
```
plot(mod1_onset, type=c("p","smooth"), col.line=1)
```



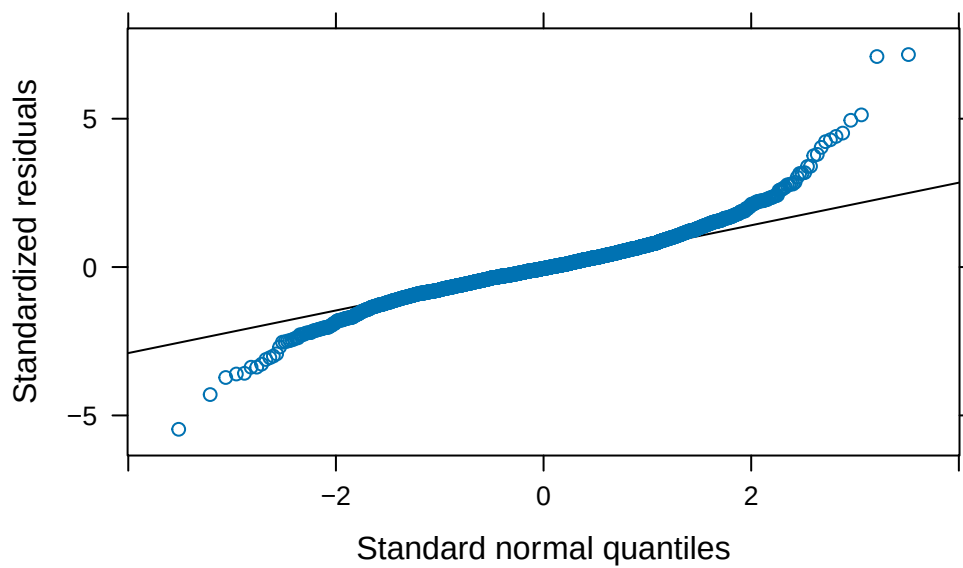
```

plot(mod1_onset,
      sqrt(abs(resid(.)))~fitted(.),
      type=c("p","smooth"), col.line=1)

```



```
lattice::qqmath(mod1_onset)
```



```
#Table
#Fixed effects
mod1_onset_table <- data.frame(
  predictor = attr(summary(mod1_onset)$coef,"dimnames")[[1]],
  B = summary(mod1_onset)$coef["Estimate"],
  beta = effectsize::standardize_parameters(mod1_onset)$Std_Coefficient,
  CI = paste0("(",round(ci[-c(1,2),1],4),", ",round(ci[-c(1,2),2],4),")"),
```

```

CI_STD = paste0("(",round(effectsize::standardize_parameters(mod1_onset)$CI_low,4),",", " ",round
pval = summary(mod1_onset)$coef[,5]
)

mod1_onset_table[,c("B","beta","pval")] <-
round(mod1_onset_table[,c("B","beta","pval")],4)
write.csv(mod1_onset_table,file = "../bohnert_sleep_git/mod1_onset_table.csv")

#Random effects
#within
sigma2 <- summary(mod1_onset)$sigma^2
#between
tau00 <- attr(summary(mod1_onset)$varcor$ID,"stddev")^2
iccvec[13] <- ICC <- tau00/(tau00 + sigma2)
N <- summary(mod1_onset)$devcomp$dims["q"]
obs <- summary(mod1_onset)$devcomp$dims["n"]
R2marg <- r.squaredGLMM(mod1_onset)[,"R2m"]
R2cond <- r.squaredGLMM(mod1_onset)[,"R2c"]
mod1_onset_table_re <- data.frame(sigma2, tau00, ICC, N, obs, R2marg, R2cond)
mod1_onset_table_re

```

	sigma2	tau00	ICC	N	obs	R2marg	R2cond
(Intercept)	3.005306	2.851818	0.4868972	314	2269	0.03915379	0.5069872

```

#####
#Model 2: COI Sub-domains
#####
mod2_onset <- lmer(act_onset.time_numeric ~ gender.v1 + age.v1 + income.v1.10000 +
cg1_edu_grp + I_frisat + z_ED_nat + z_HE_nat + z_SE_nat + (1|ID) , data = datlong)
summary(mod2_onset)

```

```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: act_onset.time_numeric ~ gender.v1 + age.v1 + income.v1.10000 +
cg1_edu_grp + I_frisat + z_ED_nat + z_HE_nat + z_SE_nat + (1 | ID)
Data: datlong

```

REML criterion at convergence: 9590.5

```

Scaled residuals:
    Min       1Q   Median       3Q      Max

```

-5.4751 -0.5064 -0.0441 0.4591 7.1566

Random effects:

Groups	Name	Variance	Std.Dev.
ID	(Intercept)	2.864	1.692
Residual		3.005	1.734

Number of obs: 2269, groups: ID, 314

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	-1.531e-01	1.143e+00	3.066e+02	-0.134	0.893512
gender.v1Male	7.533e-01	2.207e-01	3.048e+02	3.413	0.000729 ***
age.v1	3.567e-02	6.804e-02	3.063e+02	0.524	0.600448
income.v1.10000	8.856e-03	3.644e-02	3.039e+02	0.243	0.808151
cg1_edu_grp<=BS	6.678e-02	2.397e-01	3.093e+02	0.279	0.780791
cg1_edu_grp>=MS	2.011e-01	4.000e-01	3.072e+02	0.503	0.615620
I_frisat	5.902e-01	8.216e-02	1.969e+03	7.183	9.63e-13 ***
z_ED_nat	-1.636e-01	2.580e-01	3.105e+02	-0.634	0.526451
z_HE_nat	1.479e-01	2.492e-01	3.061e+02	0.594	0.553274
z_SE_nat	1.941e-01	2.175e-01	3.079e+02	0.892	0.372988

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

	(Intr)	gnd.1M	age.v1	i.1.10	c1__<=	c1__>=	I_frst	z_ED_n	z_HE_n
gender.v1M1	-0.175								
age.v1	-0.971	0.130							
inc.1.10000	-0.011	0.001	-0.108						
cg1_d_g<=BS	-0.079	-0.079	-0.045	-0.136					
cg1_d_g>=MS	0.002	-0.151	-0.062	-0.247	0.466				
I_frisat	-0.011	0.001	-0.009	0.006	-0.003	-0.008			
z_ED_nat	0.028	0.040	-0.032	0.128	0.063	0.015	0.000		
z_HE_nat	-0.072	-0.063	-0.002	0.033	-0.053	-0.030	0.004	-0.248	
z_SE_nat	0.155	-0.024	-0.067	-0.199	-0.061	-0.061	0.002	-0.700	-0.255

```
ci <- confint(mod2_onset)
```

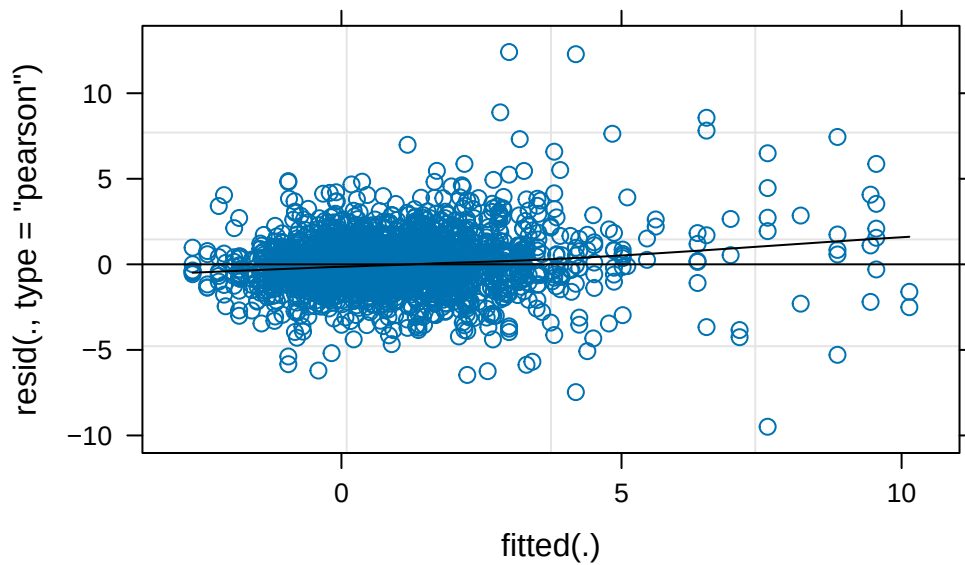
Computing profile confidence intervals ...

```
ci
```

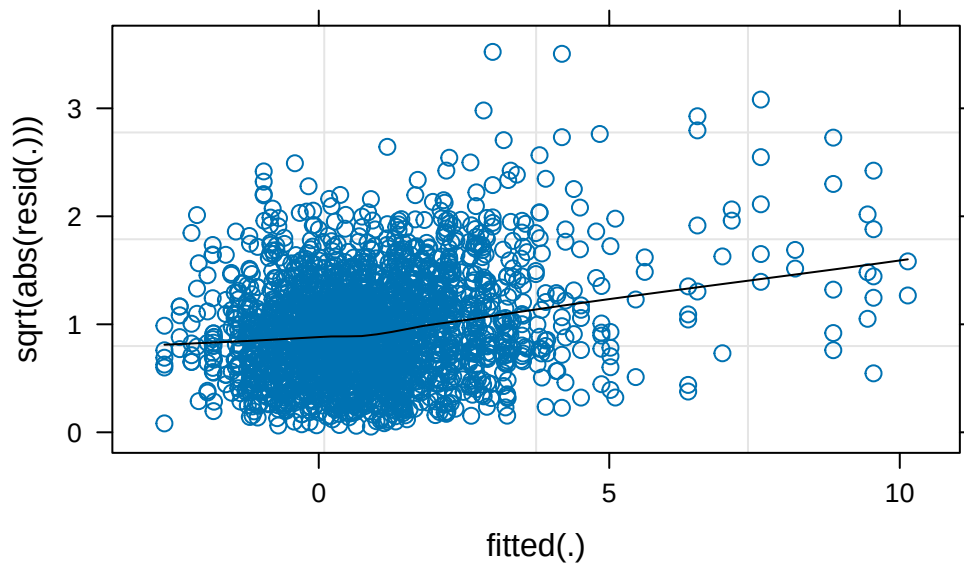
2.5 % 97.5 %

```
.sig01      1.52231185 1.82397946
.sigma      1.68026138 1.78891481
(Intercept) -2.36812554 2.06140080
gender.v1Male 0.32565906 1.18102504
age.v1      -0.09617257 0.16750202
income.v1.10000 -0.06174250 0.07947402
cg1_edu_grp<=BS -0.39763334 0.53149246
cg1_edu_grp>=MS -0.57386467 0.97647597
I_frisat     0.42907341 0.75120400
z_ED_nat     -0.66389914 0.33603268
z_HE_nat     -0.33493919 0.63096960
z_SE_nat     -0.22735882 0.61568794
```

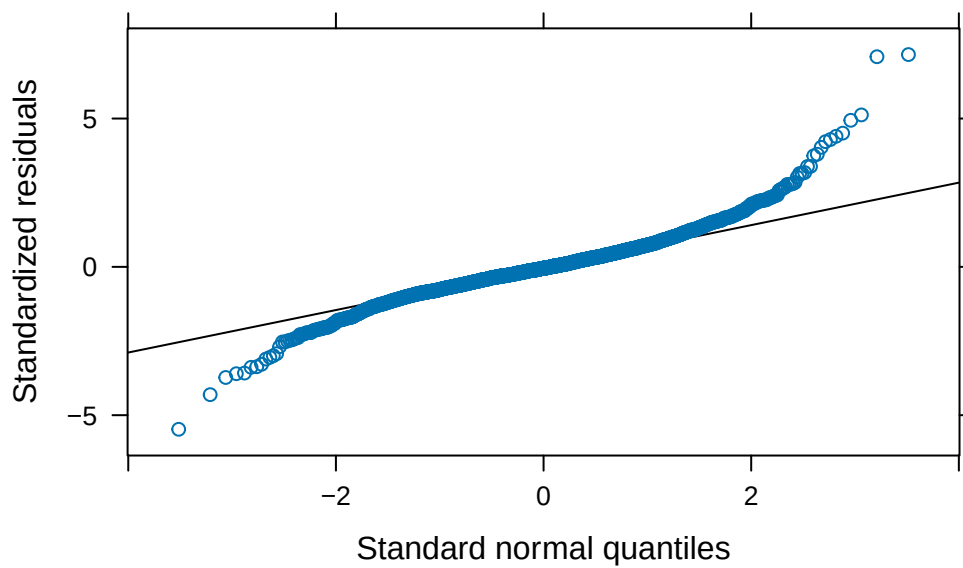
```
plot(mod2_onset, type=c("p","smooth"), col.line=1)
```



```
plot(mod2_onset,
      sqrt(abs(resid(.)))~fitted(.),
      type=c("p","smooth"), col.line=1)
```



```
lattice::qqmath(mod2_onset)
```



```
#Table
#Fixed effects
mod2_onset_table <- data.frame(
  predictor = attr(summary(mod2_onset)$coef,"dimnames")[[1]],
  B = summary(mod2_onset)$coef["Estimate"],
  beta = effectsize::standardize_parameters(mod2_onset)$Std_Coefficient,
  CI = paste0("(",round(ci[-c(1,2),1],4),", ",round(ci[-c(1,2),2],4),")"),
```

```

CI_STD = paste0("(",round(effectsize::standardize_parameters(mod2_onset)$CI_low,4),", ",round
pval = summary(mod2_onset)$coef[,5]
)

mod2_onset_table[,c("B","beta","pval")] <-
round(mod2_onset_table[,c("B","beta","pval")],4)
write.csv(mod2_onset_table,file = "../bohnert_sleep_git/mod2_onset_table.csv")

#Random effects
#within
sigma2 <- summary(mod2_onset)$sigma^2
#between
tau00 <- attr(summary(mod2_onset)$varcor$ID,"stddev")^2
iccvec[14] <- ICC <- tau00/(tau00 + sigma2)
N <- summary(mod2_onset)$devcomp$dims["q"]
obs <- summary(mod2_onset)$devcomp$dims["n"]
R2marg <- r.squaredGLMM(mod2_onset)[,"R2m"]
R2cond <- r.squaredGLMM(mod2_onset)[,"R2c"]
mod2_onset_table_re <- data.frame(sigma2, tau00, ICC, N, obs, R2marg, R2cond)
mod2_onset_table_re

```

	sigma2	tau00	ICC	N	obs	R2marg	R2cond
(Intercept)	3.005495	2.863633	0.4879145	314	2269	0.04025563	0.5085288

```

#####
#Model 3: COI Sub-domains
#####
mod3_onset <- lmer(act_onset.time_numeric ~ gender.v1 + age.v1 + income.v1.10000 +
cg1_edu_grp + I_frisat + z_ED_EC_nat +
  z_ED_EL_nat +
  z_ED_ER_nat +
  z_ED_SP_nat +
  z_HE_EP_nat +
  z_HE_HE_nat +
  z_HE_HR_nat +
  z_HE_SE_nat +
  z_SE_EI_nat +
  z_SE_EO_nat +
  z_SE_ER_nat +
  z_SE_HQ_nat +
  z_SE_SR_nat +

```

```
z_SE_WL_nat + (1|ID) , data = datlong)
summary(mod3_onset)
```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]

```
Formula: act_onset.time_numeric ~ gender.v1 + age.v1 + income.v1.10000 +
  cg1_edu_grp + I_frisat + z_ED_EC_nat + z_ED_EL_nat + z_ED_ER_nat +
  z_ED_SP_nat + z_HE_EP_nat + z_HE_HE_nat + z_HE_HR_nat + z_HE_SE_nat +
  z_SE_EI_nat + z_SE_EO_nat + z_SE_ER_nat + z_SE_HQ_nat + z_SE_SR_nat +
  z_SE_WL_nat + (1 | ID)
Data: datlong
```

REML criterion at convergence: 8305.2

Scaled residuals:

Min	1Q	Median	3Q	Max
-5.6798	-0.5240	-0.0544	0.4640	7.3227

Random effects:

Groups	Name	Variance	Std.Dev.
ID	(Intercept)	2.925	1.71
	Residual	2.889	1.70

Number of obs: 1981, groups: ID, 276

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	-0.59086	1.28248	258.38960	-0.461	0.645387
gender.v1Male	0.86102	0.24172	256.85002	3.562	0.000438 ***
age.v1	0.03182	0.07444	257.86469	0.427	0.669394
income.v1.10000	-0.01040	0.04016	256.02289	-0.259	0.795906
cg1_edu_grp<=BS	0.07226	0.26342	260.36667	0.274	0.784063
cg1_edu_grp>=MS	0.40076	0.44667	258.79470	0.897	0.370438
I_frisat	0.56451	0.08631	1716.40893	6.540	8.08e-11 ***
z_ED_EC_nat	-0.28510	0.13092	256.82473	-2.178	0.030338 *
z_ED_EL_nat	0.17086	0.29439	260.82498	0.580	0.562145
z_ED_ER_nat	-0.64851	0.51808	259.37746	-1.252	0.211784
z_ED_SP_nat	0.17810	0.27534	258.79620	0.647	0.518307
z_HE_EP_nat	1.31718	0.99264	255.18253	1.327	0.185714
z_HE_HE_nat	0.35463	0.64235	259.20426	0.552	0.581364
z_HE_HR_nat	0.07698	0.26523	257.03511	0.290	0.771873
z_HE_SE_nat	-0.06628	0.18785	260.89753	-0.353	0.724513
z_SE_EI_nat	0.95858	0.52004	259.27409	1.843	0.066427 .

z_SE_EO_nat	-0.29356	0.33663	259.23081	-0.872	0.383991
z_SE_ER_nat	-0.42487	0.35093	258.93484	-1.211	0.227114
z_SE_HQ_nat	0.55065	0.31422	257.04925	1.752	0.080895
z_SE_SR_nat	-0.05762	0.31070	257.28254	-0.185	0.853031
z_SE_WL_nat	0.13587	0.36692	258.12500	0.370	0.711461

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation matrix not shown by default, as $p = 21 > 12$.

Use `print(x, correlation=TRUE)` or

`vcov(x)` if you need it

```
ci <- confint(mod3_onset)
```

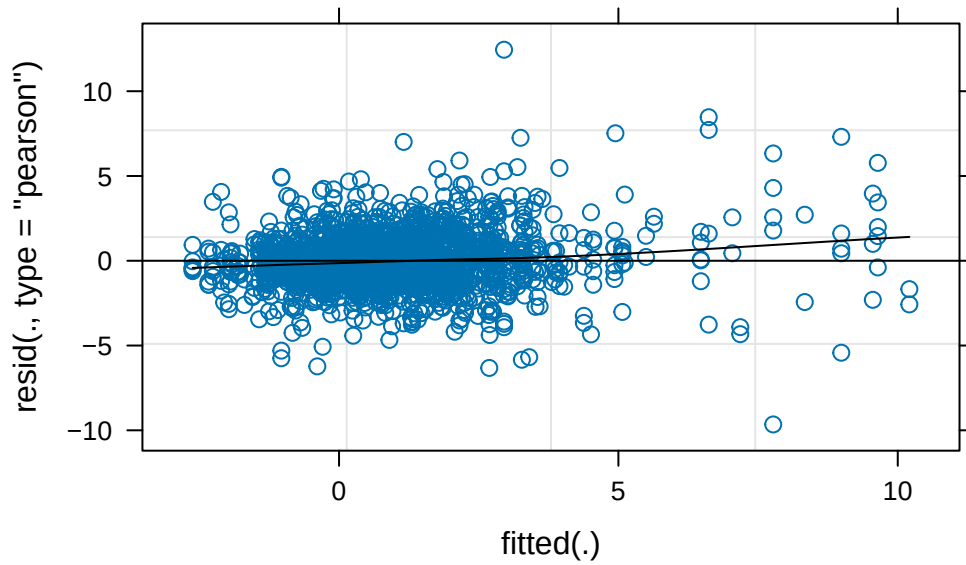
Computing profile confidence intervals ...

```
ci
```

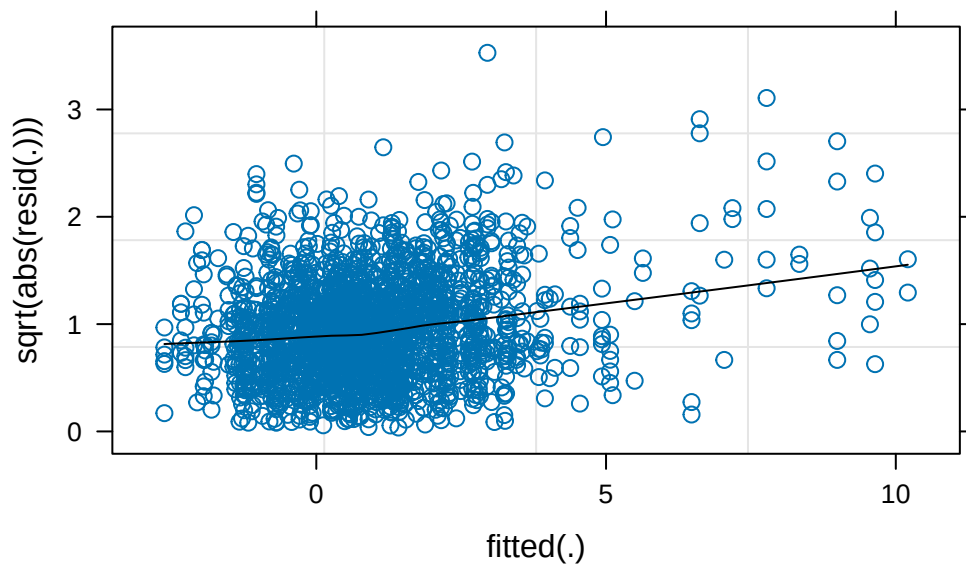
	2.5 %	97.5 %
.sig01	1.49037401	1.80680085
.sigma	1.64372437	1.75776016
(Intercept)	-3.02163687	1.83840869
gender.v1Male	0.40311761	1.31903501
age.v1	-0.10925639	0.17283938
income.v1.10000	-0.08642042	0.06574840
cg1_edu_grp<=BS	-0.42650523	0.57186717
cg1_edu_grp>=MS	-0.44452620	1.24822230
I_frisat	0.39544532	0.73381996
z_ED_EC_nat	-0.53321198	-0.03713612
z_ED_EL_nat	-0.38754613	0.72822760
z_ED_ER_nat	-1.63157225	0.33185901
z_ED_SP_nat	-0.34430326	0.69915792
z_HE_EP_nat	-0.56353129	3.19744225
z_HE_HE_nat	-0.86251957	1.57184006
z_HE_HR_nat	-0.42560116	0.57942549
z_HE_SE_nat	-0.42169553	0.29028728
z_SE_EI_nat	-0.02679865	1.94401823
z_SE_EO_nat	-0.93111935	0.34464814
z_SE_ER_nat	-1.09062418	0.23928832
z_SE_HQ_nat	-0.04442670	1.14624608

```
z_SE_SR_nat      -0.64533239  0.53203283
z_SE_WL_nat      -0.55872673  0.83173971
```

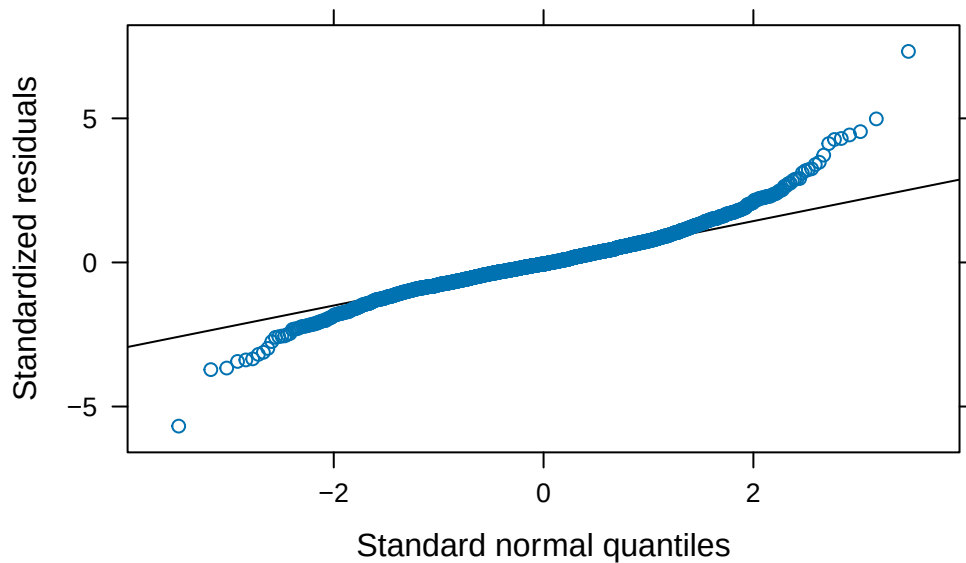
```
plot(mod3_onset, type=c("p","smooth"), col.line=1)
```



```
plot(mod3_onset,
      sqrt(abs(resid(.)))~fitted(.),
      type=c("p","smooth"), col.line=1)
```



```
lattice::qqmath(mod3_onset)
```



```
#Table
#fixed effects
mod3_onset_table <- data.frame(
  predictor = attr(summary(mod3_onset)$coef,"dimnames")[[1]],
  B = summary(mod3_onset)$coef[, "Estimate"],
  beta = effectsize::standardize_parameters(mod3_onset)$Std_Coefficient,
  CI = paste0("(",round(ci[-c(1,2),1],4),", ", " ",round(ci[-c(1,2),2],4),")"),
  CI_STD = paste0("(",round(effectsize::standardize_parameters(mod3_onset)$CI_low,4),", ", " ",round(
  pval = summary(mod3_onset)$coef[,5]
)

mod3_onset_table[,c("B","beta","pval")] <-
round(mod3_onset_table[,c("B","beta","pval")],4)
write.csv(mod3_onset_table,file = "../bohnert_sleep_git/mod3_onset_table.csv")

#Random effects
#within
sigma2 <- summary(mod3_onset)$sigma^2
#between
tau00 <- attr(summary(mod3_onset)$varcor$ID,"stddev")^2
iccvec[15] <- ICC <- tau00/(tau00 + sigma2)
N <- summary(mod3_onset)$devcomp$dims["q"]
obs <- summary(mod3_onset)$devcomp$dims["n"]
```

```
R2marg <- r.squaredGLMM(mod3_onset)[,"R2m"]
R2cond <- r.squaredGLMM(mod3_onset)[,"R2c"]
mod3_onset_table_re <- data.frame(sigma2, tau00, ICC, N, obs, R2marg, R2cond)
mod3_onset_table_re
```

	sigma2	tau00	ICC	N	obs	R2marg	R2cond
(Intercept)	2.889137	2.925301	0.5031099	276	1981	0.06940013	0.5375941

```
#####
```

```
#Model 4: Neighborhood
```

```
#####
```

```
mod4_onset <- lmer(act_onset.time_numeric ~ gender.v1 + age.v1 + income.v1.10000 +
cg1_edu_grp + I_frisat +
  p.mn.cohesion.mean.v1 + p.mn.safety.mean.v1 + p.mn.violence.mean.v1 + (1|ID) , data = dat.
summary(mod4_onset)
```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [lmerModLmerTest]

Formula: act_onset.time_numeric ~ gender.v1 + age.v1 + income.v1.10000 + cg1_edu_grp + I_frisat + p.mn.cohesion.mean.v1 + p.mn.safety.mean.v1 + p.mn.violence.mean.v1 + (1 | ID)

Data: datlong

REML criterion at convergence: 9214.8

Scaled residuals:

	Min	1Q	Median	3Q	Max
	-5.4348	-0.5110	-0.0486	0.4519	7.0707

Random effects:

Groups	Name	Variance	Std.Dev.
ID	(Intercept)	2.978	1.726
	Residual	3.061	1.750

Number of obs: 2169, groups: ID, 300

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	-1.15064	1.31339	294.78109	-0.876	0.381701
gender.v1Male	0.77032	0.22738	291.02889	3.388	0.000801 ***
age.v1	0.03783	0.07184	292.77667	0.527	0.598867
income.v1.10000	0.01742	0.03742	290.23518	0.465	0.642023

cg1_edu_grp<=BS	0.08921	0.25568	295.83724	0.349	0.727404
cg1_edu_grp>=MS	0.18823	0.42341	293.04283	0.445	0.656970
I_frisat	0.58974	0.08482	1881.97846	6.953	4.91e-12 ***
p.mn.cohesion.mean.v1	0.06034	0.16908	292.33000	0.357	0.721461
p.mn.safety.mean.v1	0.09967	0.15826	292.50452	0.630	0.529315
p.mn.violence.mean.v1	0.18858	0.17679	293.60283	1.067	0.286970

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

```
(Intr) gnd.1M age.v1 i.1.10 c1__<= c1__>= I_first p.mn.c..1
gender.v1M1 -0.100
age.v1      -0.816  0.109
inc.1.10000 0.050  0.000 -0.114
cg1_d_g<=BS -0.033 -0.066 -0.105 -0.169
cg1_d_g>=MS  0.080 -0.145 -0.095 -0.272  0.486
I_frisat    -0.011  0.001 -0.008  0.007 -0.002 -0.007
p.mn.chs..1 -0.182 -0.011 -0.039  0.056 -0.081 -0.169  0.001
p.mn.sft..1 -0.209 -0.045 -0.056 -0.161  0.093  0.076 -0.001 -0.557
p.mn.vln..1 -0.419 -0.090 -0.058 -0.023  0.045 -0.011  0.001 -0.011
p.mn.s..1
gender.v1M1
age.v1
inc.1.10000
cg1_d_g<=BS
cg1_d_g>=MS
I_frisat
p.mn.chs..1
p.mn.sft..1
p.mn.vln..1 0.575
```

```
ci <- confint(mod4_onset)
```

Computing profile confidence intervals ...

```
ci
```

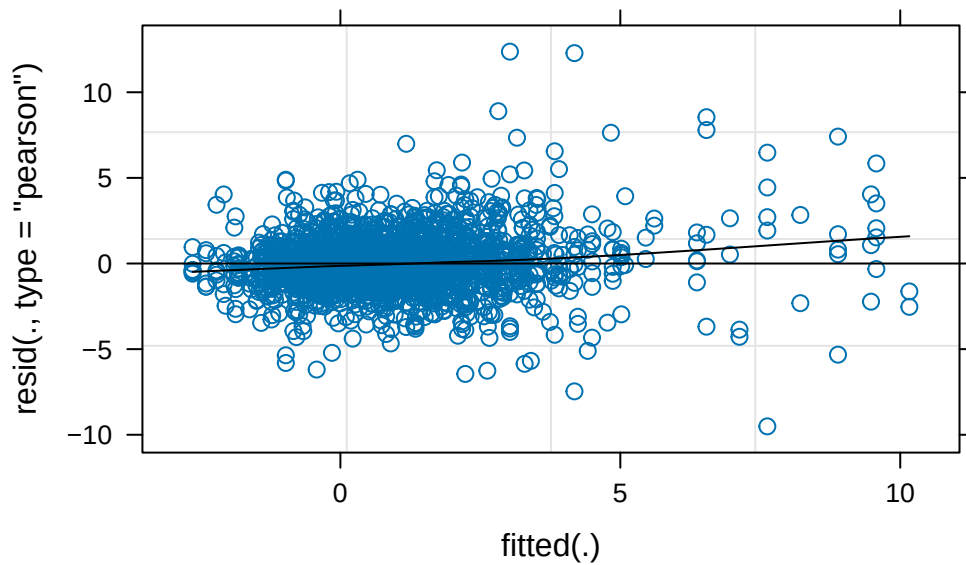
	2.5 %	97.5 %
.sig01	1.54847485	1.86205014
.sigma	1.69460337	1.80675188
(Intercept)	-3.69383426	1.39346477

```

gender.v1Male      0.33003928 1.21071058
age.v1             -0.10131903 0.17695591
income.v1.10000    -0.05504489 0.08990347
cg1_edu_grp<=BS    -0.40578148 0.58457877
cg1_edu_grp>=MS    -0.63145411 1.00854577
I_frisat           0.42341658 0.75595642
p.mn.cohesion.mean.v1 -0.26717024 0.38771199
p.mn.safety.mean.v1 -0.20685710 0.40612347
p.mn.violence.mean.v1 -0.15389117 0.53085771

```

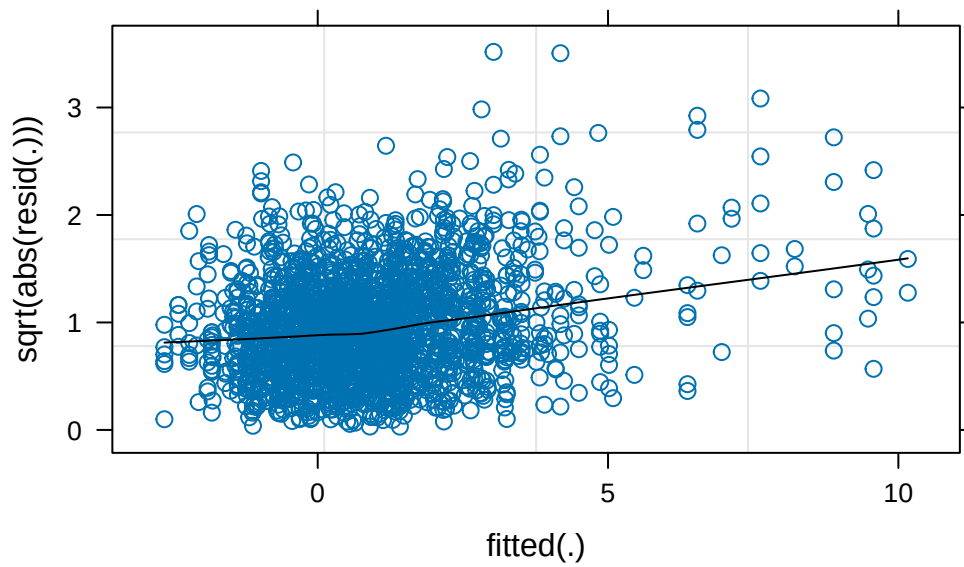
```
plot(mod4_onset, type=c("p","smooth"), col.line=1)
```



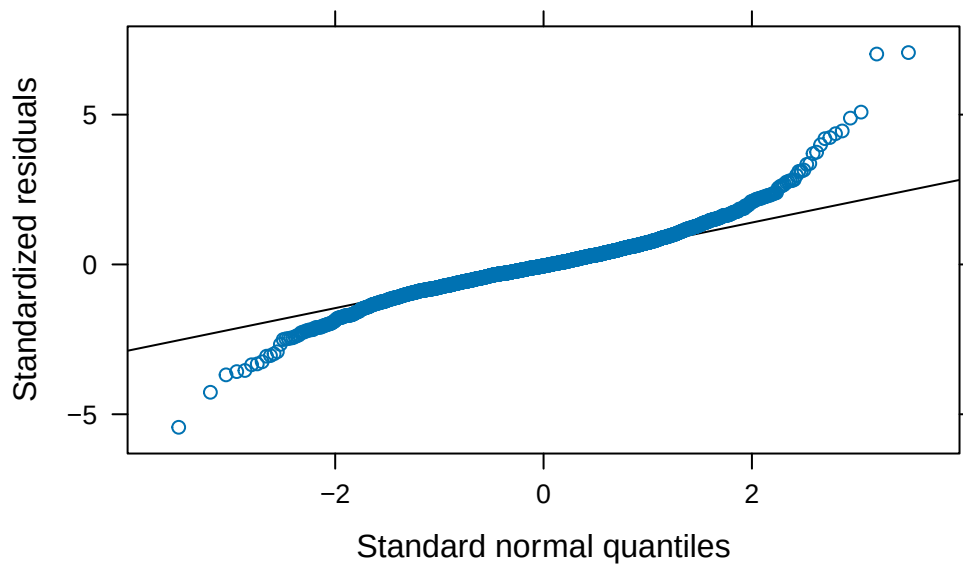
```

plot(mod4_onset,
      sqrt(abs(resid(.)))~fitted(.),
      type=c("p","smooth"), col.line=1)

```



```
lattice::qqmath(mod4_onset)
```



```
#Table
#Fixed Effects
mod4_onset_table <- data.frame(
  predictor = attr(summary(mod4_onset)$coef,"dimnames")[[1]],
  B = summary(mod4_onset)$coef["Estimate"],
  beta = effectsize::standardize_parameters(mod4_onset)$Std_Coefficient,
  CI = paste0("(",round(ci[-c(1,2),1],4),", ",round(ci[-c(1,2),2],4),")"),
```

```

CI_STD = paste0("(",round(effects::standardize_parameters(mod4_onset)$CI_low,4),", ",round
pval = summary(mod4_onset)$coef[,5]
)

mod4_onset_table[,c("B","beta","pval")] <-
round(mod4_onset_table[,c("B","beta","pval")],4)
write.csv(mod4_onset_table,file = "../bohnert_sleep_git/mod4_onset_table.csv")

#Random effects
#within
sigma2 <- summary(mod4_onset)$sigma^2
#between
tau00 <- attr(summary(mod4_onset)$varcor$ID,"stddev")^2
iccvec[16] <- ICC <- tau00/(tau00 + sigma2)
N <- summary(mod4_onset)$devcomp$dims["q"]
obs <- summary(mod4_onset)$devcomp$dims["n"]
R2marg <- r.squaredGLMM(mod4_onset)[,"R2m"]
R2cond <- r.squaredGLMM(mod4_onset)[,"R2c"]
mod4_onset_table_re <- data.frame(sigma2, tau00, ICC, N, obs, R2marg, R2cond)
mod4_onset_table_re

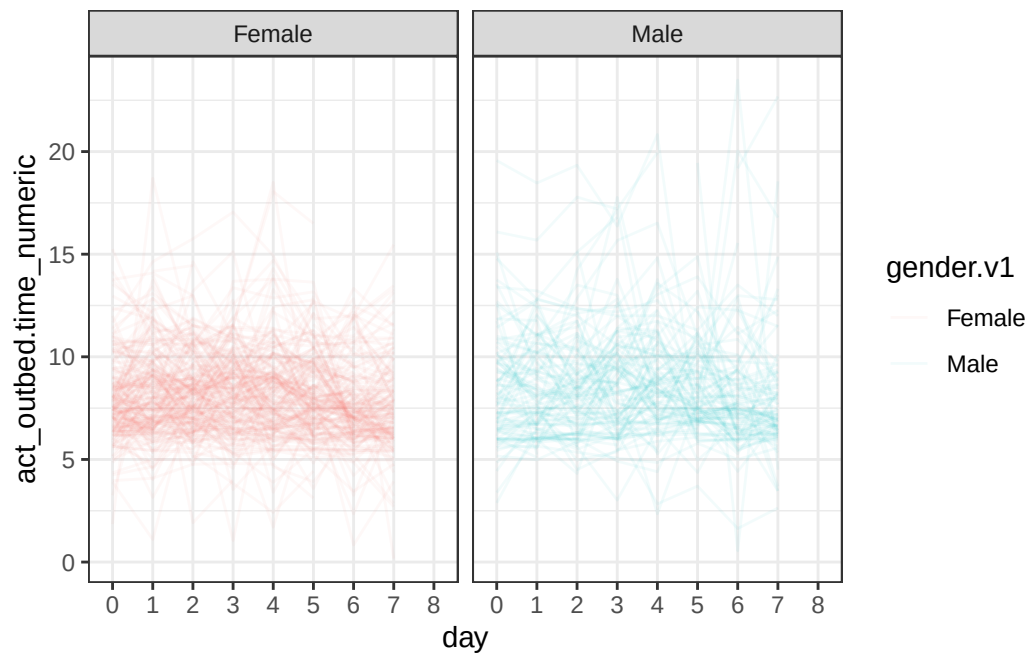
```

	sigma2	tau00	ICC	N	obs	R2marg	R2cond
(Intercept)	3.061356	2.977598	0.4930652	300	2169	0.03961334	0.5131466

Sleep Outbed

```
ggplot(aes(x = day, y = act_outbed.time_numeric, group = ID, color = gender.v1), data = dat1)
```

Warning: Removed 452 rows containing missing values or values outside the scale range (`geom\_line()`).



```
library(lme4)
#####
#act_outbed.time_numeric
#####
#####
#Model 1: COI Total
#####
mod1_outbed <- lmer(act_outbed.time_numeric ~ gender.v1 + age.v1 + income.v1.10000 +
  cg1_edu_grp + I_frisat + z_COI_nat + (1|ID) , data = datlong)
summary(mod1_outbed)
```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: act_outbed.time_numeric ~ gender.v1 + age.v1 + income.v1.10000 +
cg1_edu_grp + I_frisat + z_COI_nat + (1 | ID)
Data: datlong

REML criterion at convergence: 9513.3

Scaled residuals:

Min	1Q	Median	3Q	Max
-7.2248	-0.4709	-0.0560	0.4516	8.8813

Random effects:

Groups	Name	Variance	Std.Dev.
ID	(Intercept)	2.662	1.632
Residual		2.920	1.709

Number of obs: 2269, groups: ID, 314

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	8.33287	1.09940	305.11241	7.579	4.19e-13 ***
gender.v1Male	0.34925	0.21294	303.63867	1.640	0.1020
age.v1	-0.01770	0.06580	304.90259	-0.269	0.7881
income.v1.10000	-0.02026	0.03472	301.94085	-0.584	0.5599
cg1_edu_grp<=BS	-0.10500	0.23113	307.93722	-0.454	0.6500
cg1_edu_grp>=MS	0.13300	0.38662	305.86152	0.344	0.7311
I_frisat	1.23865	0.08098	1966.04459	15.295	< 2e-16 ***
z_COI_nat	0.25574	0.14286	306.62648	1.790	0.0744 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

	(Intr)	gnd.1M	age.v1	i.1.10	c1__<=	c1__>=	I_frst
gender.v1M1	-0.181						
age.v1	-0.975	0.132					
inc.1.10000	-0.001	-0.002	-0.110				
cg1_d_g<=BS	-0.082	-0.084	-0.045	-0.147			
cg1_d_g>=MS	0.001	-0.154	-0.061	-0.255	0.465		
I_frisat	-0.011	0.001	-0.009	0.006	-0.003	-0.008	
z_COI_nat	0.249	-0.037	-0.170	-0.111	-0.054	-0.110	0.007

```
ci <- confint(mod1_outbed)
```

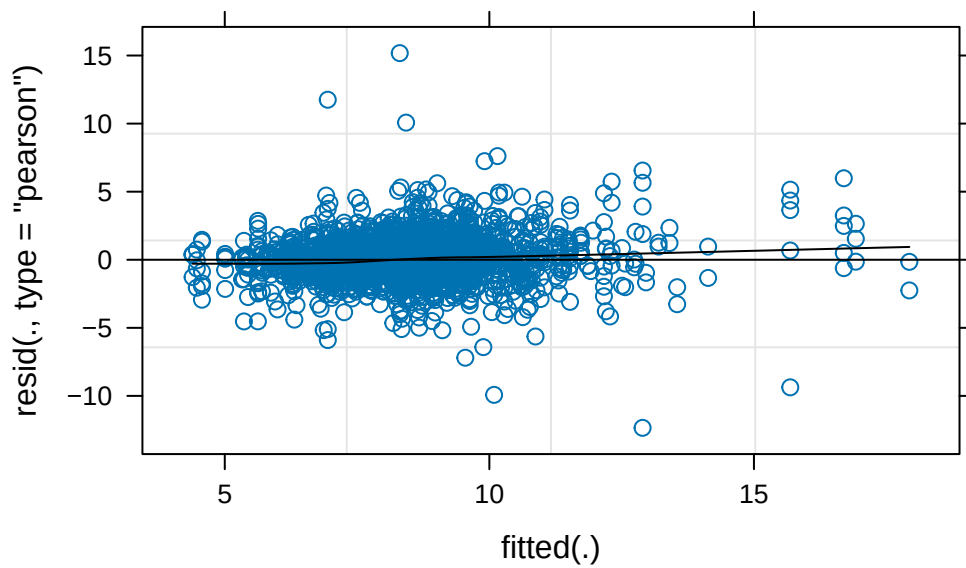
Computing profile confidence intervals ...

```
ci
```

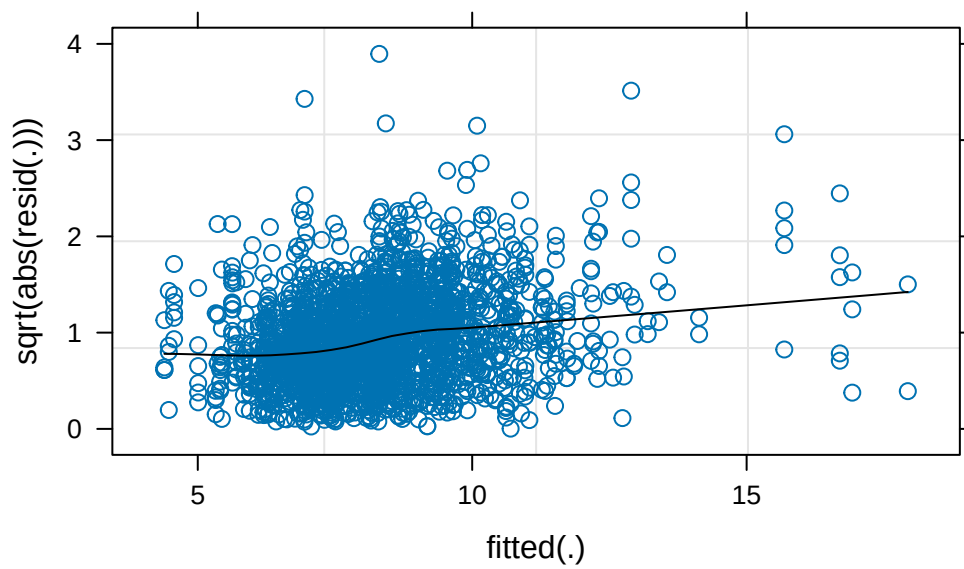
	2.5 %	97.5 %
.sig01	1.47140285	1.76646957
.sigma	1.65620018	1.76339456
(Intercept)	6.19656706	10.47069814
gender.v1Male	-0.06463854	0.76317185
age.v1	-0.14565952	0.11013760

```
income.v1.10000 -0.08773557 0.04723155
cg1_edu_grp<=BS -0.55415042 0.34442966
cg1_edu_grp>=MS -0.61833980 0.88473068
I_frisat        1.07990436 1.39741137
z_COI_nat       -0.02212034 0.53330833
```

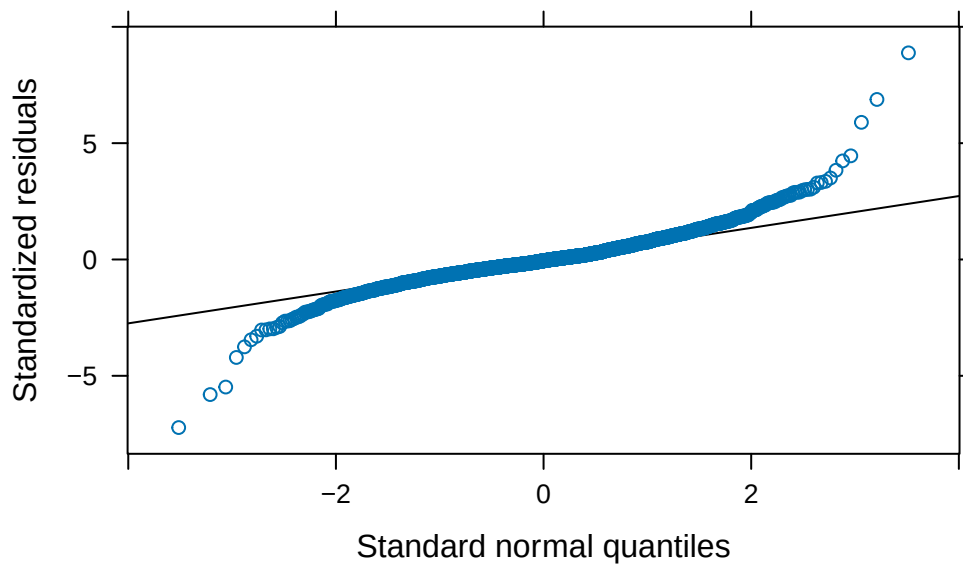
```
plot(mod1_outbed, type=c("p","smooth"), col.line=1)
```



```
plot(mod1_outbed,
      sqrt(abs(resid(.)))~fitted(.),
      type=c("p","smooth"), col.line=1)
```



```
lattice::qqmath(mod1_outbed)
```



```
mod1_outbed_table <- data.frame(
  predictor = attr(summary(mod1_outbed)$coef,"dimnames")[[1]],
  B = summary(mod1_outbed)$coef["Estimate"],
  beta = effectsize::standardize_parameters(mod1_outbed)$Std_Coefficient,
  CI = paste0("(",round(ci[-c(1,2),1],4),", ", " ",round(ci[-c(1,2),2],4),")"),
  CI_STD = paste0("(",round(effectsize::standardize_parameters(mod1_outbed)$CI_low,4),",", " ",rou
  pval = summary(mod1_outbed)$coef[,5]
```

```
)

mod1_outbed_table[,c("B","beta","pval")] <-
round(mod1_outbed_table[,c("B","beta","pval")],4)
write.csv(mod1_outbed_table,file = "../bohnert_sleep_git/mod1_outbed_table.csv")

#Random effects
#within
sigma2 <- summary(mod1_outbed)$sigma^2
#between
tau00 <- attr(summary(mod1_outbed)$varcor$ID,"stddev")^2
iccvec[17] <- ICC <- tau00/(tau00 + sigma2)
N <- summary(mod1_outbed)$devcomp$dims["q"]
obs <- summary(mod1_outbed)$devcomp$dims["n"]
R2marg <- r.squaredGLMM(mod1_outbed)[,"R2m"]
R2cond <- r.squaredGLMM(mod1_outbed)[,"R2c"]
mod1_outbed_table_re <- data.frame(sigma2, tau00, ICC, N, obs, R2marg, R2cond)
mod1_outbed_table_re
```

	sigma2	tau00	ICC	N	obs	R2marg	R2cond
(Intercept)	2.920053	2.662081	0.4768931	314	2269	0.06329162	0.5100014

```
#####
#Model 2: COI Sub-domains
#####
mod2_outbed <- lmer(act_outbed.time_numeric ~ gender.v1 + age.v1 + income.v1.10000 +
cg1_edu_grp + I_frisat + z_ED_nat + z_HE_nat + z_SE_nat + (1|ID) , data = datlong)
summary(mod2_outbed)
```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]

Formula: act_outbed.time_numeric ~ gender.v1 + age.v1 + income.v1.10000 +
cg1_edu_grp + I_frisat + z_ED_nat + z_HE_nat + z_SE_nat + (1 | ID)
Data: datlong

REML criterion at convergence: 9515.2

Scaled residuals:

Min	1Q	Median	3Q	Max
-7.2204	-0.4724	-0.0590	0.4539	8.8793

Random effects:

Groups	Name	Variance	Std.Dev.
ID	(Intercept)	2.678	1.636
	Residual	2.920	1.709

Number of obs: 2269, groups: ID, 314

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	8.25083	1.10812	303.33043	7.446	1.01e-12 ***
gender.v1Male	0.34178	0.21398	301.49818	1.597	0.111
age.v1	-0.01699	0.06596	303.01252	-0.258	0.797
income.v1.10000	-0.01798	0.03533	300.58623	-0.509	0.611
cg1_edu_grp<=BS	-0.11021	0.23244	306.02786	-0.474	0.636
cg1_edu_grp>=MS	0.12996	0.38785	303.95227	0.335	0.738
I_frisat	1.23883	0.08098	1966.01410	15.298	< 2e-16 ***
z_ED_nat	0.07623	0.25016	307.31963	0.305	0.761
z_HE_nat	0.20529	0.24164	302.73645	0.850	0.396
z_SE_nat	0.06024	0.21091	304.59045	0.286	0.775

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

	(Intr)	gnd.1M	age.v1	i.1.10	c1__<=	c1__>=	I_frst	z_ED_n	z_HE_n
gender.v1M1	-0.175								
age.v1	-0.971	0.130							
inc.1.10000	-0.011	0.001	-0.108						
cg1_d_g<=BS	-0.079	-0.079	-0.045	-0.136					
cg1_d_g>=MS	0.002	-0.151	-0.062	-0.247	0.466				
I_frisat	-0.012	0.001	-0.009	0.006	-0.003	-0.008			
z_ED_nat	0.027	0.040	-0.031	0.128	0.062	0.015	0.000		
z_HE_nat	-0.072	-0.063	-0.002	0.033	-0.053	-0.030	0.004	-0.248	
z_SE_nat	0.155	-0.025	-0.067	-0.199	-0.060	-0.061	0.002	-0.701	-0.255

```
ci <- confint(mod2_outbed)
```

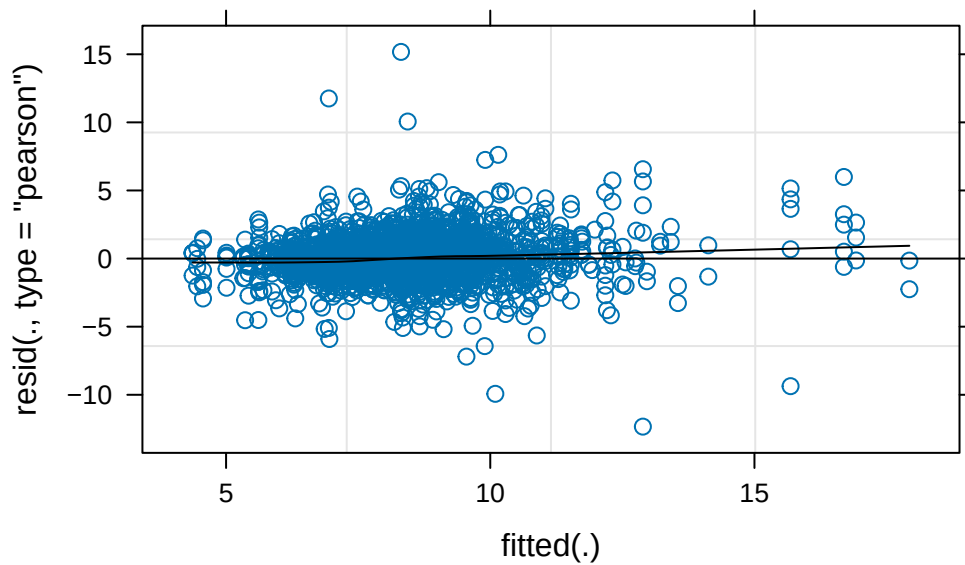
Computing profile confidence intervals ...

```
ci
```

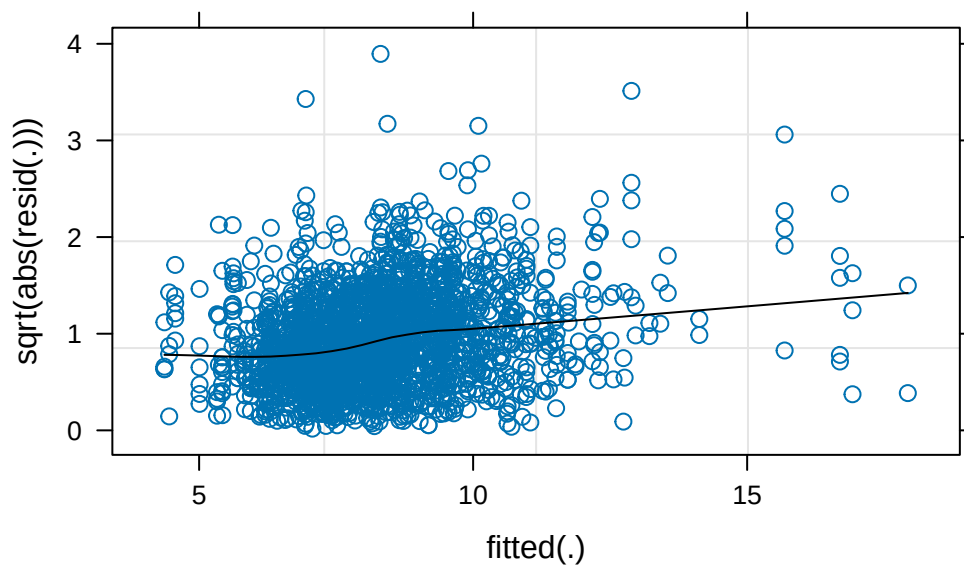
	2.5 %	97.5 %
.sig01	1.47015884	1.76501378

.sigma	1.65616845	1.76335885
(Intercept)	6.10521349	10.39918580
gender.v1Male	-0.07276611	0.75636445
age.v1	-0.14489434	0.11071257
income.v1.10000	-0.08641201	0.05046969
cg1_edu_grp<=BS	-0.56038645	0.34035681
cg1_edu_grp>=MS	-0.62119843	0.88169534
I_frisat	1.08009540	1.39759733
z_ED_nat	-0.40897001	0.56047510
z_HE_nat	-0.26299231	0.67332461
z_SE_nat	-0.34816449	0.46910657

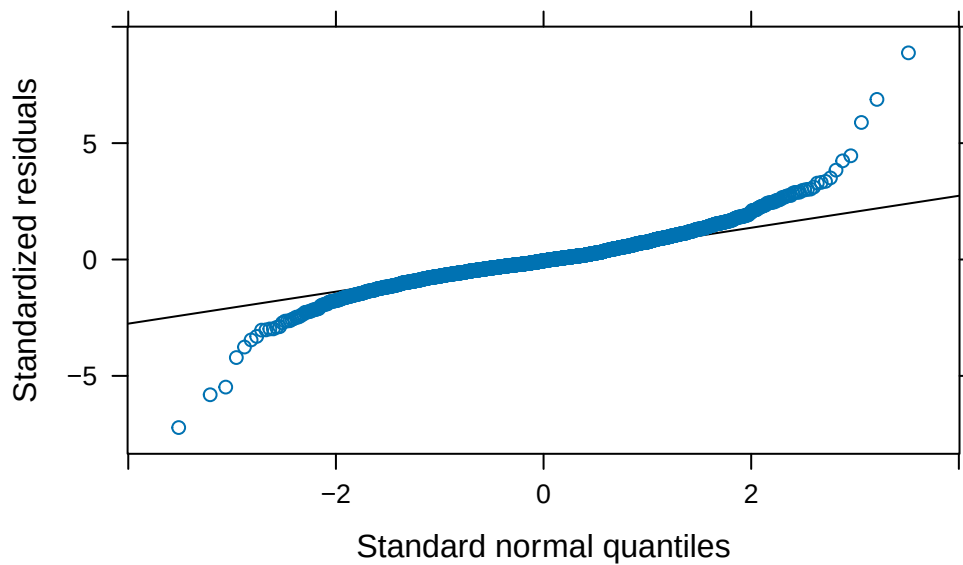
```
plot(mod2_outbed, type=c("p","smooth"), col.line=1)
```



```
plot(mod2_outbed,
      sqrt(abs(resid(.)))~fitted(.),
      type=c("p","smooth"), col.line=1)
```



```
lattice::qqmath(mod2_outbed)
```



```
mod2_outbed_table <- data.frame(
  predictor = attr(summary(mod2_outbed)$coef,"dimnames")[[1]],
  B = summary(mod2_outbed)$coef["Estimate"],
  beta = effectsize::standardize_parameters(mod2_outbed)$Std_Coefficient,
  CI = paste0("(",round(ci[-c(1,2),1],4),", ",round(ci[-c(1,2),2],4),")"),
  CI_STD = paste0("(",round(effectsize::standardize_parameters(mod2_outbed)$CI_low,4),", ",round(
  pval = summary(mod2_outbed)$coef[,5]
```



```
)

mod2_outbed_table[,c("B","beta","pval")] <-
round(mod2_outbed_table[,c("B","beta","pval")],4)
write.csv(mod2_outbed_table,file = "../bohnert_sleep_git/mod2_outbed_table.csv")

#Random effects
#within
sigma2 <- summary(mod2_outbed)$sigma^2
#between
tau00 <- attr(summary(mod2_outbed)$varcor$ID,"stddev")^2
iccvec[18] <- ICC <- tau00/(tau00 + sigma2)
N <- summary(mod2_outbed)$devcomp$dims["q"]
obs <- summary(mod2_outbed)$devcomp$dims["n"]
R2marg <- r.squaredGLMM(mod2_outbed)[,"R2m"]
R2cond <- r.squaredGLMM(mod2_outbed)[,"R2c"]
mod2_outbed_table_re <- data.frame(sigma2, tau00, ICC, N, obs, R2marg, R2cond)
mod2_outbed_table_re
```

	sigma2	tau00	ICC	N	obs	R2marg	R2cond
(Intercept)	2.919934	2.677901	0.4783815	314	2269	0.06381035	0.5116662

```
#####
#Model 3: COI Sub-domains
#####
mod3_outbed <- lmer(act_outbed.time_numeric ~ gender.v1 + age.v1 + income.v1.10000 +
cgl1_edu_grp + I_frisat + z_ED_EC_nat +
  z_ED_EL_nat +
  z_ED_ER_nat +
  z_ED_SP_nat +
  z_HE_EP_nat +
  z_HE_HE_nat +
  z_HE_HR_nat +
  z_HE_SE_nat +
  z_SE_EI_nat +
  z_SE_EO_nat +
  z_SE_ER_nat +
  z_SE_HQ_nat +
  z_SE_SR_nat +
  z_SE_WL_nat + (1|ID) , data = datlong)
summary(mod3_outbed)
```

```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: act_outbed.time_numeric ~ gender.v1 + age.v1 + income.v1.10000 +
  cg1_edu_grp + I_frisat + z_ED_EC_nat + z_ED_EL_nat + z_ED_ER_nat +
  z_ED_SP_nat + z_HE_EP_nat + z_HE_HE_nat + z_HE_HR_nat + z_HE_SE_nat +
  z_SE_EI_nat + z_SE_EO_nat + z_SE_ER_nat + z_SE_HQ_nat + z_SE_SR_nat +
  z_SE_WL_nat + (1 | ID)
Data: datlong

```

REML criterion at convergence: 8357.5

Scaled residuals:

Min	1Q	Median	3Q	Max
-7.1771	-0.4849	-0.0465	0.4613	8.8017

Random effects:

Groups	Name	Variance	Std.Dev.
ID	(Intercept)	2.751	1.659
	Residual	3.001	1.732

Number of obs: 1981, groups: ID, 276

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	7.79805	1.25193	255.85471	6.229	1.92e-09 ***
gender.v1Male	0.35597	0.23593	254.28942	1.509	0.1326
age.v1	-0.01664	0.07267	255.32579	-0.229	0.8191
income.v1.10000	-0.04736	0.03920	253.41732	-1.208	0.2281
cg1_edu_grp<=BS	-0.15789	0.25719	257.91073	-0.614	0.5398
cg1_edu_grp>=MS	0.23053	0.43605	256.34505	0.529	0.5975
I_frisat	1.23417	0.08795	1714.86472	14.032	< 2e-16 ***
z_ED_EC_nat	-0.21554	0.12778	254.15351	-1.687	0.0929 .
z_ED_EL_nat	0.48846	0.28744	258.47344	1.699	0.0905 .
z_ED_ER_nat	-0.73785	0.50578	256.89722	-1.459	0.1458
z_ED_SP_nat	0.11990	0.26879	256.30320	0.446	0.6559
z_HE_EP_nat	1.56793	0.96873	252.53078	1.619	0.1068
z_HE_HE_nat	0.77274	0.62709	256.74519	1.232	0.2190
z_HE_HR_nat	0.07611	0.25888	254.46152	0.294	0.7690
z_HE_SE_nat	0.04939	0.18341	258.46107	0.269	0.7879
z_SE_EI_nat	0.74145	0.50769	256.80065	1.460	0.1454
z_SE_EO_nat	-0.23529	0.32864	256.79903	-0.716	0.4747
z_SE_ER_nat	-0.37887	0.34258	256.44492	-1.106	0.2698
z_SE_HQ_nat	0.19911	0.30670	254.47094	0.649	0.5168
z_SE_SR_nat	-0.20176	0.30327	254.70504	-0.665	0.5065

```

z_SE_WL_nat      0.42310    0.35817  255.63994    1.181    0.2386
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Correlation matrix not shown by default, as $p = 21 > 12$.
 Use `print(x, correlation=TRUE)` or
`vcov(x)` if you need it

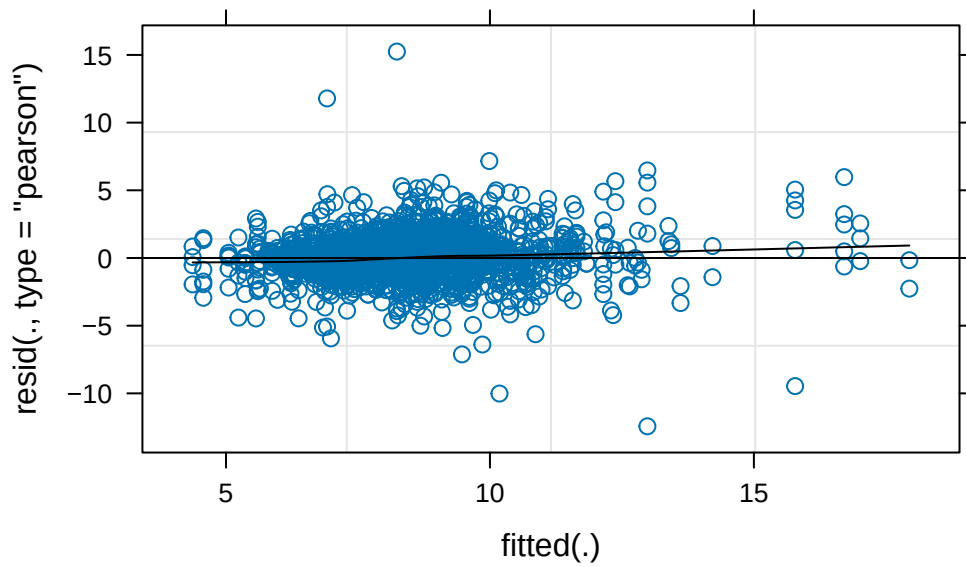
```
ci <- confint(mod3_outbed)
```

Computing profile confidence intervals ...

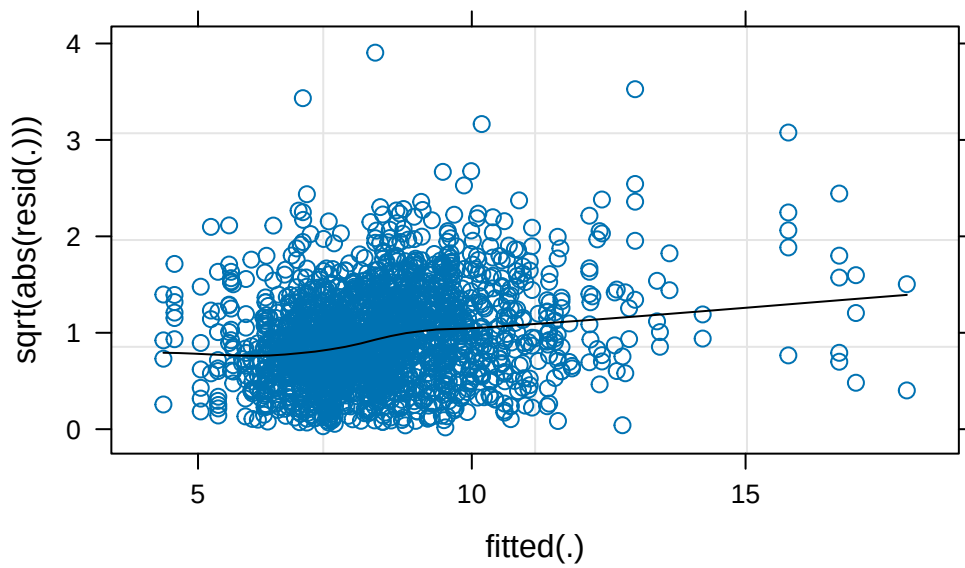
```
ci
```

	2.5 %	97.5 %
.sig01	1.44099044	1.75359050
.sigma	1.67521260	1.79152622
(Intercept)	5.43316234	10.17599265
gender.v1Male	-0.09081750	0.80285999
age.v1	-0.15471572	0.12056731
income.v1.10000	-0.12153357	0.02692902
cg1_edu_grp<=BS	-0.64480999	0.32961994
cg1_edu_grp>=MS	-0.59454498	1.05738999
I_frisat	1.06200641	1.40682574
z_ED_EC_nat	-0.45724307	0.02678403
z_ED_EL_nat	-0.05774703	1.03136490
z_ED_ER_nat	-1.69654070	0.21963030
z_ED_SP_nat	-0.39030622	0.62800563
z_HE_EP_nat	-0.26654831	3.40254083
z_HE_HE_nat	-0.41631786	1.95944295
z_HE_HR_nat	-0.41516252	0.56549101
z_HE_SE_nat	-0.29773572	0.39720103
z_SE_EI_nat	-0.21968962	1.70368926
z_SE_EO_nat	-0.85739185	0.38766803
z_SE_ER_nat	-1.02869583	0.26916696
z_SE_HQ_nat	-0.38204269	0.77973641
z_SE_SR_nat	-0.77514601	0.37367400
z_SE_WL_nat	-0.25424526	1.10262756

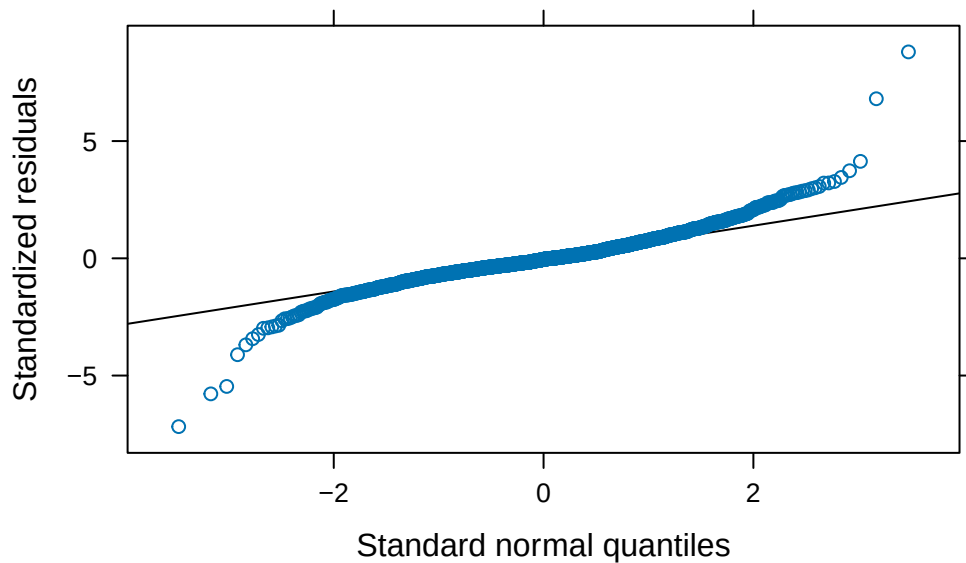
```
plot(mod3_outbed, type=c("p","smooth"), col.line=1)
```



```
plot(mod3_outbed,
      sqrt(abs(resid(.)))~fitted(.),
      type=c("p","smooth"), col.line=1)
```



```
lattice::qqmath(mod3_outbed)
```



```

mod3_outbed_table <- data.frame(
  predictor = attr(summary(mod3_outbed)$coef,"dimnames")[[1]],
  B = summary(mod3_outbed)$coef[, "Estimate"],
  beta = effectsize::standardize_parameters(mod3_outbed)$Std_Coefficient,
  CI = paste0("(",round(ci[-c(1,2),1],4),", ",round(ci[-c(1,2),2],4),")"),
  CI_STD = paste0("(",round(effectsize::standardize_parameters(mod3_outbed)$CI_low,4),", ",round(effectsize::standardize_parameters(mod3_outbed)$CI_high,4),")"),
  pval = summary(mod3_outbed)$coef[,5]
)

mod3_outbed_table[,c("B","beta","pval")] <-
round(mod3_outbed_table[,c("B","beta","pval")],4)
write.csv(mod3_outbed_table,file = "../bohnert_sleep_git/mod3_outbed_table.csv")

#Random effects
#within
sigma2 <- summary(mod3_outbed)$sigma^2
#between
tau00 <- attr(summary(mod3_outbed)$varcor$ID,"stddev")^2
iccvec[19] <- ICC <- tau00/(tau00 + sigma2)
N <- summary(mod3_outbed)$devcomp$dims["q"]
obs <- summary(mod3_outbed)$devcomp$dims["n"]
R2marg <- r.squaredGLMM(mod3_outbed)[,"R2m"]
R2cond <- r.squaredGLMM(mod3_outbed)[,"R2c"]
mod3_outbed_table_re <- data.frame(sigma2, tau00, ICC, N, obs, R2marg, R2cond)
mod3_outbed_table_re

```

	sigma2	tau00	ICC	N	obs	R2marg	R2cond
(Intercept)	3.000702	2.751455	0.4783345	276	1981	0.08832613	0.5244112

```
#####
#Model 4: Neighborhood
#####
mod4_outbed <- lmer(act_outbed.time_numeric ~ gender.v1 + age.v1 + income.v1.10000 +
  cg1_edu_grp + I_frisat +
  p.mn.cohesion.mean.v1 +
  p.mn.safety.mean.v1 +
  p.mn.violence.mean.v1 + (1|ID) , data = datlong)
summary(mod4_outbed)
```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]

Formula: act_outbed.time_numeric ~ gender.v1 + age.v1 + income.v1.10000 +
cg1_edu_grp + I_frisat + p.mn.cohesion.mean.v1 + p.mn.safety.mean.v1 +
p.mn.violence.mean.v1 + (1 | ID)
Data: datlong

REML criterion at convergence: 9136.3

Scaled residuals:

Min	1Q	Median	3Q	Max
-7.1881	-0.4724	-0.0587	0.4432	8.8388

Random effects:

Groups	Name	Variance	Std.Dev.
ID	(Intercept)	2.755	1.660
Residual		2.968	1.723

Number of obs: 2169, groups: ID, 300

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	6.935e+00	1.267e+00	2.921e+02	5.472	9.6e-08 ***
gender.v1Male	3.490e-01	2.194e-01	2.883e+02	1.591	0.113
age.v1	1.184e-02	6.932e-02	2.901e+02	0.171	0.865
income.v1.10000	-1.223e-02	3.611e-02	2.875e+02	-0.339	0.735
cg1_edu_grp<=BS	-6.574e-02	2.467e-01	2.932e+02	-0.266	0.790
cg1_edu_grp>=MS	2.045e-01	4.085e-01	2.903e+02	0.500	0.617
I_frisat	1.232e+00	8.351e-02	1.880e+03	14.748	< 2e-16 ***
p.mn.cohesion.mean.v1	2.611e-03	1.631e-01	2.896e+02	0.016	0.987

```
p.mn.safety.mean.v1    7.717e-02  1.527e-01  2.898e+02   0.505    0.614
p.mn.violence.mean.v1  2.475e-01  1.706e-01  2.909e+02   1.451    0.148
---
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

```
(Intr) gnd.1M age.v1 i.1.10 c1__<= c1__>= I_frst p.mn.c..1
gender.v1M1 -0.100
age.v1      -0.816  0.109
inc.1.10000 0.050  0.000 -0.114
cg1_d_g<=BS -0.033 -0.066 -0.105 -0.169
cg1_d_g>=MS  0.080 -0.145 -0.095 -0.272  0.486
I_friscat   -0.011  0.001 -0.009  0.007 -0.002 -0.007
p.mn.chs..1 -0.182 -0.011 -0.039  0.056 -0.081 -0.169  0.001
p.mn.sft..1 -0.209 -0.045 -0.056 -0.161  0.093  0.076 -0.001 -0.557
p.mn.vln..1 -0.419 -0.090 -0.058 -0.023  0.046 -0.011  0.001 -0.011
p.mn.s..1
gender.v1M1
age.v1
inc.1.10000
cg1_d_g<=BS
cg1_d_g>=MS
I_friscat
p.mn.chs..1
p.mn.sft..1
p.mn.vln..1  0.575
```

```
ci <- confint(mod4_outbed)
```

Computing profile confidence intervals ...

```
ci
```

	2.5 %	97.5 %
.sig01	1.48742372	1.79239993
.sigma	1.66861904	1.77913237
(Intercept)	4.48307736	9.39184039
gender.v1Male	-0.07568294	0.77391272
age.v1	-0.12252148	0.14596073
income.v1.10000	-0.08213332	0.05769623
cg1_edu_grp<=BS	-0.54331130	0.41226798

```

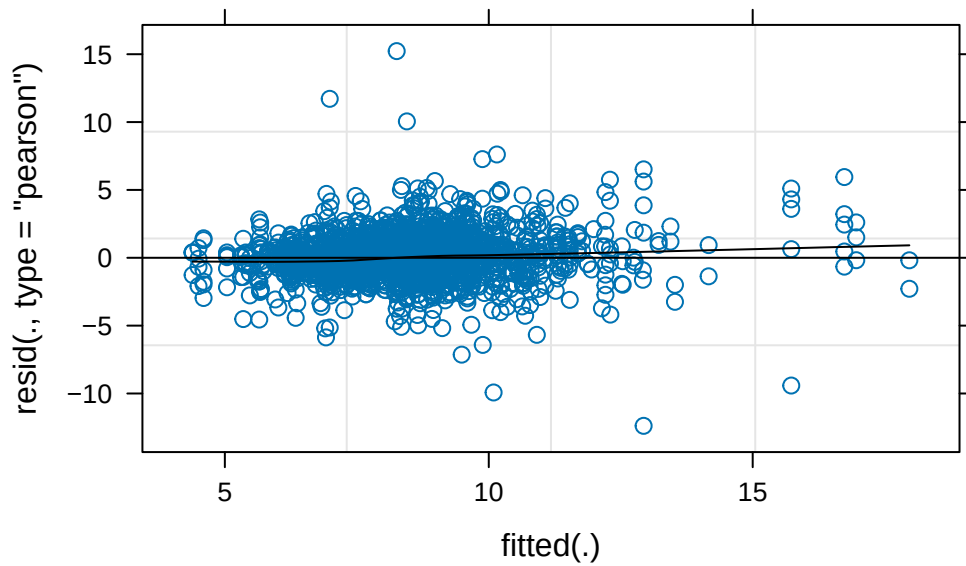
cg1_edu_grp>=MS      -0.58631102  0.99593761
I_frisat             1.06794774  1.39537372
p.mn.cohesion.mean.v1 -0.31347473  0.31833762
p.mn.safety.mean.v1   -0.21854779  0.37283356
p.mn.violence.mean.v1 -0.08317494  0.57749782

```

```

plot(mod4_outbed, type=c("p","smooth"), col.line=1)

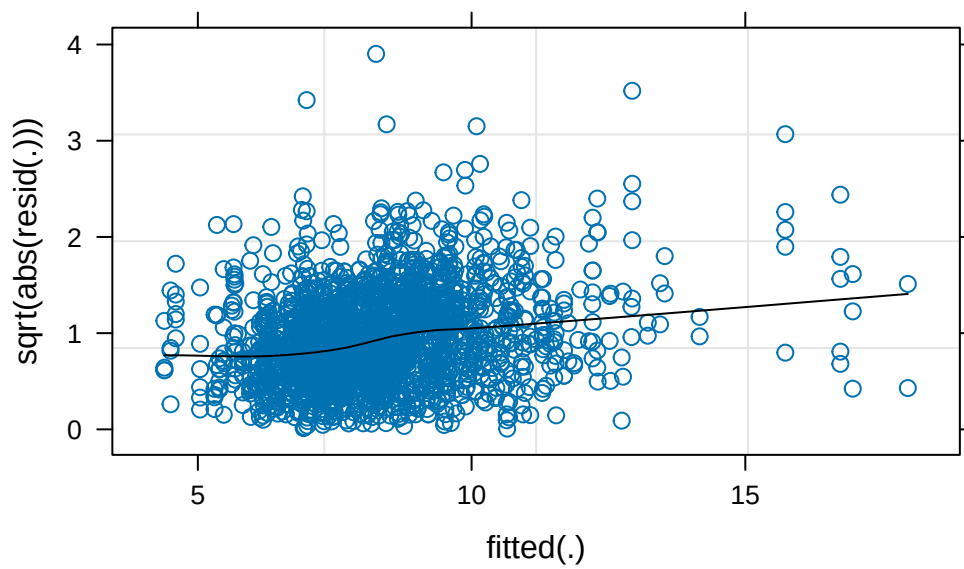
```



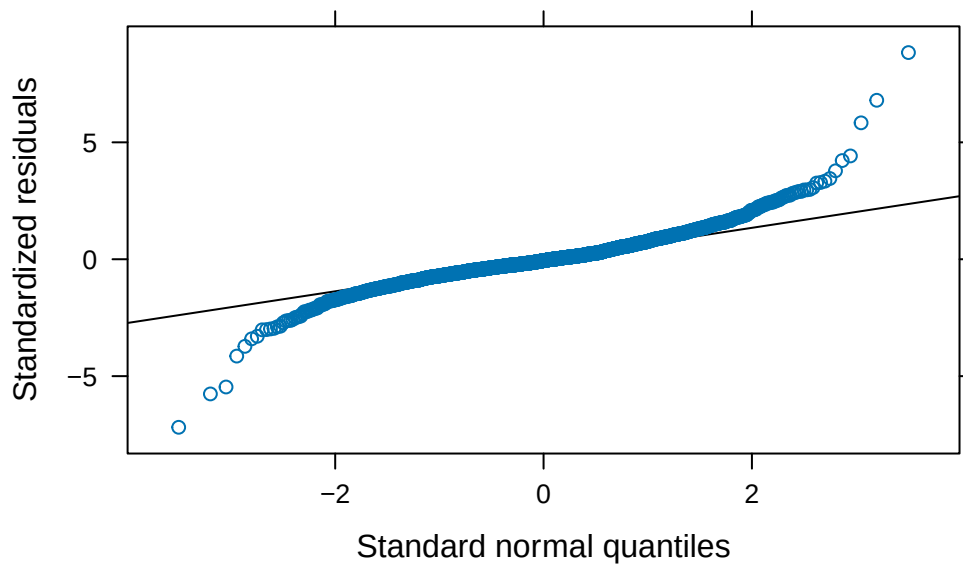
```

plot(mod4_outbed,
      sqrt(abs(resid(.)))~fitted(.),
      type=c("p","smooth"), col.line=1)

```

```
lattice::qqmath(mod4_outbed)
```



```
#Table
#Fixed Effects
mod4_outbed_table <- data.frame(
  predictor = attr(summary(mod4_outbed)$coef,"dimnames")[[1]],
  B = summary(mod4_outbed)$coef[, "Estimate"],
  beta = effectsize::standardize_parameters(mod4_outbed)$Std_Coefficient,
  CI = paste0("(",round(ci[-c(1,2),1],4),", ",round(ci[-c(1,2),2],4),")"),
```

```

CI_STD = paste0("(",round(effectsize::standardize_parameters(mod4_outbed)$CI_low,4),",", " ",rou
pval = summary(mod4_outbed)$coef[,5]
)

```

```

mod4_outbed_table[,c("B","beta","pval")] <-
round(mod4_outbed_table[,c("B","beta","pval")],4)
write.csv(mod4_outbed_table,file = "../bohnert_sleep_git/mod4_outbed_table.csv")

```

```

#Random effects
#within
sigma2 <- summary(mod4_outbed)$sigma^2
#between
tau00 <- attr(summary(mod4_outbed)$varcor$ID,"stddev")^2
iccvec[20] <- ICC <- tau00/(tau00 + sigma2)
N <- summary(mod4_outbed)$devcomp$dims["q"]
obs <- summary(mod4_outbed)$devcomp$dims["n"]
R2marg <- r.squaredGLMM(mod4_outbed)[,"R2m"]
R2cond <- r.squaredGLMM(mod4_outbed)[,"R2c"]
mod4_outbed_table_re <- data.frame(sigma2, tau00, ICC, N, obs, R2marg, R2cond)
mod4_outbed_table_re

```

	sigma2	tau00	ICC	N	obs	R2marg	R2cond
(Intercept)	2.968199	2.75511	0.4813842	300	2169	0.06103719	0.513039

```

#Output the entire image
save.image("../bohnert_sleep/bohnert_sleep_20250322.rda")

```

```

iccmat <- matrix(iccvec, ncol = 4, byrow = T)
row.names(iccmat) <- c("Total Sleep","Efficiency","Awakening","Awake","Inbed")
colnames(iccmat) <- paste0("Model ",1:4)
iccmat

```

	Model 1	Model 2	Model 3	Model 4
Total Sleep	0.2188655	0.2195362	0.2016540	0.2215956
Efficiency	0.4687089	0.4687906	0.4564512	0.4710724
Awakening	0.3724133	0.3725027	0.3588628	0.3876270
Awake	0.4868972	0.4879145	0.5031099	0.4930652
Inbed	0.4768931	0.4783815	0.4783345	0.4813842

```
write.csv(iccmat,file = "../iccmatrix.csv", row.names = FALSE)
```

Weekly Level

```
library(lme4)
#####
#act_outbed.min_sd
#####
dat <- dat %>% mutate(income.v1.10000 = income.v1/10000)
#####
#Model 1: COI Total
#####
mod1_totsleep_sd <- lm(act_totsleep.min_sd ~ gender.v1 + age.v1 + income.v1.10000 +
cg1_edu_grp + z_COI_nat, data = dat)
summary(mod1_totsleep_sd)
```

Call:

```
lm(formula = act_totsleep.min_sd ~ gender.v1 + age.v1 + income.v1.10000 +
    cg1_edu_grp + z_COI_nat, data = dat)
```

Residuals:

Min	1Q	Median	3Q	Max
-63.28	-24.96	-5.71	21.22	150.41

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	85.4926	21.0531	4.061	6.22e-05 ***
gender.v1Male	-0.6763	4.0695	-0.166	0.868
age.v1	-0.1398	1.2585	-0.111	0.912
income.v1.10000	-0.3886	0.6641	-0.585	0.559
cg1_edu_grp<=BS	5.2081	4.4057	1.182	0.238
cg1_edu_grp>=MS	3.1718	7.3723	0.430	0.667
z_COI_nat	-1.5028	2.7204	-0.552	0.581

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

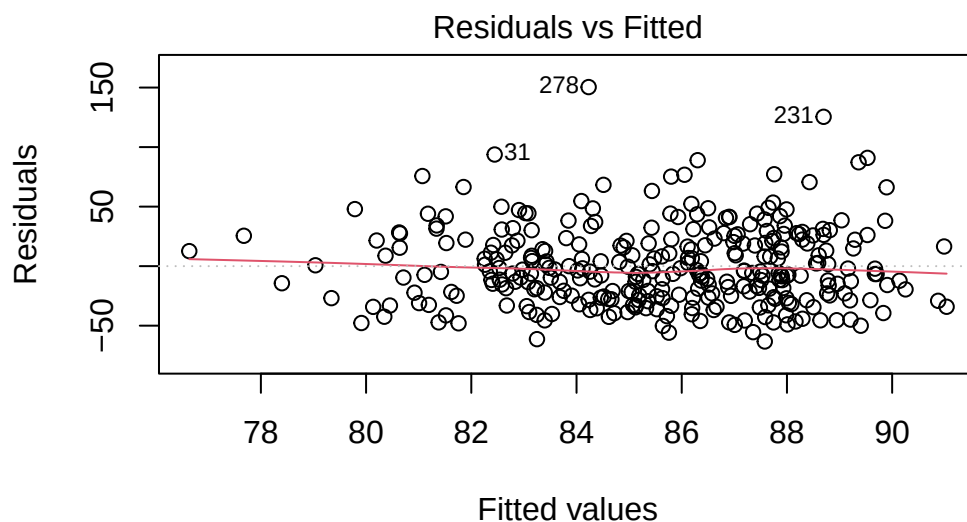
Residual standard error: 33.5 on 306 degrees of freedom
(11 observations deleted due to missingness)

Multiple R-squared: 0.006602, Adjusted R-squared: -0.01288
 F-statistic: 0.3389 on 6 and 306 DF, p-value: 0.916

```
ci <- confint(mod1_totsleep_sd)
ci
```

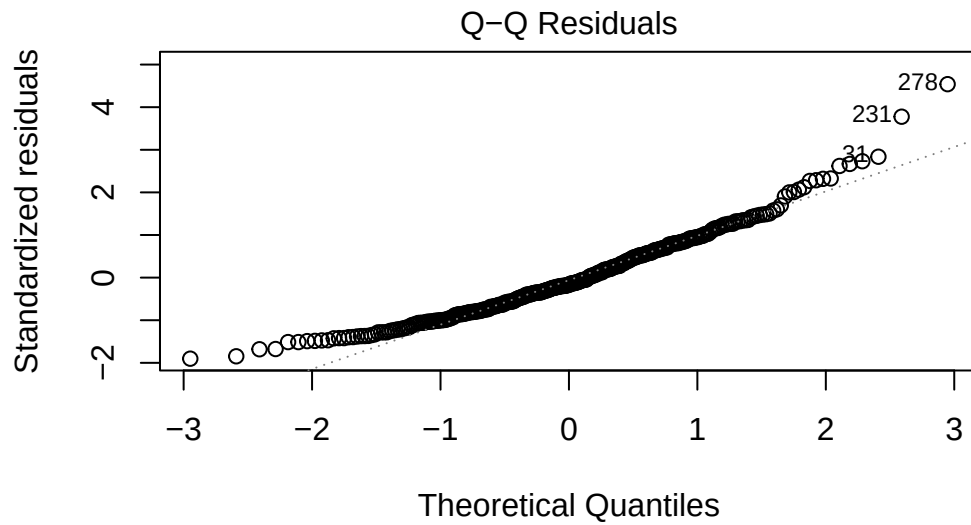
	2.5 %	97.5 %
(Intercept)	44.065424	126.9198706
gender.v1Male	-8.684051	7.3314830
age.v1	-2.616145	2.3366102
income.v1.10000	-1.695303	0.9180734
cg1_edu_grp<=BS	-3.461299	13.8774205
cg1_edu_grp>=MS	-11.334990	17.6786885
z_COI_nat	-6.855773	3.8502655

```
plot(mod1_totsleep_sd,1)
```



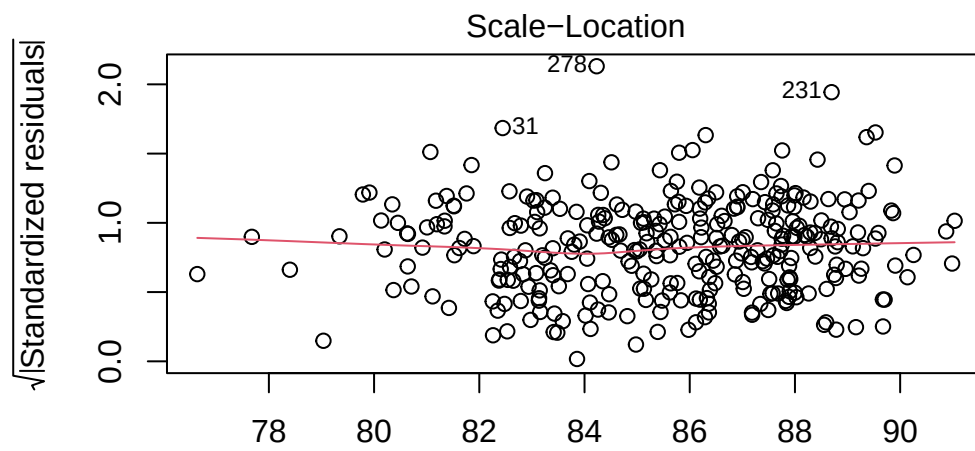
n(act_totsleep.min_sd ~ gender.v1 + age.v1 + income.v1.10000 + cg1_edu

```
plot(mod1_totsleep_sd,2)
```



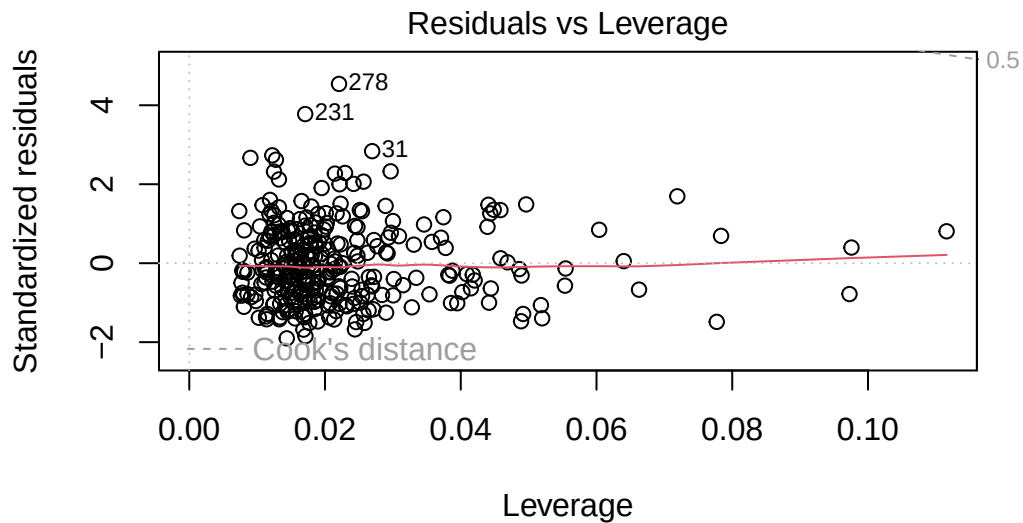
$n(\text{act_totsleep.min_sd} \sim \text{gender.v1} + \text{age.v1} + \text{income.v1.10000} + \text{cg1_edu})$

```
plot(mod1_totsleep_sd,3)
```



$n(\text{act_totsleep.min_sd} \sim \text{gender.v1} + \text{age.v1} + \text{income.v1.10000} + \text{cg1_edu})$

```
plot(mod1_totsleep_sd,5)
```



```
lm(act_totsleep.min_sd ~ gender.v1 + age.v1 + income.v1.10000 + cg1_edu
```

```
effectsize::standardize_parameters(mod1_totsleep_sd)
```

```
# Standardization method: refit
```

Parameter	Std. Coef.	95% CI
(Intercept)	-0.10	[-0.32, 0.13]
gender v1 [Male]	-0.02	[-0.26, 0.22]
age v1	-6.57e-03	[-0.12, 0.11]
income v1 10000	-0.04	[-0.15, 0.08]
cg1_edu_grp [<=BS]	0.16	[-0.10, 0.42]
cg1_edu_grp [>=MS]	0.10	[-0.34, 0.53]
z COI nat	-0.03	[-0.15, 0.08]

```
#####
#Model 2: COI Sub-domains
#####
mod2_totsleep_sd <- lm(act_totsleep.min_sd ~ gender.v1 + age.v1 + income.v1.10000 +
cg1_edu_grp + z_ED_nat + z_HE_nat + z_SE_nat, data = dat)
summary(mod2_totsleep_sd)
```

Call:

```
lm(formula = act_totsleep.min_sd ~ gender.v1 + age.v1 + income.v1.10000 +
    cg1_edu_grp + z_ED_nat + z_HE_nat + z_SE_nat, data = dat)
```

Residuals:

Min	1Q	Median	3Q	Max
-62.903	-24.269	-5.685	21.614	150.060

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	86.759847	21.239064	4.085	5.65e-05 ***
gender.v1Male	-0.555080	4.090562	-0.136	0.892
age.v1	-0.155601	1.262136	-0.123	0.902
income.v1.10000	-0.410481	0.675535	-0.608	0.544
cg1_edu_grp<=BS	5.285435	4.430620	1.193	0.234
cg1_edu_grp>=MS	3.205795	7.397274	0.433	0.665
z_ED_nat	0.035206	4.754386	0.007	0.994
z_HE_nat	-2.675828	4.626616	-0.578	0.563
z_SE_nat	-0.007061	4.017227	-0.002	0.999

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 33.6 on 304 degrees of freedom

(11 observations deleted due to missingness)

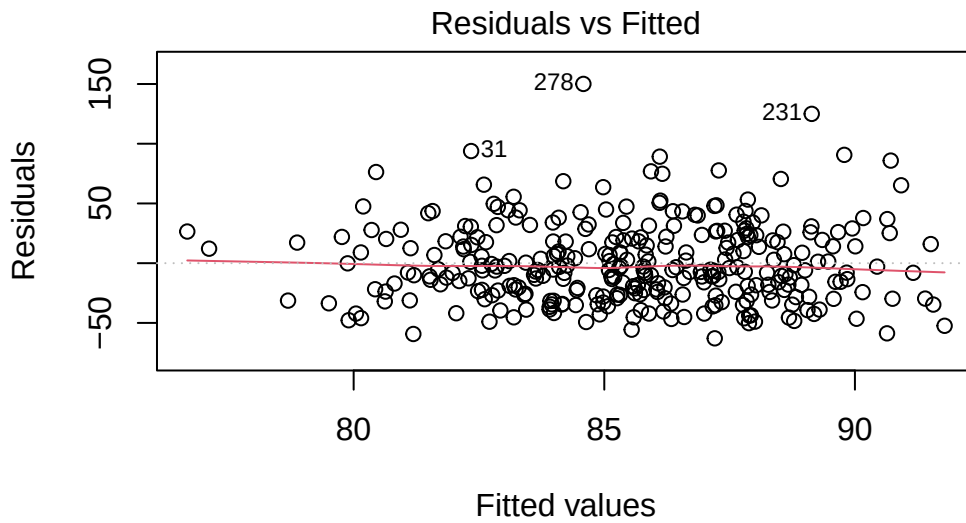
Multiple R-squared: 0.00748, Adjusted R-squared: -0.01864

F-statistic: 0.2864 on 8 and 304 DF, p-value: 0.9702

```
ci <- confint(mod2_totsleep_sd)
ci
```

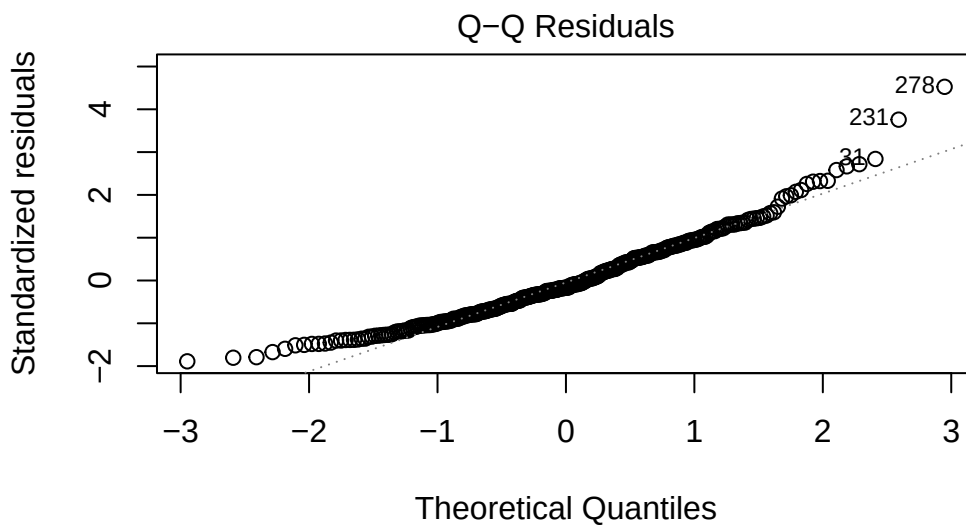
	2.5 %	97.5 %
(Intercept)	44.965657	128.5540370
gender.v1Male	-8.604480	7.4943197
age.v1	-2.639230	2.3280278
income.v1.10000	-1.739799	0.9188358
cg1_edu_grp<=BS	-3.433131	14.0040015
cg1_edu_grp>=MS	-11.350546	17.7621360
z_ED_nat	-9.320466	9.3908790
z_HE_nat	-11.780074	6.4284182
z_SE_nat	-7.912153	7.8980305

```
plot(mod2_totsleep_sd,1)
```



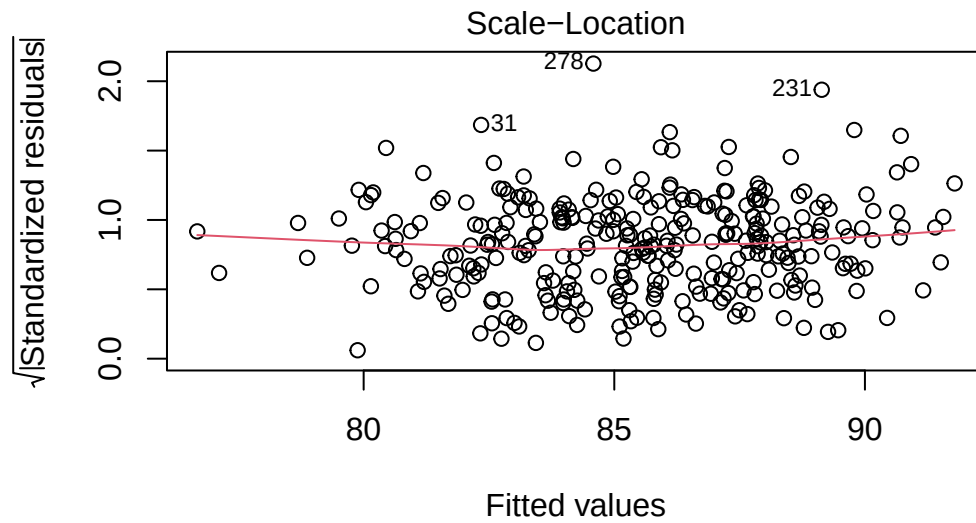
```
n(act_totsleep.min_sd ~ gender.v1 + age.v1 + income.v1.10000 + cg1_edu,
```

```
plot(mod2_totsleep_sd,2)
```



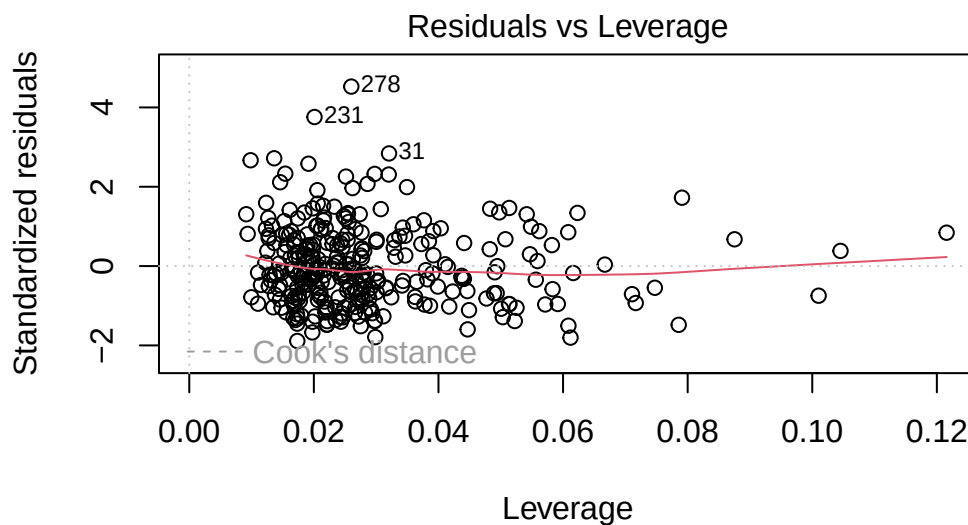
```
n(act_totsleep.min_sd ~ gender.v1 + age.v1 + income.v1.10000 + cg1_edu,
```

```
plot(mod2_totsleep_sd,3)
```

```
n(act_totsleep.min_sd ~ gender.v1 + age.v1 + income.v1.10000 + cg1_edu,
```

```
plot(mod2_totsleep_sd,5)
```



```
n(act_totsleep.min_sd ~ gender.v1 + age.v1 + income.v1.10000 + cg1_edu,
```

```
effectsize::standardize_parameters(mod2_totsleep_sd)
```

```
# Standardization method: refit
```

Parameter	Std. Coef.	95% CI

(Intercept)		-0.10		[-0.33, 0.12]
gender v1 [Male]		-0.02		[-0.26, 0.23]
age v1		-7.31e-03		[-0.12, 0.11]
income v1 10000		-0.04		[-0.16, 0.08]
cg1 edu grp [<=BS]		0.16		[-0.10, 0.42]
cg1 edu grp [>=MS]		0.10		[-0.34, 0.53]
z ED nat		7.69e-04		[-0.20, 0.21]
z HE nat		-0.04		[-0.19, 0.11]
z SE nat		-1.87e-04		[-0.21, 0.21]

```
#####
#Model 3: COI Sub-domains
#####
mod3_totsleep_sd <- lm(act_totsleep.min_sd ~ gender.v1 + age.v1 + income.v1.10000 +
cg1_edu_grp + z_ED_EC_nat +
  z_ED_EL_nat +
  z_ED_ER_nat +
  z_ED_SP_nat +
  z_HE_EP_nat +
  z_HE_HE_nat +
  z_HE_HR_nat +
  z_HE_SE_nat +
  z_SE_EI_nat +
  z_SE_EO_nat +
  z_SE_ER_nat +
  z_SE_HQ_nat +
  z_SE_SR_nat +
  z_SE_WL_nat , data = dat)
summary(mod3_totsleep_sd)
```

Call:

```
lm(formula = act_totsleep.min_sd ~ gender.v1 + age.v1 + income.v1.10000 +
  cg1_edu_grp + z_ED_EC_nat + z_ED_EL_nat + z_ED_ER_nat + z_ED_SP_nat +
  z_HE_EP_nat + z_HE_HE_nat + z_HE_HR_nat + z_HE_SE_nat + z_SE_EI_nat +
  z_SE_EO_nat + z_SE_ER_nat + z_SE_HQ_nat + z_SE_SR_nat + z_SE_WL_nat,
  data = dat)
```

Residuals:

Min	1Q	Median	3Q	Max
-58.394	-23.926	-3.165	20.881	135.237

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	84.529456	23.573010	3.586	0.000403	***
gender.v1Male	-4.599797	4.430248	-1.038	0.300127	
age.v1	0.003352	1.366172	0.002	0.998044	
income.v1.10000	-0.474055	0.736322	-0.644	0.520274	
cg1_edu_grp<=BS	5.661303	4.823846	1.174	0.241647	
cg1_edu_grp>=MS	4.985358	8.173849	0.610	0.542461	
z_ED_EC_nat	1.978492	2.413261	0.820	0.413072	
z_ED_EL_nat	2.475011	5.369557	0.461	0.645239	
z_ED_ER_nat	-4.490924	9.467552	-0.474	0.635657	
z_ED_SP_nat	9.173549	5.031249	1.823	0.069426	.
z_HE_EP_nat	-22.439742	18.219668	-1.232	0.219225	
z_HE_HE_nat	2.465273	11.738652	0.210	0.833825	
z_HE_HR_nat	0.142759	4.858407	0.029	0.976581	
z_HE_SE_nat	-4.699647	3.440585	-1.366	0.173160	
z_SE_EI_nat	-2.398716	9.500883	-0.252	0.800879	
z_SE_EO_nat	-5.477195	6.152236	-0.890	0.374156	
z_SE_ER_nat	10.504368	6.419692	1.636	0.103016	
z_SE_HQ_nat	1.771437	5.760717	0.308	0.758712	
z_SE_SR_nat	-12.393106	5.696632	-2.176	0.030510	*
z_SE_WL_nat	7.122337	6.711245	1.061	0.289578	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 33.48 on 255 degrees of freedom

(49 observations deleted due to missingness)

Multiple R-squared: 0.07512, Adjusted R-squared: 0.006205

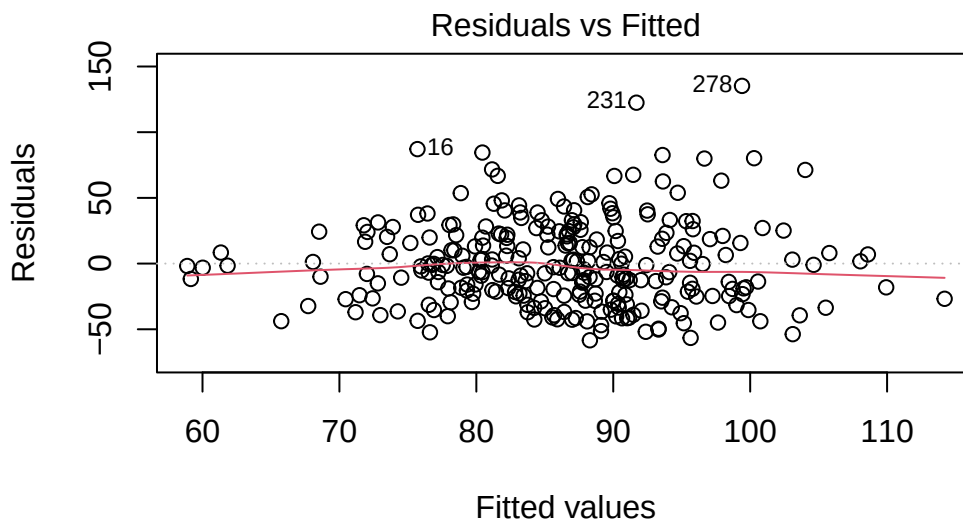
F-statistic: 1.09 on 19 and 255 DF, p-value: 0.3612

```
ci <- confint(mod3_totsleep_sd)
ci
```

	2.5 %	97.5 %
(Intercept)	38.1068774	130.952034
gender.v1Male	-13.3243302	4.124736
age.v1	-2.6870652	2.693769
income.v1.10000	-1.9241009	0.975991
cg1_edu_grp<=BS	-3.8383488	15.160954
cg1_edu_grp>=MS	-11.1114893	21.082205
z_ED_EC_nat	-2.7739694	6.730953
z_ED_EL_nat	-8.0993152	13.049337

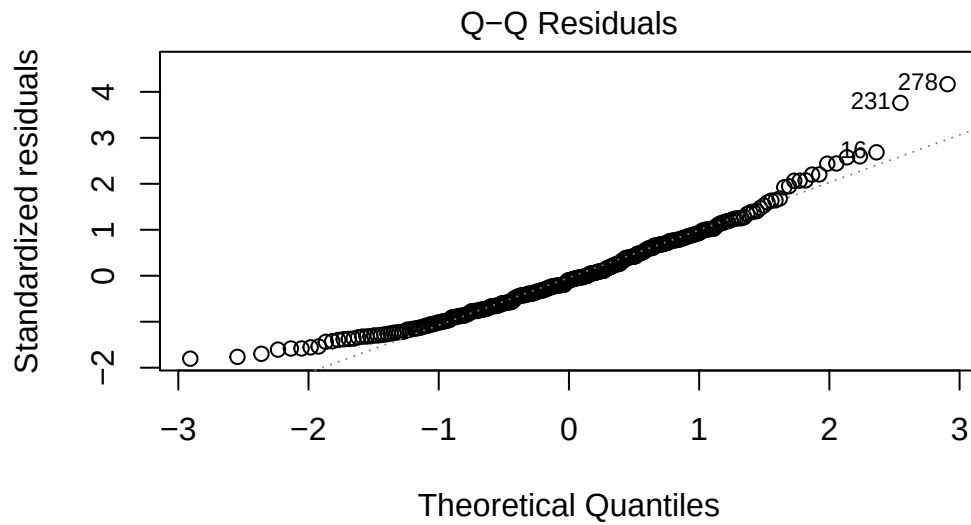
z_ED_ER_nat	-23.1354742	14.153627
z_ED_SP_nat	-0.7345417	19.081640
z_HE_EP_nat	-58.3199259	13.440442
z_HE_HE_nat	-20.6517791	25.582325
z_HE_HR_nat	-9.4249528	9.710472
z_HE_SE_nat	-11.4752275	2.075933
z_SE_EI_nat	-21.1089053	16.311473
z_SE_EO_nat	-17.5928585	6.638468
z_SE_ER_nat	-2.1379989	23.146736
z_SE_HQ_nat	-9.5732045	13.116078
z_SE_SR_nat	-23.6115431	-1.174670
z_SE_WL_nat	-6.0941884	20.338863

```
plot(mod3_totsleep_sd,1)
```



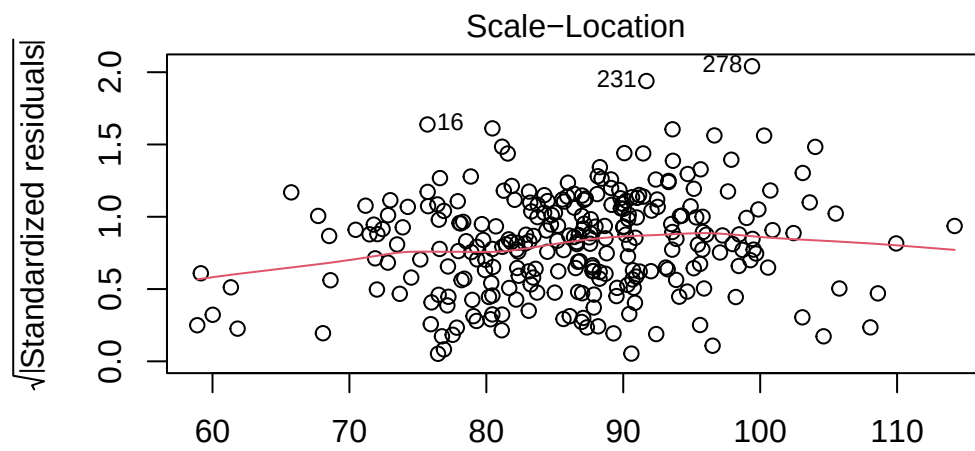
```
n(act_totsleep.min_sd ~ gender.v1 + age.v1 + income.v1.10000 + cg1_edu,
```

```
plot(mod3_totsleep_sd,2)
```



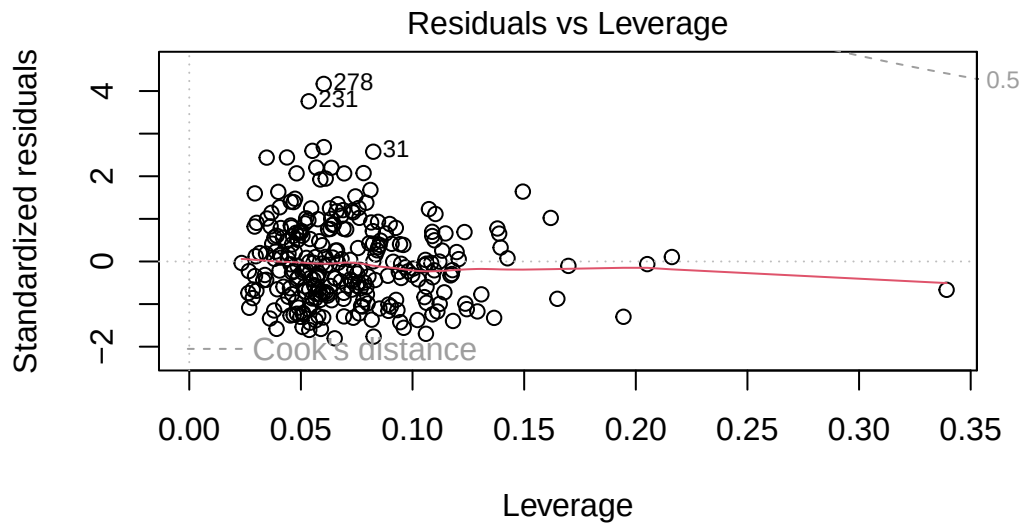
$n(\text{act_totlsleep.min_sd} \sim \text{gender.v1} + \text{age.v1} + \text{income.v1.10000} + \text{cg1_edu})$

```
plot(mod3_totlsleep_sd,3)
```



$n(\text{act_totlsleep.min_sd} \sim \text{gender.v1} + \text{age.v1} + \text{income.v1.10000} + \text{cg1_edu})$

```
plot(mod3_totlsleep_sd,5)
```



```
n(act_totsleep.min_sd ~ gender.v1 + age.v1 + income.v1.10000 + cg1_edu.
```

```
effectsize::standardize_parameters(mod3_totsleep_sd)
```

```
# Standardization method: refit
```

Parameter	Std. Coef.	95% CI
(Intercept)	-0.07	[-0.31, 0.17]
gender v1 [Male]	-0.14	[-0.40, 0.12]
age v1	1.57e-04	[-0.13, 0.13]
income v1 10000	-0.04	[-0.17, 0.09]
cg1_edu_grp [<=BS]	0.17	[-0.11, 0.45]
cg1_edu_grp [>=MS]	0.15	[-0.33, 0.63]
z ED EC nat	0.05	[-0.08, 0.18]
z ED EL nat	0.05	[-0.16, 0.25]
z ED ER nat	-0.09	[-0.46, 0.28]
z ED SP nat	0.19	[-0.02, 0.40]
z HE EP nat	-0.09	[-0.23, 0.05]
z HE HE nat	0.02	[-0.16, 0.20]
z HE HR nat	2.40e-03	[-0.16, 0.16]
z HE SE nat	-0.12	[-0.30, 0.05]
z SE EI nat	-0.06	[-0.49, 0.38]
z SE EO nat	-0.14	[-0.46, 0.17]
z SE ER nat	0.32	[-0.06, 0.70]
z SE HQ nat	0.03	[-0.18, 0.24]
z SE SR nat	-0.31	[-0.60, -0.03]
z SE WL nat	0.13	[-0.11, 0.38]

```
#####
#Model 4: Neighborhood
#####
mod4_totsleep_sd <- lm(act_totsleep.min_sd ~ gender.v1 + age.v1 + income.v1.10000 +
  cg1_edu_grp +
  p.mn.cohesion.mean.v1 +
  p.mn.safety.mean.v1 +
  p.mn.violence.mean.v1, data = dat)
summary(mod4_totsleep_sd)
```

Call:

```
lm(formula = act_totsleep.min_sd ~ gender.v1 + age.v1 + income.v1.10000 +
  cg1_edu_grp + p.mn.cohesion.mean.v1 + p.mn.safety.mean.v1 +
  p.mn.violence.mean.v1, data = dat)
```

Residuals:

Min	1Q	Median	3Q	Max
-60.375	-23.320	-5.227	20.407	152.461

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	83.3057	23.8177	3.498	0.000543 ***
gender.v1Male	-0.2239	4.1131	-0.054	0.956629
age.v1	0.2850	1.2996	0.219	0.826573
income.v1.10000	-0.2632	0.6771	-0.389	0.697805
cg1_edu_grp<=BS	3.9125	4.6125	0.848	0.397008
cg1_edu_grp>=MS	2.5105	7.6442	0.328	0.742835
p.mn.cohesion.mean.v1	-2.4545	3.0535	-0.804	0.422155
p.mn.safety.mean.v1	0.3832	2.8575	0.134	0.893408
p.mn.violence.mean.v1	1.4024	3.1980	0.439	0.661346

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 33.39 on 290 degrees of freedom

(25 observations deleted due to missingness)

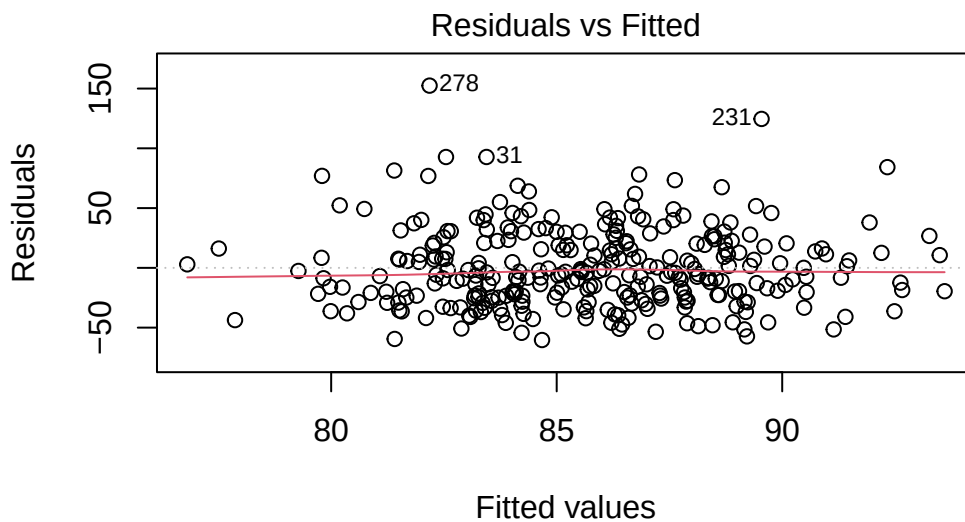
Multiple R-squared: 0.008925, Adjusted R-squared: -0.01841

F-statistic: 0.3265 on 8 and 290 DF, p-value: 0.9556

```
ci <- confint(mod4_totsleep_sd)
ci
```

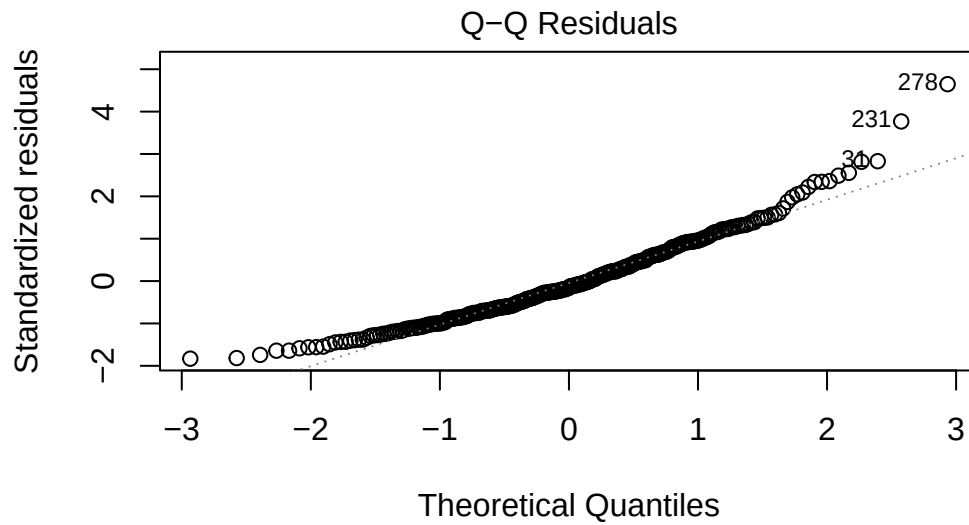
	2.5 %	97.5 %
(Intercept)	36.428258	130.183229
gender.v1Male	-8.319151	7.871390
age.v1	-2.272928	2.842944
income.v1.10000	-1.595937	1.069561
cg1_edu_grp<=BS	-5.165717	12.990621
cg1_edu_grp>=MS	-12.534727	17.555697
p.mn.cohesion.mean.v1	-8.464273	3.555323
p.mn.safety.mean.v1	-5.240899	6.007351
p.mn.violence.mean.v1	-4.891965	7.696696

```
plot(mod4_totsleep_sd,1)
```



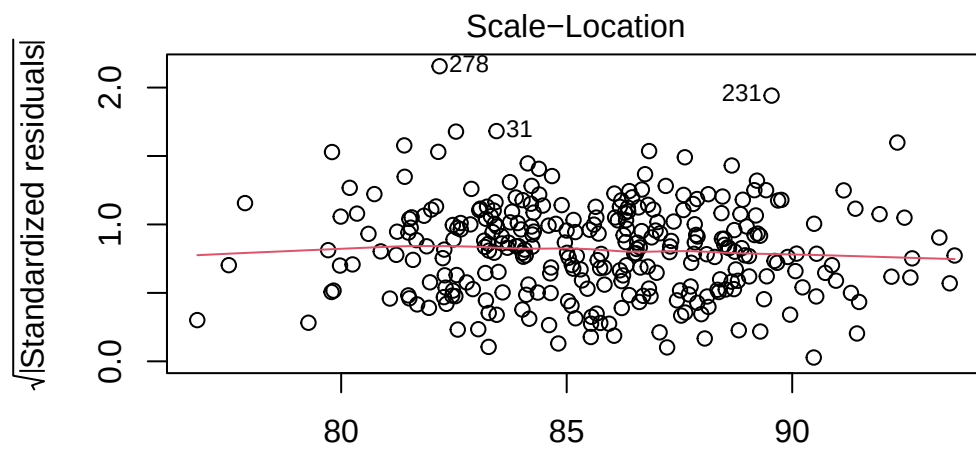
```
n(act_totsleep.min_sd ~ gender.v1 + age.v1 + income.v1.10000 + cg1_edu
```

```
plot(mod4_totsleep_sd,2)
```

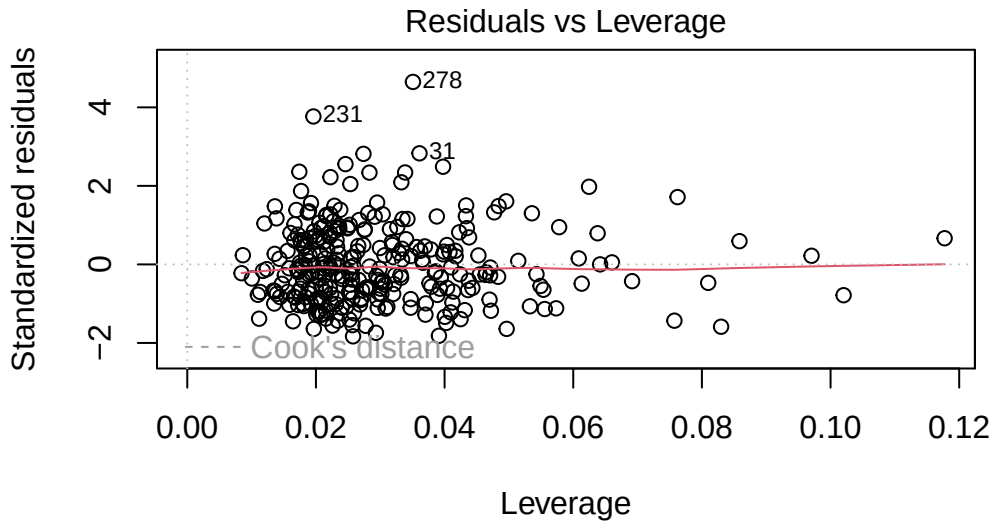
```
n(act_totsleep.min_sd ~ gender.v1 + age.v1 + income.v1.10000 + cg1_edu.
```

```
plot(mod4_totsleep_sd,3)
```



```
n(act_totsleep.min_sd ~ gender.v1 + age.v1 + income.v1.10000 + cg1_edu.
```

```
plot(mod4_totsleep_sd,5)
```



```
n(act_totsleep.min_sd ~ gender.v1 + age.v1 + income.v1.10000 + cg1_edu
```

```
effectsize::standardize_parameters(mod4_totsleep_sd)
```

```
# Standardization method: refit
```

Parameter	Std. Coef.	95% CI
(Intercept)	-0.08	[-0.32, 0.16]
gender v1 [Male]	-6.77e-03	[-0.25, 0.24]
age v1	0.01	[-0.11, 0.13]
income v1 10000	-0.02	[-0.15, 0.10]
cg1 edu grp [<=BS]	0.12	[-0.16, 0.39]
cg1 edu grp [>=MS]	0.08	[-0.38, 0.53]
p mn cohesion mean v1	-0.07	[-0.22, 0.09]
p mn safety mean v1	0.01	[-0.18, 0.21]
p mn violence mean v1	0.04	[-0.12, 0.19]

```
library(lme4)
#####
#act_outbed.time.min_sd
#####
dat <- dat %>% mutate(income.v1.10000 = income.v1/10000)
#####
#Model 1: COI Total
#####
mod1_outbed_sd <- lm(act_outbed.time.min_sd ~ gender.v1 + age.v1 + income.v1.10000 +
```

```
cg1_edu_grp + z_COI_nat, data = dat)
summary(mod1_outbed_sd)
```

Call:

```
lm(formula = act_outbed.time.min_sd ~ gender.v1 + age.v1 + income.v1.10000 +
    cg1_edu_grp + z_COI_nat, data = dat)
```

Residuals:

Min	1Q	Median	3Q	Max
-129.18	-67.58	-37.29	0.69	432.60

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	158.750	72.724	2.183	0.0298 *
gender.v1Male	22.076	14.057	1.570	0.1173
age.v1	-1.671	4.347	-0.384	0.7010
income.v1.10000	-3.260	2.294	-1.421	0.1563
cg1_edu_grp<=BS	1.941	15.219	0.128	0.8986
cg1_edu_grp>=MS	17.257	25.466	0.678	0.4985
z_COI_nat	4.295	9.397	0.457	0.6479

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 115.7 on 306 degrees of freedom

(11 observations deleted due to missingness)

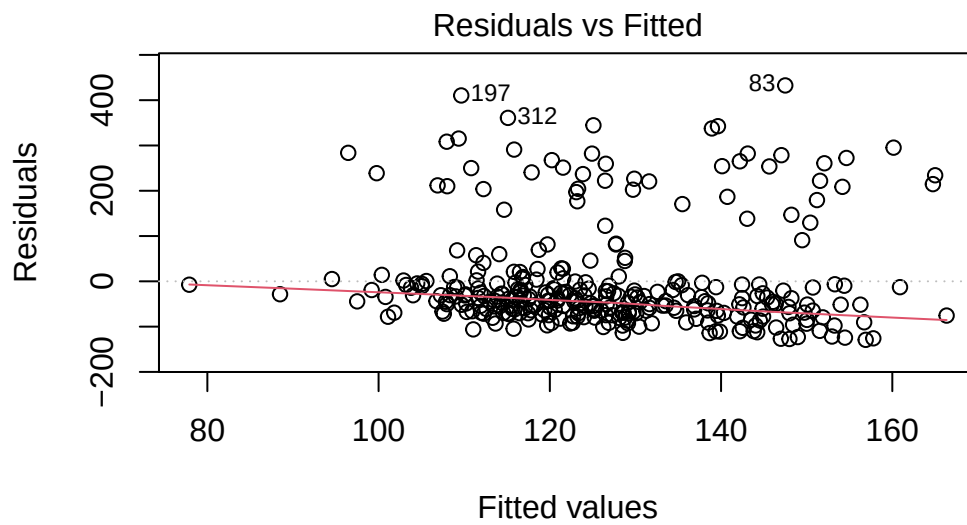
Multiple R-squared: 0.01762, Adjusted R-squared: -0.001644

F-statistic: 0.9146 on 6 and 306 DF, p-value: 0.4845

```
ci <- confint(mod1_outbed_sd)
ci
```

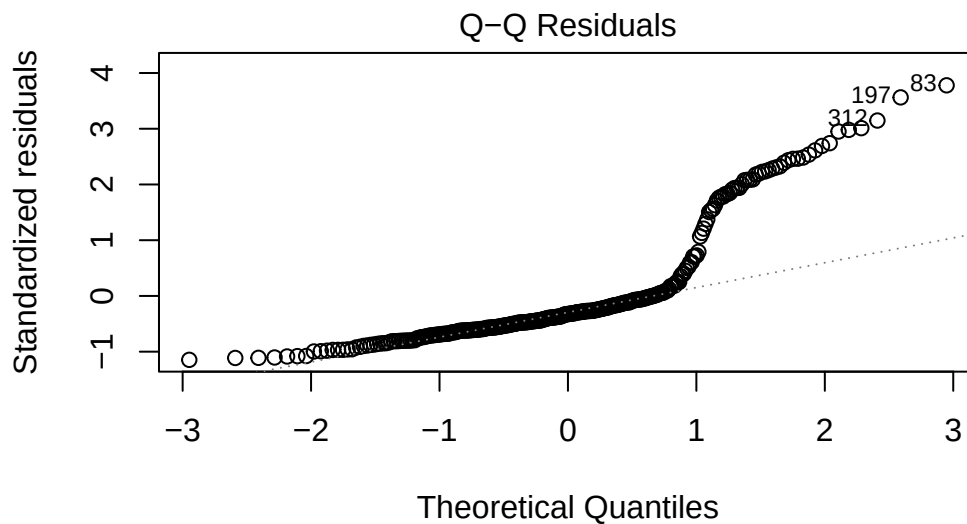
	2.5 %	97.5 %
(Intercept)	15.647776	301.852363
gender.v1Male	-5.585150	49.737402
age.v1	-10.225117	6.883212
income.v1.10000	-7.773684	1.253718
cg1_edu_grp<=BS	-28.005692	31.887546
cg1_edu_grp>=MS	-32.854234	67.367880
z_COI_nat	-14.195796	22.786135

```
plot(mod1_outbed_sd,1)
```



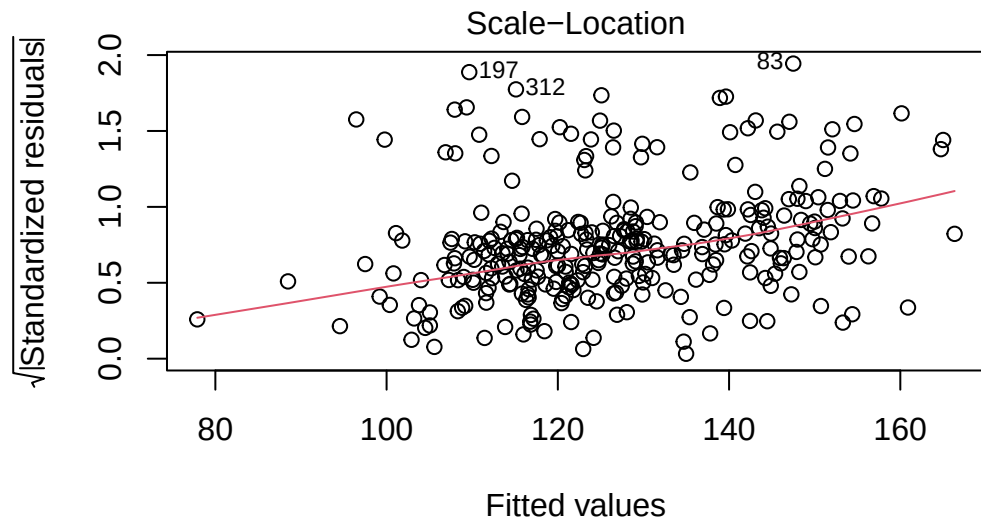
```
l(act_outbed.time.min_sd ~ gender.v1 + age.v1 + income.v1.10000 + cg1_
```

```
plot(mod1_outbed_sd,2)
```



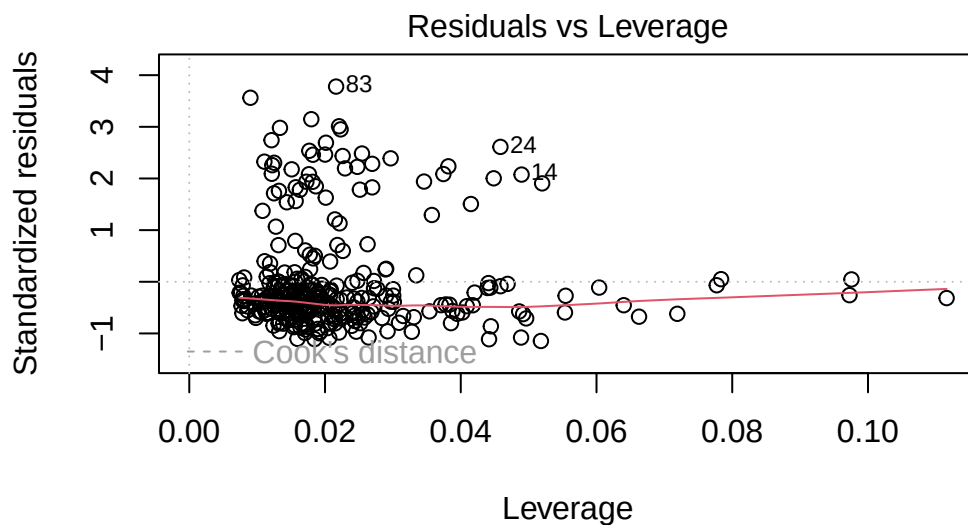
```
l(act_outbed.time.min_sd ~ gender.v1 + age.v1 + income.v1.10000 + cg1_
```

```
plot(mod1_outbed_sd,3)
```



```
l(act_outbed.time.min_sd ~ gender.v1 + age.v1 + income.v1.10000 + cg1_)
```

```
plot(mod1_outbed_sd,5)
```



```
l(act_outbed.time.min_sd ~ gender.v1 + age.v1 + income.v1.10000 + cg1_)
```

```
effectsize::standardize_parameters(mod1_outbed_sd)
```

```
# Standardization method: refit
```

Parameter	Std. Coef.	95% CI

(Intercept)		-0.09		[-0.32, 0.13]
gender v1 [Male]		0.19		[-0.05, 0.43]
age v1		-0.02		[-0.14, 0.09]
income v1 10000		-0.09		[-0.20, 0.03]
cg1_edu_grp [<=BS]		0.02		[-0.24, 0.28]
cg1_edu_grp [>=MS]		0.15		[-0.28, 0.58]
z COI nat		0.03		[-0.09, 0.14]

```
#####
#Model 2: COI Sub-domains
#####
mod2_outbed_sd <- lm(act_outbed.time.min_sd ~ gender.v1 + age.v1 + income.v1.10000 +
cg1_edu_grp + z_ED_nat + z_HE_nat + z_SE_nat, data = dat)
summary(mod2_outbed_sd)
```

Call:

```
lm(formula = act_outbed.time.min_sd ~ gender.v1 + age.v1 + income.v1.10000 +
    cg1_edu_grp + z_ED_nat + z_HE_nat + z_SE_nat, data = dat)
```

Residuals:

Min	1Q	Median	3Q	Max
-136.70	-68.72	-38.59	0.74	427.13

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	161.2303	73.2482	2.201	0.0285 *
gender.v1Male	21.4504	14.1073	1.521	0.1294
age.v1	-1.6633	4.3528	-0.382	0.7026
income.v1.10000	-3.6702	2.3298	-1.575	0.1162
cg1_edu_grp<=BS	0.7835	15.2801	0.051	0.9591
cg1_edu_grp>=MS	16.3116	25.5113	0.639	0.5231
z_ED_nat	-16.4688	16.3967	-1.004	0.3160
z_HE_nat	2.7387	15.9560	0.172	0.8638
z_SE_nat	15.3971	13.8544	1.111	0.2673

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 115.9 on 304 degrees of freedom

(11 observations deleted due to missingness)

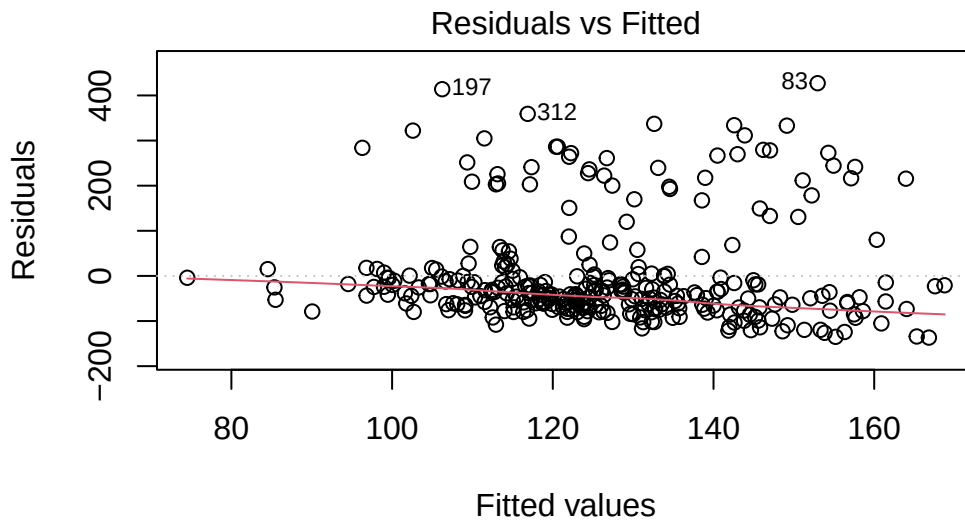
Multiple R-squared: 0.02164, Adjusted R-squared: -0.004104

F-statistic: 0.8406 on 8 and 304 DF, p-value: 0.5675

```
ci <- confint(mod2_outbed_sd)
ci
```

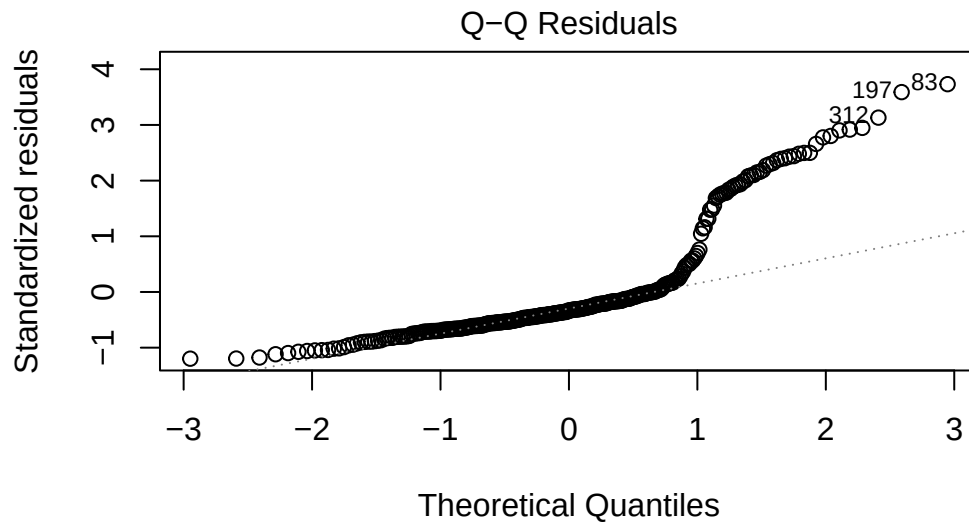
	2.5 %	97.5 %
(Intercept)	17.092672	305.3678447
gender.v1Male	-6.309939	49.2107487
age.v1	-10.228730	6.9020850
income.v1.10000	-8.254727	0.9142308
cg1_edu_grp<=BS	-29.284648	30.8516089
cg1_edu_grp>=MS	-33.889578	66.5126964
z_ED_nat	-48.734147	15.7965477
z_HE_nat	-28.659496	34.1369882
z_SE_nat	-11.865598	42.6597234

```
plot(mod2_outbed_sd,1)
```



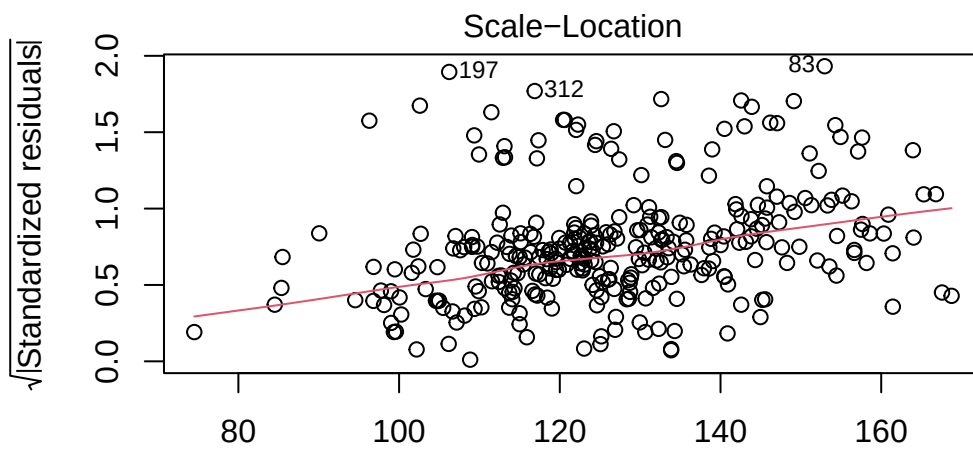
```
lm(act_outbed.time.min_sd ~ gender.v1 + age.v1 + income.v1.10000 + cg1_educat
```

```
plot(mod2_outbed_sd,2)
```



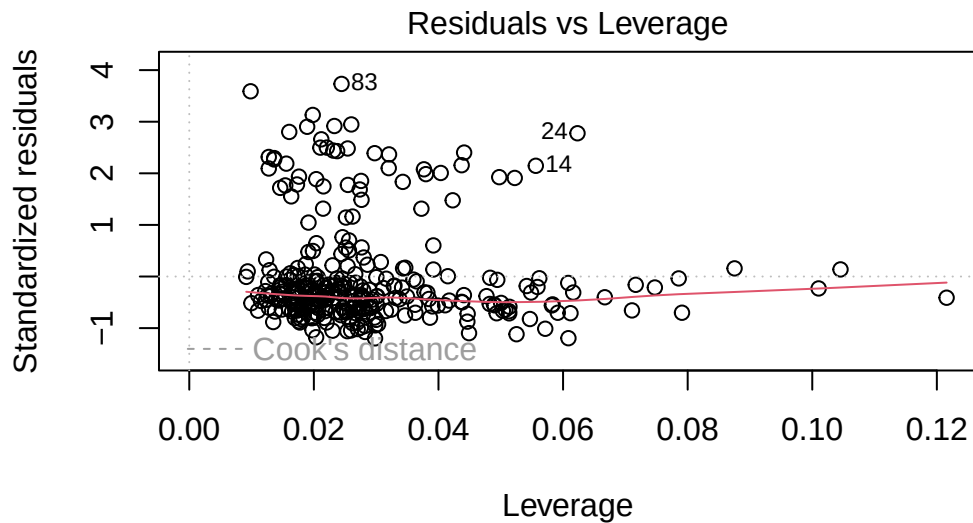
```
1(act_outbed.time.min_sd ~ gender.v1 + age.v1 + income.v1.10000 + cg1_1)
```

```
plot(mod2_outbed_sd,3)
```



```
1(act_outbed.time.min_sd ~ gender.v1 + age.v1 + income.v1.10000 + cg1_1)
```

```
plot(mod2_outbed_sd,5)
```

```
lm(act_outbed.time.min_sd ~ gender.v1 + age.v1 + income.v1.10000 + cg1_educat
```

```
effectsize::standardize_parameters(mod2_outbed_sd)
```

```
# Standardization method: refit
```

Parameter	Std. Coef.	95% CI
(Intercept)	-0.08	[-0.31, 0.14]
gender v1 [Male]	0.19	[-0.05, 0.43]
age v1	-0.02	[-0.14, 0.09]
income v1 10000	-0.10	[-0.22, 0.02]
cg1 edu grp [<=BS]	6.78e-03	[-0.25, 0.27]
cg1 edu grp [>=MS]	0.14	[-0.29, 0.58]
z ED nat	-0.10	[-0.31, 0.10]
z HE nat	0.01	[-0.14, 0.16]
z SE nat	0.12	[-0.09, 0.33]

```
#####
#Model 3: COI Sub-domains
#####
mod3_outbed_sd <- lm(act_outbed.time.min_sd ~ gender.v1 + age.v1 + income.v1.10000 +
  cg1_edu_grp + z_ED_EC_nat +
    z_ED_EL_nat +
    z_ED_ER_nat +
    z_ED_SP_nat +
    z_HE_EP_nat +
```

```

z_HE_HE_nat +
z_HE_HR_nat +
z_HE_SE_nat +
z_SE_EI_nat +
z_SE_EO_nat +
z_SE_ER_nat +
z_SE_HQ_nat +
z_SE_SR_nat +
z_SE_WL_nat , data = dat)
summary(mod3_outbed_sd)

```

Call:

```

lm(formula = act_outbed.time.min_sd ~ gender.v1 + age.v1 + income.v1.10000 +
    cg1_edu_grp + z_ED_EC_nat + z_ED_EL_nat + z_ED_ER_nat + z_ED_SP_nat +
    z_HE_EP_nat + z_HE_HE_nat + z_HE_HR_nat + z_HE_SE_nat + z_SE_EI_nat +
    z_SE_EO_nat + z_SE_ER_nat + z_SE_HQ_nat + z_SE_SR_nat + z_SE_WL_nat,
    data = dat)

```

Residuals:

	Min	1Q	Median	3Q	Max
	-152.60	-69.64	-38.87	10.45	442.90

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	173.4664	84.4451	2.054	0.041 *
gender.v1Male	20.0116	15.8704	1.261	0.208
age.v1	-2.7228	4.8940	-0.556	0.578
income.v1.10000	-3.8675	2.6377	-1.466	0.144
cg1_edu_grp<=BS	2.0896	17.2804	0.121	0.904
cg1_edu_grp>=MS	20.0654	29.2810	0.685	0.494
z_ED_EC_nat	0.9852	8.6450	0.114	0.909
z_ED_EL_nat	3.1323	19.2353	0.163	0.871
z_ED_ER_nat	-23.7461	33.9154	-0.700	0.484
z_ED_SP_nat	3.9687	18.0233	0.220	0.826
z_HE_EP_nat	-14.4578	65.2680	-0.222	0.825
z_HE_HE_nat	23.1996	42.0511	0.552	0.582
z_HE_HR_nat	8.9200	17.4042	0.513	0.609
z_HE_SE_nat	-7.2363	12.3251	-0.587	0.558
z_SE_EI_nat	6.3814	34.0348	0.187	0.851
z_SE_EO_nat	-17.7628	22.0390	-0.806	0.421
z_SE_ER_nat	15.9065	22.9971	0.692	0.490

z_SE_HQ_nat	18.9433	20.6365	0.918	0.360
z_SE_SR_nat	-4.8968	20.4069	-0.240	0.811
z_SE_WL_nat	8.1512	24.0416	0.339	0.735

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 119.9 on 255 degrees of freedom

(49 observations deleted due to missingness)

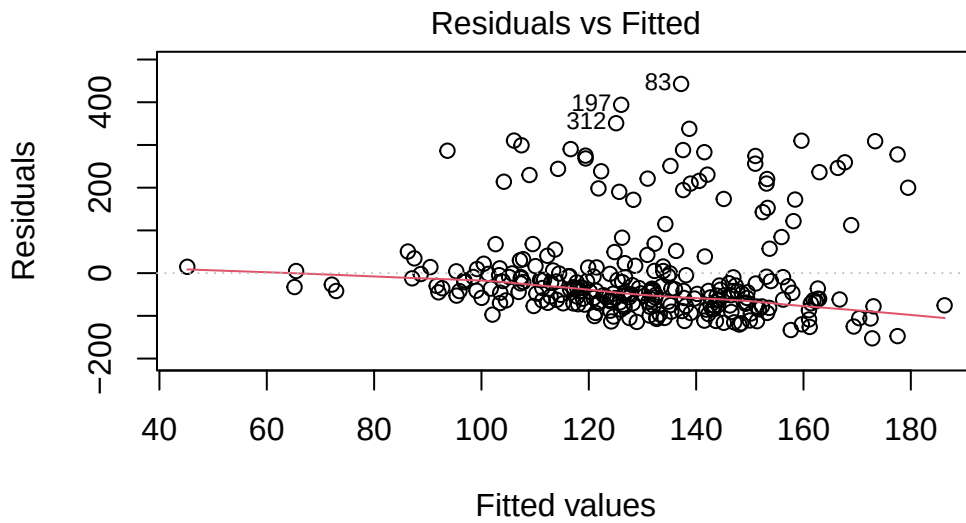
Multiple R-squared: 0.03579, Adjusted R-squared: -0.03606

F-statistic: 0.4981 on 19 and 255 DF, p-value: 0.9621

```
ci <- confint(mod3_outbed_sd)
ci
```

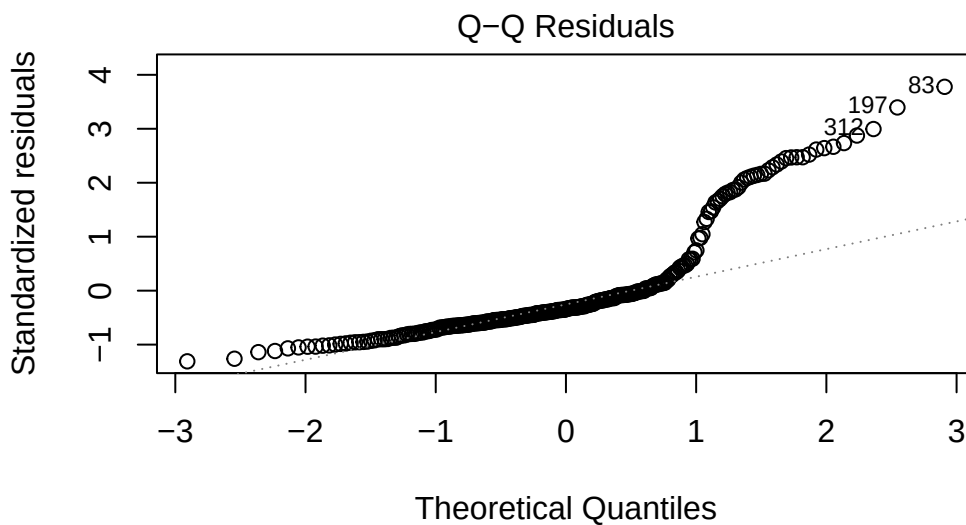
	2.5 %	97.5 %
(Intercept)	7.167714	339.765142
gender.v1Male	-11.242112	51.265345
age.v1	-12.360671	6.914989
income.v1.10000	-9.061994	1.326950
cg1_edu_grp<=BS	-31.940839	36.120001
cg1_edu_grp>=MS	-37.597982	77.728871
z_ED_EC_nat	-16.039457	18.009845
z_ED_EL_nat	-34.747922	41.012491
z_ED_ER_nat	-90.536117	43.043915
z_ED_SP_nat	-31.524886	39.462244
z_HE_EP_nat	-142.990662	114.075132
z_HE_HE_nat	-59.612156	106.011390
z_HE_HR_nat	-25.354258	43.194207
z_HE_SE_nat	-31.508315	17.035747
z_SE_EI_nat	-60.643786	73.406521
z_SE_EO_nat	-61.164487	25.638931
z_SE_ER_nat	-29.382040	61.194981
z_SE_HQ_nat	-21.696434	59.582950
z_SE_SR_nat	-45.084379	35.290804
z_SE_WL_nat	-39.194152	55.496462

```
plot(mod3_outbed_sd,1)
```



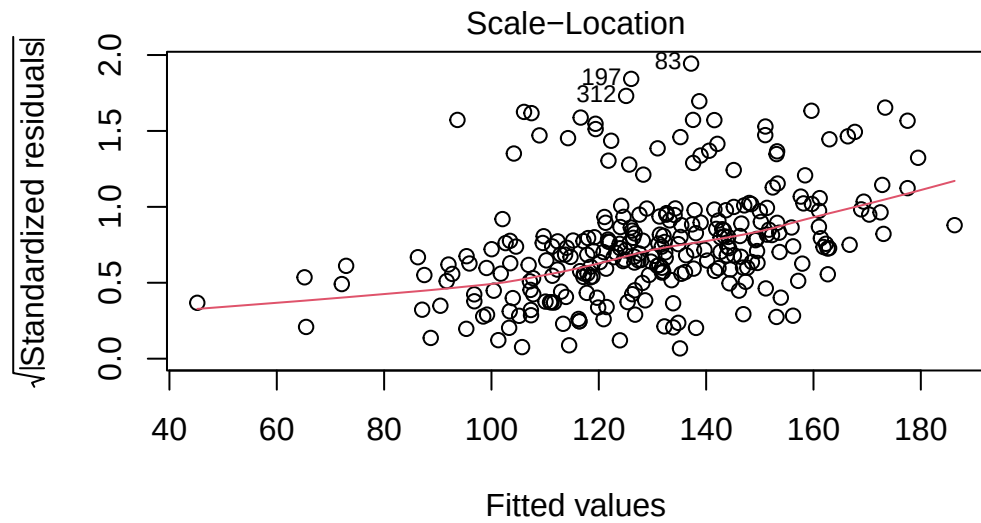
```
l(act_outbed.time.min_sd ~ gender.v1 + age.v1 + income.v1.10000 + cg1_1)
```

```
plot(mod3_outbed_sd,2)
```



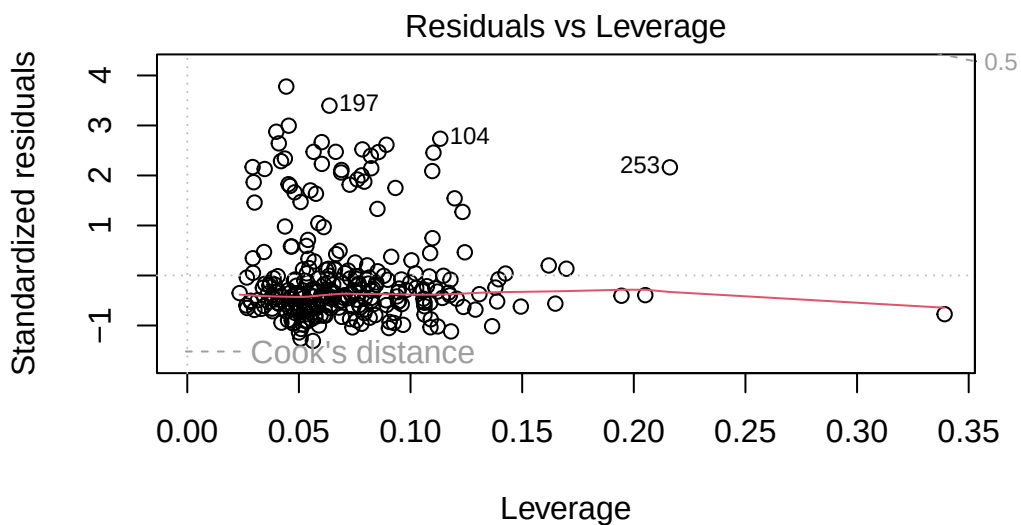
```
l(act_outbed.time.min_sd ~ gender.v1 + age.v1 + income.v1.10000 + cg1_1)
```

```
plot(mod3_outbed_sd,3)
```



```
l(act_outbed.time.min_sd ~ gender.v1 + age.v1 + income.v1.10000 + cg1_1)
```

```
plot(mod3_outbed_sd,5)
```



```
l(act_outbed.time.min_sd ~ gender.v1 + age.v1 + income.v1.10000 + cg1_1)
```

```
effectsize::standardize_parameters(mod3_outbed_sd)
```

```
# Standardization method: refit
```

Parameter	Std. Coef.	95% CI

(Intercept)		-0.09		[-0.33, 0.16]
gender v1 [Male]		0.17		[-0.10, 0.44]
age v1		-0.04		[-0.16, 0.09]
income v1 10000		-0.10		[-0.23, 0.03]
cg1 edu grp [<=BS]		0.02		[-0.27, 0.31]
cg1 edu grp [>=MS]		0.17		[-0.32, 0.66]
z ED EC nat		7.70e-03		[-0.13, 0.14]
z ED EL nat		0.02		[-0.19, 0.23]
z ED ER nat		-0.13		[-0.51, 0.24]
z ED SP nat		0.02		[-0.19, 0.24]
z HE EP nat		-0.02		[-0.16, 0.13]
z HE HE nat		0.05		[-0.13, 0.23]
z HE HR nat		0.04		[-0.12, 0.21]
z HE SE nat		-0.05		[-0.23, 0.13]
z SE EI nat		0.04		[-0.40, 0.48]
z SE EO nat		-0.13		[-0.45, 0.19]
z SE ER nat		0.14		[-0.25, 0.53]
z SE HQ nat		0.10		[-0.11, 0.31]
z SE SR nat		-0.04		[-0.33, 0.26]
z SE WL nat		0.04		[-0.21, 0.29]

```
#####
#Model 4: Neighborhood
#####
mod4_outbed_sd <- lm(act_outbed.time.min_sd ~ gender.v1 + age.v1 + income.v1.10000 +
  cg1_edu_grp +
  p.mn.cohesion.mean.v1 +
  p.mn.safety.mean.v1 +
  p.mn.violence.mean.v1, data = dat)
summary(mod4_outbed_sd)
```

Call:

```
lm(formula = act_outbed.time.min_sd ~ gender.v1 + age.v1 + income.v1.10000 +
  cg1_edu_grp + p.mn.cohesion.mean.v1 + p.mn.safety.mean.v1 +
  p.mn.violence.mean.v1, data = dat)
```

Residuals:

Min	1Q	Median	3Q	Max
-136.65	-67.72	-39.94	2.75	425.95

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	156.7347	83.7323	1.872	0.0622 .
gender.v1Male	23.7163	14.4597	1.640	0.1021
age.v1	0.4634	4.5690	0.101	0.9193
income.v1.10000	-3.0945	2.3805	-1.300	0.1947
cg1_edu_grp<=BS	-2.7084	16.2154	-0.167	0.8675
cg1_edu_grp>=MS	15.9525	26.8737	0.594	0.5532
p.mn.cohesion.mean.v1	-5.4737	10.7347	-0.510	0.6105
p.mn.safety.mean.v1	0.9619	10.0458	0.096	0.9238
p.mn.violence.mean.v1	-8.0441	11.2429	-0.715	0.4749

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 117.4 on 290 degrees of freedom

(25 observations deleted due to missingness)

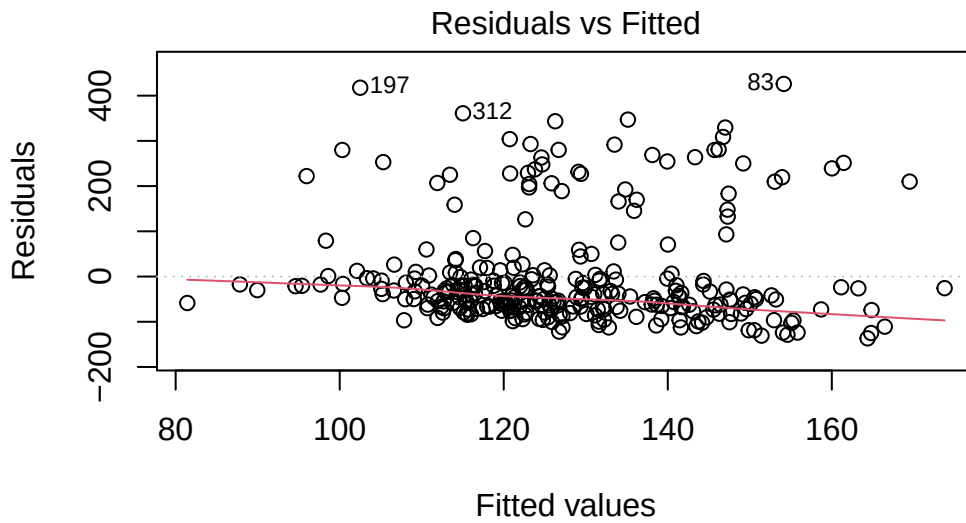
Multiple R-squared: 0.01855, Adjusted R-squared: -0.008527

F-statistic: 0.6851 on 8 and 290 DF, p-value: 0.7047

```
ci <- confint(mod4_outbed_sd)
ci
```

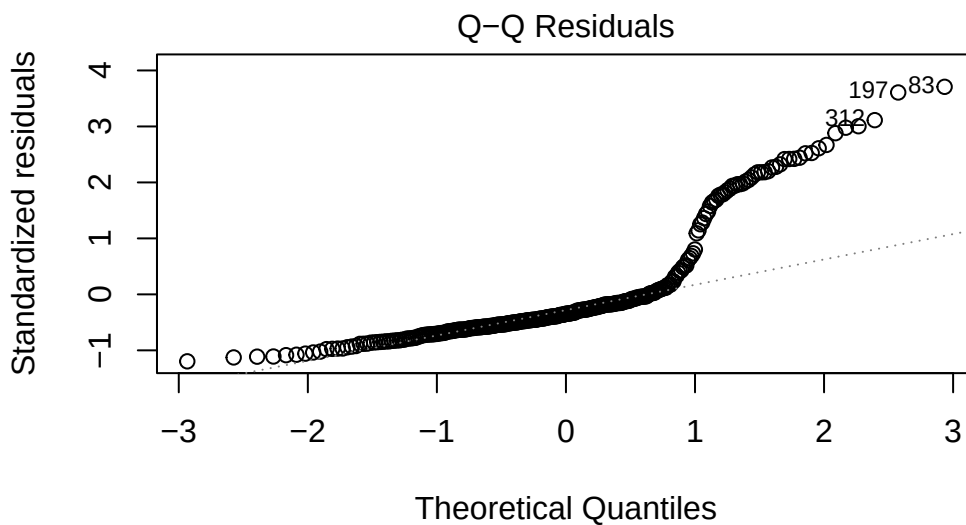
	2.5 %	97.5 %
(Intercept)	-8.065361	321.534787
gender.v1Male	-4.743026	52.175606
age.v1	-8.529105	9.455992
income.v1.10000	-7.779885	1.590802
cg1_edu_grp<=BS	-34.623149	29.206340
cg1_edu_grp>=MS	-36.939658	68.844690
p.mn.cohesion.mean.v1	-26.601467	15.654004
p.mn.safety.mean.v1	-18.809967	20.733802
p.mn.violence.mean.v1	-30.172155	14.083891

```
plot(mod4_outbed_sd,1)
```



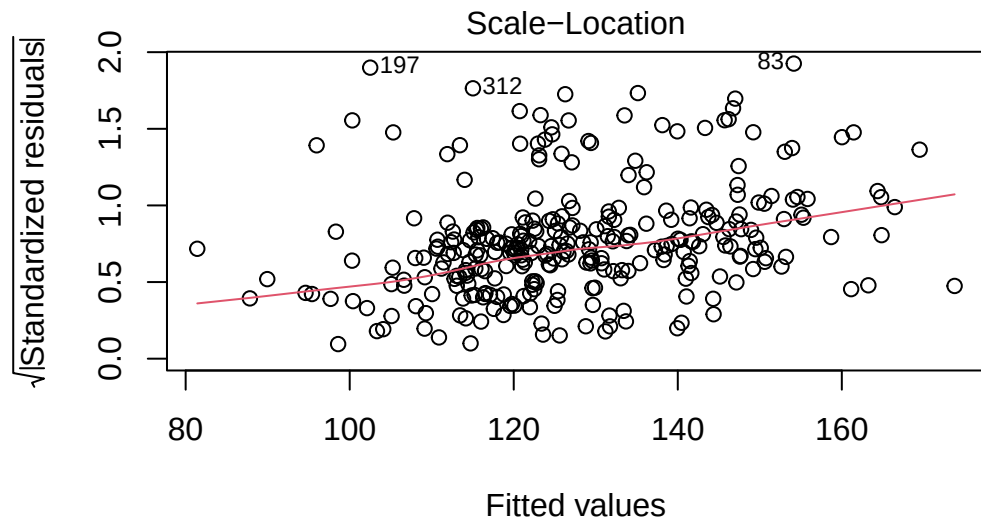
```
l(act_outbed.time.min_sd ~ gender.v1 + age.v1 + income.v1.10000 + cg1_1)
```

```
plot(mod4_outbed_sd,2)
```



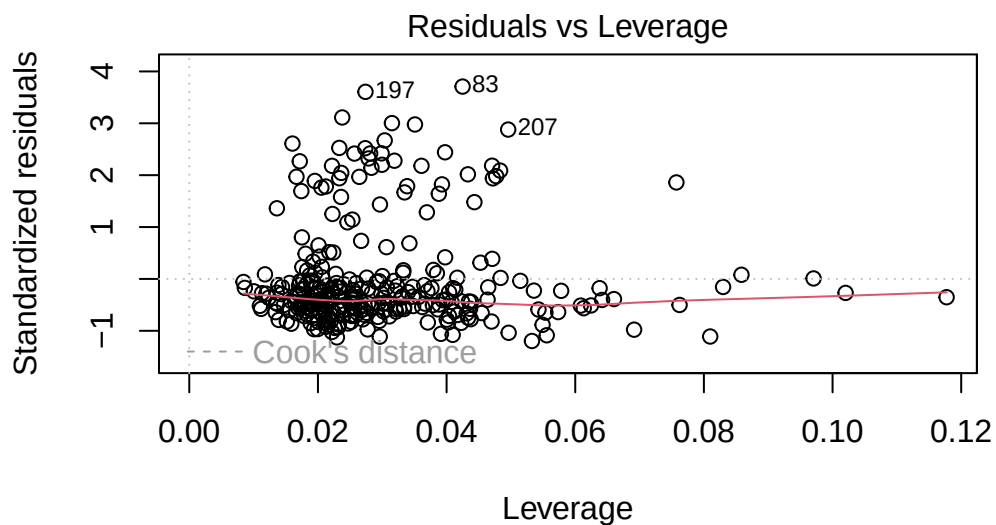
```
l(act_outbed.time.min_sd ~ gender.v1 + age.v1 + income.v1.10000 + cg1_1)
```

```
plot(mod4_outbed_sd,3)
```

```
l(act_outbed.time.min_sd ~ gender.v1 + age.v1 + income.v1.10000 + cg1_1)
```

```
plot(mod4_outbed_sd,5)
```



```
l(act_outbed.time.min_sd ~ gender.v1 + age.v1 + income.v1.10000 + cg1_1)
```

```
effectsize::standardize_parameters(mod4_outbed_sd)
```

```
# Standardization method: refit
```

Parameter	Std. Coef.	95% CI

(Intercept)		-0.07		[-0.31, 0.17]
gender v1 [Male]		0.20		[-0.04, 0.45]
age v1		6.08e-03		[-0.11, 0.12]
income v1 10000		-0.08		[-0.20, 0.04]
cg1_edu_grp [<=BS]		-0.02		[-0.30, 0.25]
cg1_edu_grp [>=MS]		0.14		[-0.32, 0.59]
p_mn_cohesion_mean_v1		-0.04		[-0.20, 0.12]
p_mn_safety_mean_v1		9.44e-03		[-0.18, 0.20]
p_mn_violence_mean_v1		-0.06		[-0.22, 0.10]

```
library(lme4)
#####
#act_onset.time.min_sd
#####
dat <- dat %>% mutate(income.v1.10000 = income.v1/10000)
#####
#Model 1: COI Total
#####
mod1_onset_sd <- lm(act_onset.time.min_sd ~ gender.v1 + age.v1 + income.v1.10000 +
cg1_edu_grp + z_COI_nat, data = dat)
summary(mod1_outbed_sd)
```

Call:

```
lm(formula = act_outbed.time.min_sd ~ gender.v1 + age.v1 + income.v1.10000 +
    cg1_edu_grp + z_COI_nat, data = dat)
```

Residuals:

Min	1Q	Median	3Q	Max
-129.18	-67.58	-37.29	0.69	432.60

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	158.750	72.724	2.183	0.0298 *
gender.v1Male	22.076	14.057	1.570	0.1173
age.v1	-1.671	4.347	-0.384	0.7010
income.v1.10000	-3.260	2.294	-1.421	0.1563
cg1_edu_grp<=BS	1.941	15.219	0.128	0.8986
cg1_edu_grp>=MS	17.257	25.466	0.678	0.4985
z_COI_nat	4.295	9.397	0.457	0.6479

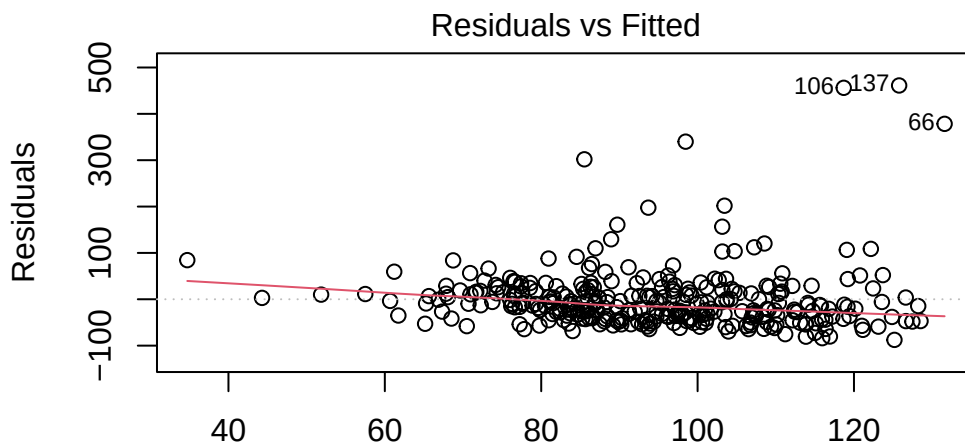
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 115.7 on 306 degrees of freedom
 (11 observations deleted due to missingness)
 Multiple R-squared: 0.01762, Adjusted R-squared: -0.001644
 F-statistic: 0.9146 on 6 and 306 DF, p-value: 0.4845

```
ci <- confint(mod1_onset_sd)
ci
```

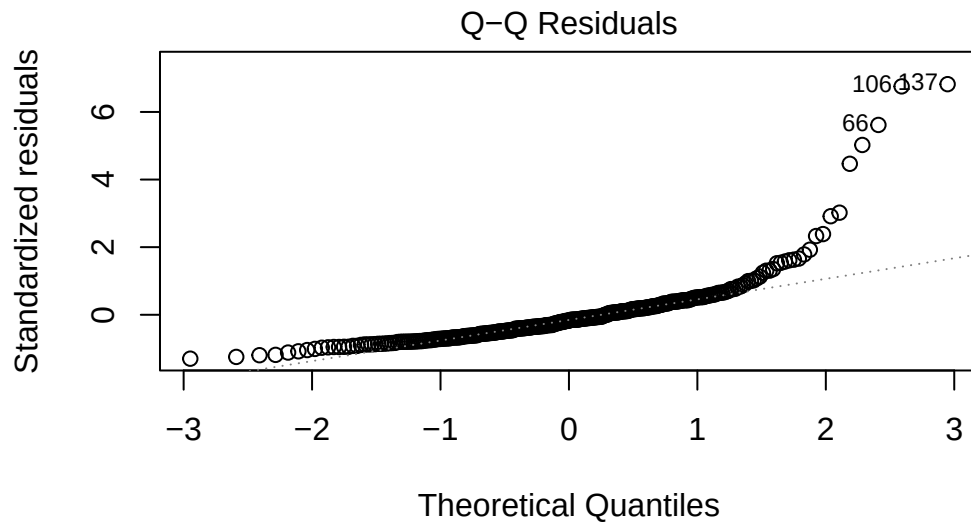
	2.5 %	97.5 %
(Intercept)	-0.8030890	167.7225032
gender.v1Male	6.3882418	38.9637708
age.v1	-4.9801678	5.0937146
income.v1.10000	-5.6414117	-0.3258149
cg1_edu_grp<=BS	-0.7533771	34.5135053
cg1_edu_grp>=MS	-16.3325904	42.6811081
z_COI_nat	-17.1857558	4.5902815

```
plot(mod1_onset_sd,1)
```



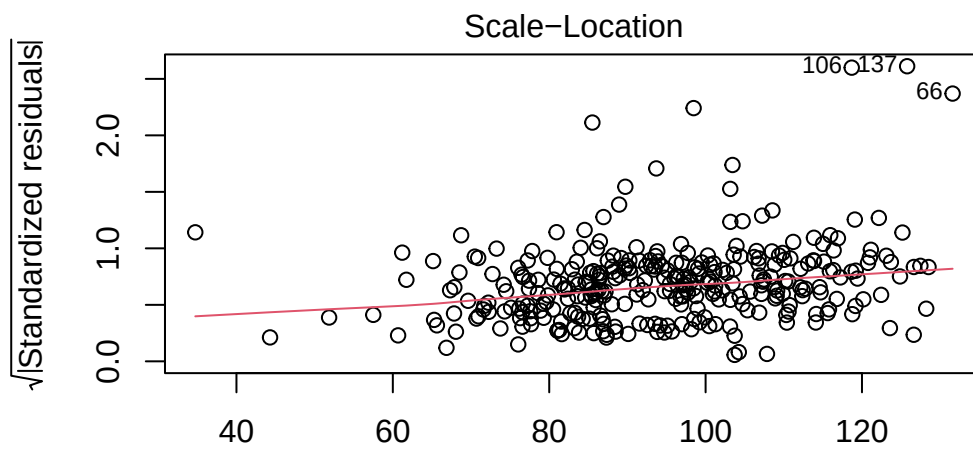
Fitted values
 $\eta(\text{act_onset.time.min_sd} \sim \text{gender.v1} + \text{age.v1} + \text{income.v1.10000} + \text{cg1_edu_grp})$

```
plot(mod1_onset_sd,2)
```



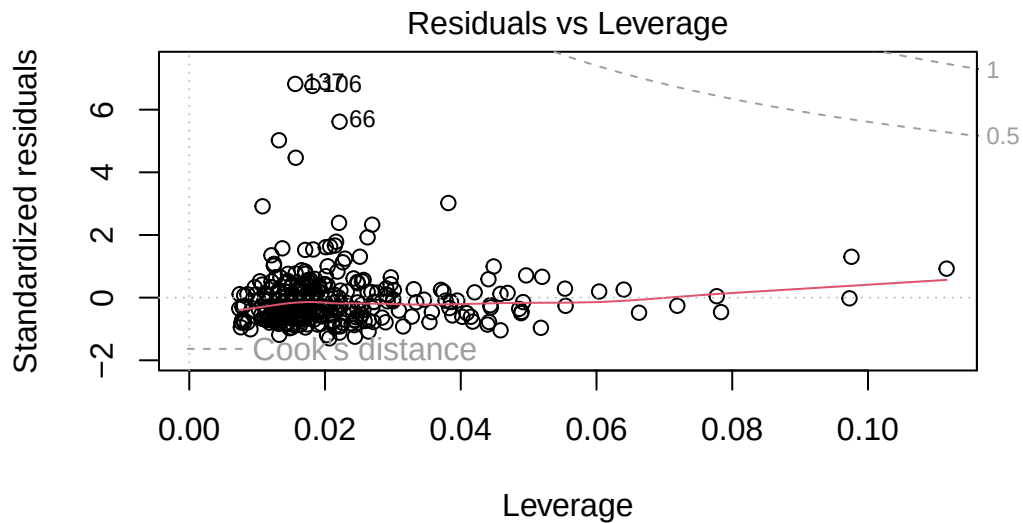
1(act_onset.time.min_sd ~ gender.v1 + age.v1 + income.v1.10000 + cg1_er

```
plot(mod1_onset_sd,3)
```



1(act_onset.time.min_sd ~ gender.v1 + age.v1 + income.v1.10000 + cg1_er

```
plot(mod1_onset_sd,5)
```



```
1(act_onset.time.min_sd ~ gender.v1 + age.v1 + income.v1.10000 + cg1_e
```

```
effectsize::standardize_parameters(mod1_onset_sd)
```

```
# Standardization method: refit
```

Parameter	Std. Coef.	95% CI
(Intercept)	-0.28	[-0.50, -0.06]
gender v1 [Male]	0.33	[0.09, 0.56]
age v1	1.28e-03	[-0.11, 0.11]
income v1 10000	-0.13	[-0.25, -0.01]
cg1 edu grp [<=BS]	0.24	[-0.01, 0.50]
cg1 edu grp [>=MS]	0.19	[-0.24, 0.62]
z COI nat	-0.07	[-0.18, 0.05]

```
#####
#Model 2: COI Sub-domains
#####
mod2_onset_sd <- lm(act_onset.time.min_sd ~ gender.v1 + age.v1 + income.v1.10000 +
cg1_edu_grp + z_ED_nat + z_HE_nat + z_SE_nat, data = dat)
summary(mod2_onset_sd)
```

Call:

```
lm(formula = act_onset.time.min_sd ~ gender.v1 + age.v1 + income.v1.10000 +
    cg1_edu_grp + z_ED_nat + z_HE_nat + z_SE_nat, data = dat)
```

Residuals:

Min	1Q	Median	3Q	Max
-103.21	-36.01	-11.68	19.09	445.98

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	88.22675	42.80803	2.061	0.04015 *
gender.v1Male	22.10090	8.24466	2.681	0.00775 **
age.v1	0.05638	2.54388	0.022	0.98233
income.v1.10000	-3.53226	1.36156	-2.594	0.00994 **
cg1_edu_grp<=BS	15.57836	8.93006	1.744	0.08209 .
cg1_edu_grp>=MS	12.10198	14.90945	0.812	0.41760
z_ED_nat	-23.41259	9.58262	-2.443	0.01513 *
z_HE_nat	-2.44926	9.32510	-0.263	0.79300
z_SE_nat	14.21305	8.09685	1.755	0.08020 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 67.72 on 304 degrees of freedom

(11 observations deleted due to missingness)

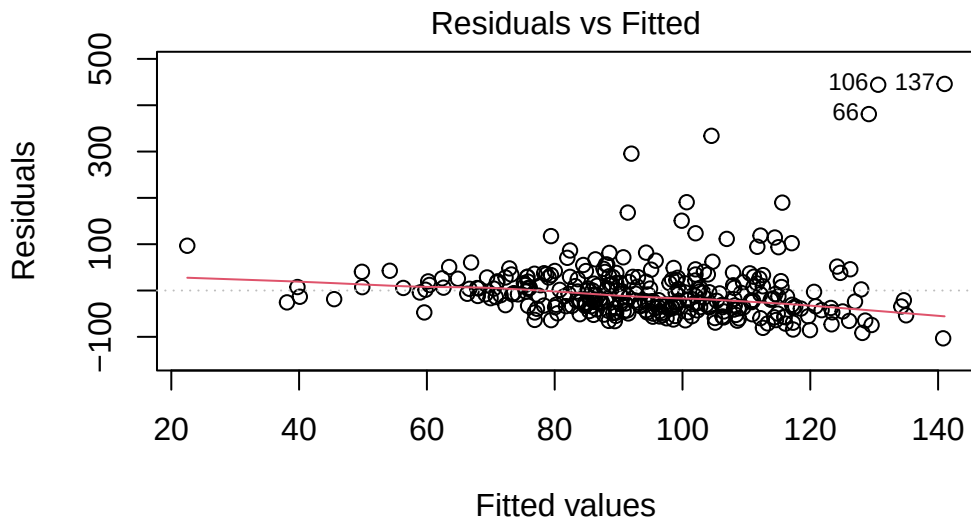
Multiple R-squared: 0.07234, Adjusted R-squared: 0.04793

F-statistic: 2.963 on 8 and 304 DF, p-value: 0.003299

```
ci <- confint(mod2_onset_sd)
ci
```

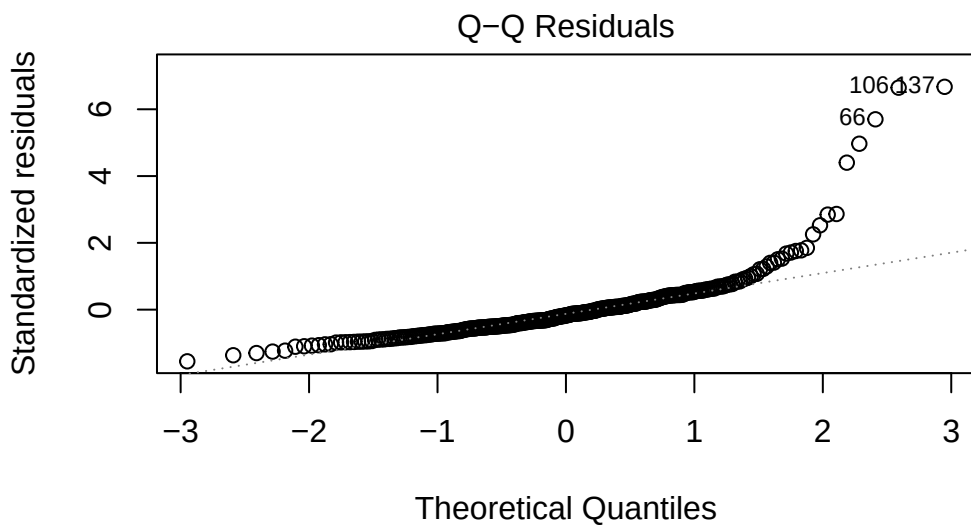
	2.5 %	97.5 %
(Intercept)	3.989182	172.464309
gender.v1Male	5.877069	38.324729
age.v1	-4.949452	5.062220
income.v1.10000	-6.211543	-0.852978
cg1_edu_grp<=BS	-1.994197	33.150916
cg1_edu_grp>=MS	-17.236799	41.440768
z_ED_nat	-42.269255	-4.555924
z_HE_nat	-20.799166	15.900649
z_SE_nat	-1.719924	30.146020

```
plot(mod2_onset_sd,1)
```



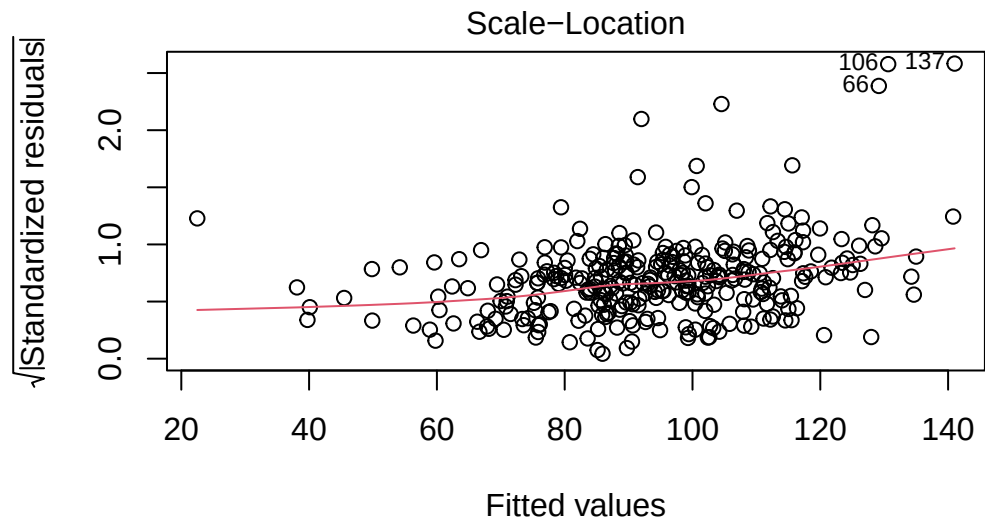
$\text{act_onset.time.min_sd} \sim \text{gender.v1} + \text{age.v1} + \text{income.v1.10000} + \text{cg1_er}$

```
plot(mod2_onset_sd, 2)
```



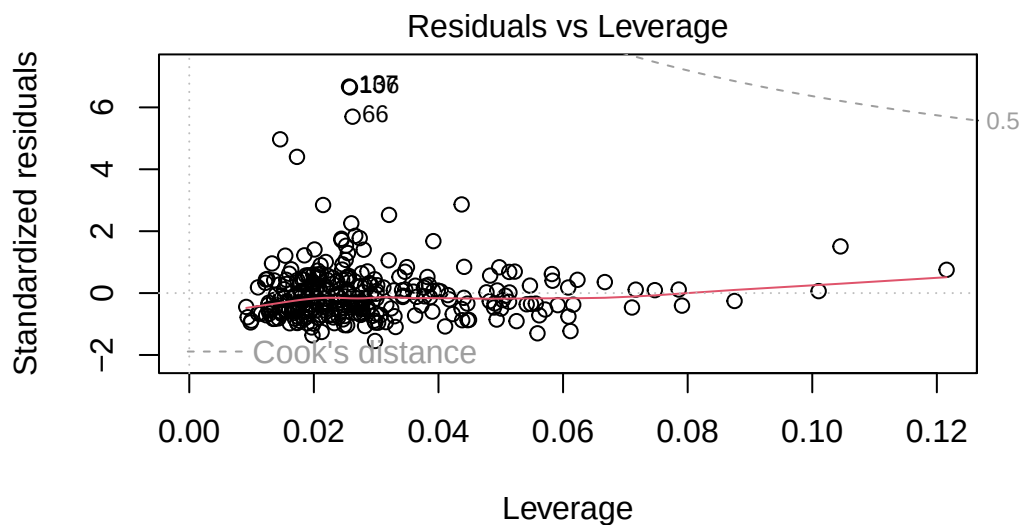
$\text{act_onset.time.min_sd} \sim \text{gender.v1} + \text{age.v1} + \text{income.v1.10000} + \text{cg1_er}$

```
plot(mod2_onset_sd, 3)
```



```
1(act_onset.time.min_sd ~ gender.v1 + age.v1 + income.v1.10000 + cg1_er
```

```
plot(mod2_onset_sd,5)
```



```
1(act_onset.time.min_sd ~ gender.v1 + age.v1 + income.v1.10000 + cg1_er
```

```
effectsize::standardize_parameters(mod2_onset_sd)
```

```
# Standardization method: refit
```

Parameter	Std. Coef.	95% CI

(Intercept)		-0.27		[-0.49, -0.05]
gender v1 [Male]		0.32		[0.08, 0.55]
age v1		1.27e-03		[-0.11, 0.11]
income v1 10000		-0.15		[-0.27, -0.04]
cg1 edu grp [<=BS]		0.22		[-0.03, 0.48]
cg1 edu grp [>=MS]		0.17		[-0.25, 0.60]
z ED nat		-0.25		[-0.44, -0.05]
z HE nat		-0.02		[-0.17, 0.13]
z SE nat		0.18		[-0.02, 0.38]

```
#####
#Model 3: COI Sub-domains
#####
mod3_onset_sd <- lm(act_onset.time.min_sd ~ gender.v1 + age.v1 + income.v1.10000 +
cg1_edu_grp + z_ED_EC_nat +
  z_ED_EL_nat +
  z_ED_ER_nat +
  z_ED_SP_nat +
  z_HE_EP_nat +
  z_HE_HE_nat +
  z_HE_HR_nat +
  z_HE_SE_nat +
  z_SE_EI_nat +
  z_SE_EO_nat +
  z_SE_ER_nat +
  z_SE_HQ_nat +
  z_SE_SR_nat +
  z_SE_WL_nat , data = dat)
summary(mod3_onset_sd)
```

Call:

```
lm(formula = act_onset.time.min_sd ~ gender.v1 + age.v1 + income.v1.10000 +
  cg1_edu_grp + z_ED_EC_nat + z_ED_EL_nat + z_ED_ER_nat + z_ED_SP_nat +
  z_HE_EP_nat + z_HE_HE_nat + z_HE_HR_nat + z_HE_SE_nat + z_SE_EI_nat +
  z_SE_EO_nat + z_SE_ER_nat + z_SE_HQ_nat + z_SE_SR_nat + z_SE_WL_nat,
  data = dat)
```

Residuals:

Min	1Q	Median	3Q	Max
-119.45	-35.34	-11.27	16.85	427.40

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	96.3696	48.1894	2.000	0.04658	*
gender.v1Male	22.9575	9.0566	2.535	0.01185	*
age.v1	-0.6008	2.7928	-0.215	0.82985	
income.v1.10000	-4.3647	1.5052	-2.900	0.00406	**
cg1_edu_grp<=BS	15.7409	9.8612	1.596	0.11167	
cg1_edu_grp>=MS	4.8645	16.7095	0.291	0.77120	
z_ED_EC_nat	-3.7437	4.9333	-0.759	0.44864	
z_ED_EL_nat	14.5633	10.9768	1.327	0.18578	
z_ED_ER_nat	-29.0613	19.3541	-1.502	0.13445	
z_ED_SP_nat	-18.5397	10.2852	-1.803	0.07264	.
z_HE_EP_nat	55.3025	37.2458	1.485	0.13883	
z_HE_HE_nat	0.3283	23.9969	0.014	0.98909	
z_HE_HR_nat	-10.1081	9.9319	-1.018	0.30976	
z_HE_SE_nat	-8.3380	7.0335	-1.185	0.23693	
z_SE_EI_nat	7.4013	19.4223	0.381	0.70347	
z_SE_EO_nat	11.0881	12.5768	0.882	0.37881	
z_SE_ER_nat	-3.8547	13.1235	-0.294	0.76921	
z_SE_HQ_nat	17.1355	11.7764	1.455	0.14688	
z_SE_SR_nat	-9.2246	11.6454	-0.792	0.42903	
z_SE_WL_nat	20.1013	13.7195	1.465	0.14411	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 68.44 on 255 degrees of freedom

(49 observations deleted due to missingness)

Multiple R-squared: 0.1168, Adjusted R-squared: 0.05105

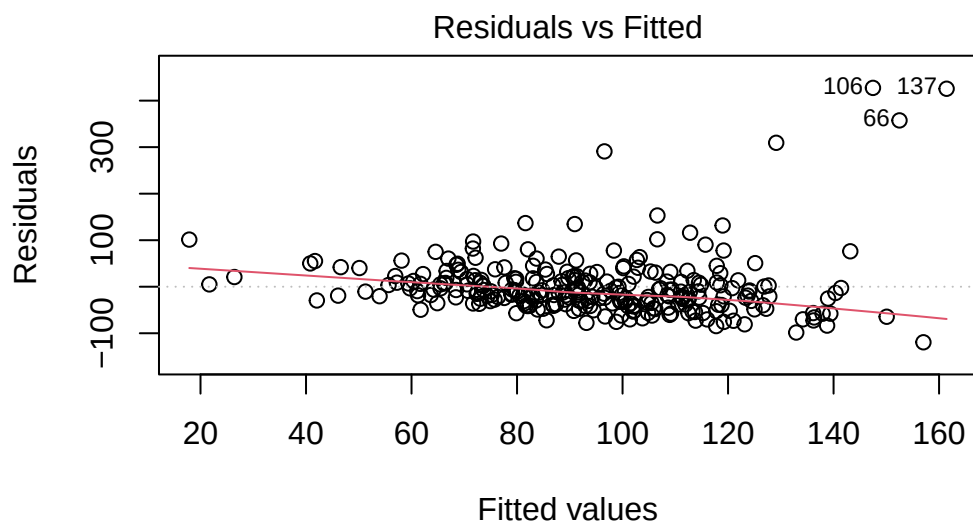
F-statistic: 1.776 on 19 and 255 DF, p-value: 0.02586

```
ci <- confint(mod3_onset_sd)
ci
```

	2.5 %	97.5 %
(Intercept)	1.469682	191.269424
gender.v1Male	5.122301	40.792749
age.v1	-6.100696	4.899134
income.v1.10000	-7.329002	-1.400457
cg1_edu_grp<=BS	-3.678915	35.160621
cg1_edu_grp>=MS	-28.041680	37.770636
z_ED_EC_nat	-13.458925	5.971618
z_ED_EL_nat	-7.053362	36.180005

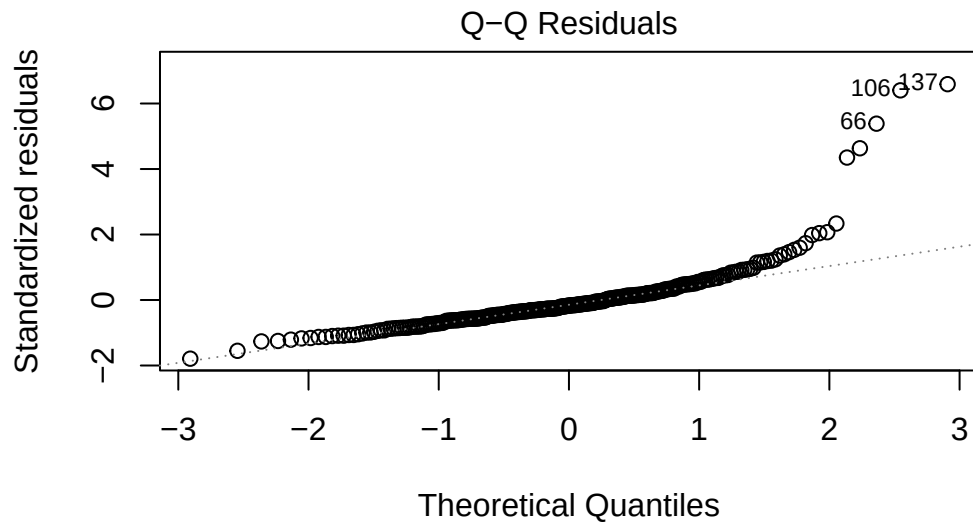
z_ED_ER_nat	-67.175621	9.053038
z_ED_SP_nat	-38.794473	1.714977
z_HE_EP_nat	-18.045918	128.651011
z_HE_HE_nat	-46.928965	47.585615
z_HE_HR_nat	-29.666984	9.450820
z_HE_SE_nat	-22.189082	5.513028
z_SE_EI_nat	-30.847170	45.649854
z_SE_EO_nat	-13.679491	35.855667
z_SE_ER_nat	-29.698964	21.989635
z_SE_HQ_nat	-6.055907	40.326911
z_SE_SR_nat	-32.157975	13.708853
z_SE_WL_nat	-6.916775	47.119283

```
plot(mod3_onset_sd,1)
```

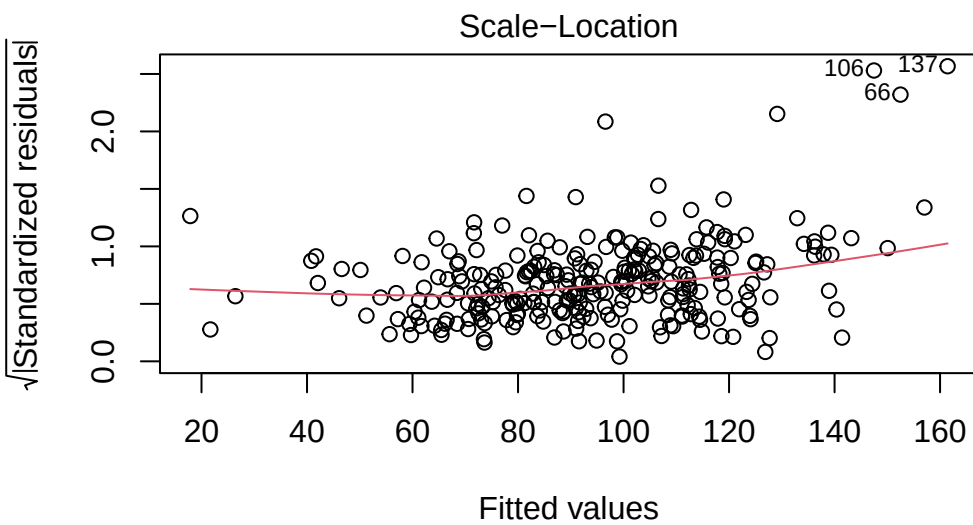


```
l(act_onset.time.min_sd ~ gender.v1 + age.v1 + income.v1.10000 + cg1_e
```

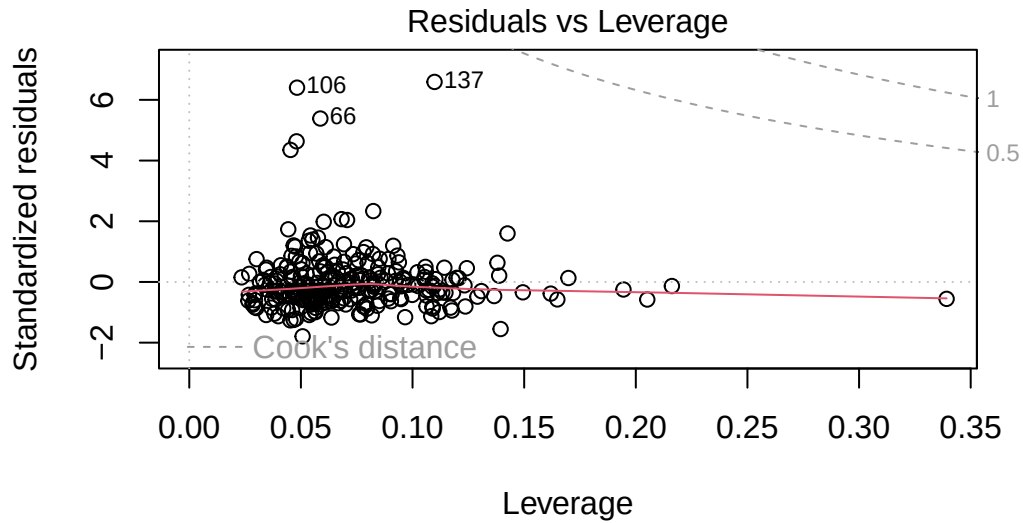
```
plot(mod3_onset_sd,2)
```



```
plot(mod3_onset_sd,3)
```



```
plot(mod3_onset_sd,5)
```



```
1(act_onset.time.min_sd ~ gender.v1 + age.v1 + income.v1.10000 + cg1_e
```

```
effectsize::standardize_parameters(mod3_onset_sd)
```

```
# Standardization method: refit
```

Parameter	Std. Coef.	95% CI
(Intercept)	-0.26	[-0.50, -0.02]
gender v1 [Male]	0.33	[0.07, 0.58]
age v1	-0.01	[-0.14, 0.11]
income v1 10000	-0.18	[-0.31, -0.06]
cg1 edu grp [<=BS]	0.22	[-0.05, 0.50]
cg1 edu grp [>=MS]	0.07	[-0.40, 0.54]
z ED EC nat	-0.05	[-0.18, 0.08]
z ED EL nat	0.14	[-0.07, 0.34]
z ED ER nat	-0.28	[-0.64, 0.09]
z ED SP nat	-0.19	[-0.39, 0.02]
z HE EP nat	0.10	[-0.03, 0.24]
z HE HE nat	1.20e-03	[-0.17, 0.17]
z HE HR nat	-0.08	[-0.24, 0.08]
z HE SE nat	-0.10	[-0.28, 0.07]
z SE EI nat	0.08	[-0.34, 0.50]
z SE EO nat	0.14	[-0.17, 0.44]
z SE ER nat	-0.06	[-0.43, 0.32]
z SE HQ nat	0.15	[-0.05, 0.36]
z SE SR nat	-0.11	[-0.39, 0.17]
z SE WL nat	0.18	[-0.06, 0.42]

```
#####
#Model 4: Neighborhood
#####
mod4_onset_sd <- lm(act_onset.time.min_sd ~ gender.v1 + age.v1 + income.v1.10000 +
cg1_edu_grp +
  p.mn.cohesion.mean.v1 +
  p.mn.safety.mean.v1 +
  p.mn.violence.mean.v1, data = dat)
summary(mod4_onset_sd)
```

Call:

```
lm(formula = act_onset.time.min_sd ~ gender.v1 + age.v1 + income.v1.10000 +
  cg1_edu_grp + p.mn.cohesion.mean.v1 + p.mn.safety.mean.v1 +
  p.mn.violence.mean.v1, data = dat)
```

Residuals:

Min	1Q	Median	3Q	Max
-87.37	-39.29	-12.70	17.57	460.29

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	79.9058	49.6900	1.608	0.1089
gender.v1Male	21.1277	8.5810	2.462	0.0144 *
age.v1	0.2432	2.7114	0.090	0.9286
income.v1.10000	-3.0006	1.4127	-2.124	0.0345 *
cg1_edu_grp<=BS	15.9816	9.6228	1.661	0.0978 .
cg1_edu_grp>=MS	12.4044	15.9479	0.778	0.4373
p.mn.cohesion.mean.v1	-2.1724	6.3704	-0.341	0.7333
p.mn.safety.mean.v1	0.3727	5.9616	0.063	0.9502
p.mn.violence.mean.v1	5.0568	6.6720	0.758	0.4491

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 69.67 on 290 degrees of freedom

(25 observations deleted due to missingness)

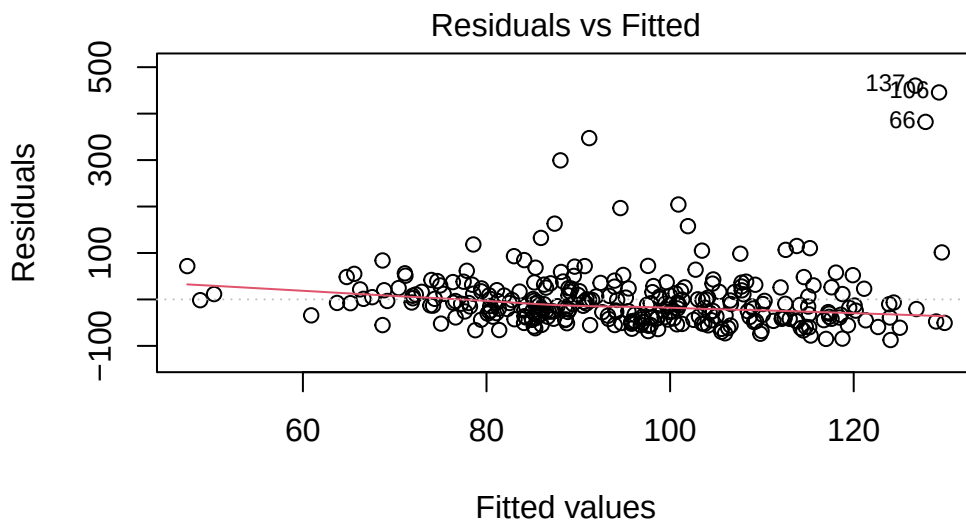
Multiple R-squared: 0.05088, Adjusted R-squared: 0.0247

F-statistic: 1.943 on 8 and 290 DF, p-value: 0.05359

```
ci <- confint(mod4_onset_sd)
ci
```

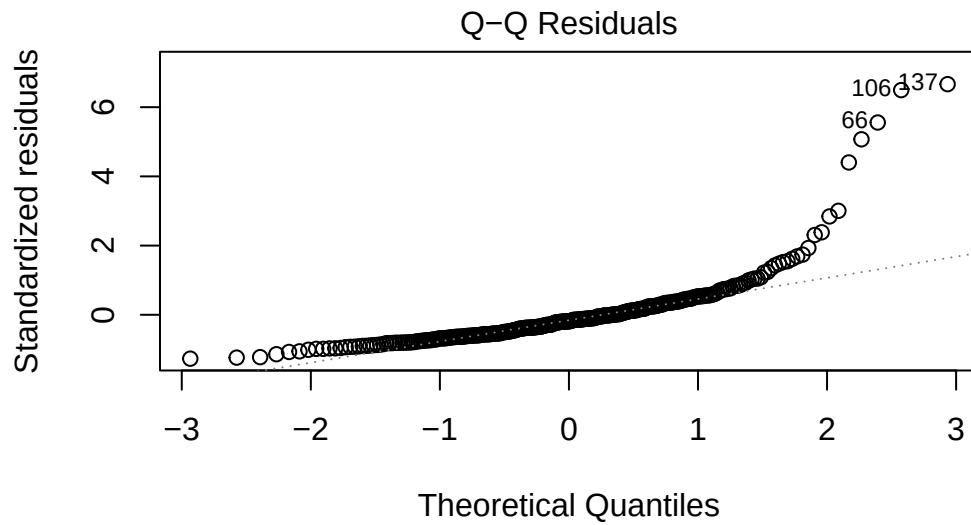
	2.5 %	97.5 %
(Intercept)	-17.892960	177.7045104
gender.v1Male	4.238798	38.0165149
age.v1	-5.093317	5.5797353
income.v1.10000	-5.781041	-0.2201133
cg1_edu_grp<=BS	-2.957884	34.9210023
cg1_edu_grp>=MS	-18.983891	43.7926309
p.mn.cohesion.mean.v1	-14.710444	10.3655863
p.mn.safety.mean.v1	-11.360745	12.1060563
p.mn.violence.mean.v1	-8.074858	18.1883912

```
plot(mod4_onset_sd,1)
```



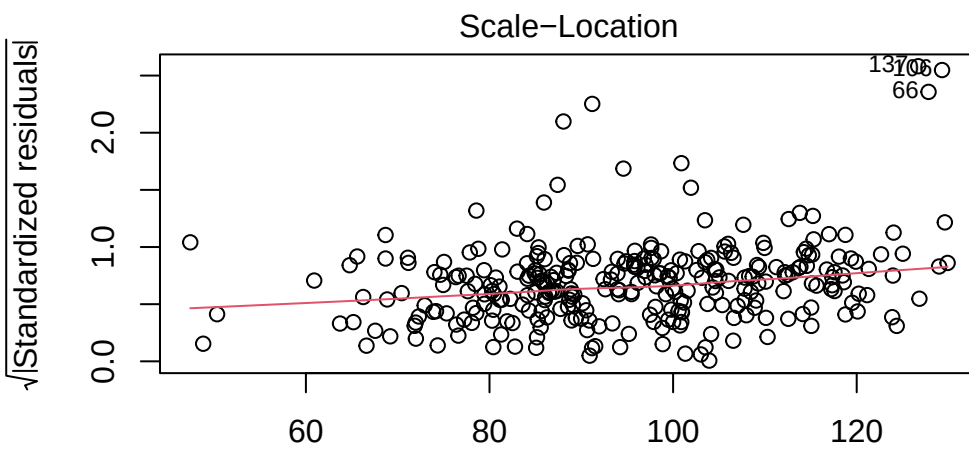
```
l(act_onset.time.min_sd ~ gender.v1 + age.v1 + income.v1.10000 + cg1_e
```

```
plot(mod4_onset_sd,2)
```



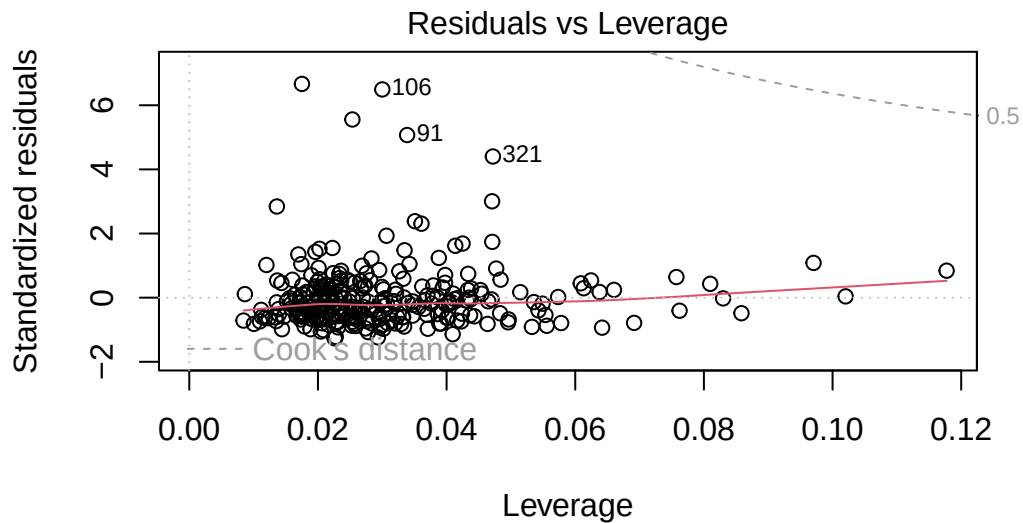
$\text{act_onset.time.min_sd} \sim \text{gender.v1} + \text{age.v1} + \text{income.v1.10000} + \text{cg1_er}$

```
plot(mod4_onset_sd,3)
```



$\text{act_onset.time.min_sd} \sim \text{gender.v1} + \text{age.v1} + \text{income.v1.10000} + \text{cg1_er}$

```
plot(mod4_onset_sd,5)
```

```
1(act_onset.time.min_sd ~ gender.v1 + age.v1 + income.v1.10000 + cg1_e
```

```
effectsize::standardize_parameters(mod4_onset_sd)
```

```
# Standardization method: refit
```

Parameter	Std. Coef.	95% CI
(Intercept)	-0.27	[-0.51, -0.03]
gender v1 [Male]	0.30	[0.06, 0.54]
age v1	5.29e-03	[-0.11, 0.12]
income v1 10000	-0.13	[-0.25, -0.01]
cg1 edu grp [<=BS]	0.23	[-0.04, 0.50]
cg1 edu grp [>=MS]	0.18	[-0.27, 0.62]
p mn cohesion mean v1	-0.03	[-0.18, 0.13]
p mn safety mean v1	6.06e-03	[-0.18, 0.20]
p mn violence mean v1	0.06	[-0.10, 0.22]

```
#Random effects for neighborhood
```

```
mod1_totsleep_commarea <- lmer(act_totsleep.min ~ gender.v1 + age.v1 + income.v1.10000 + c
```

```
boundary (singular) fit: see help('isSingular')
```

```
summary(mod1_totsleep_commarea)
```

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
```

```
Formula:
```

```
act_totsleep.min ~ gender.v1 + age.v1 + income.v1.10000 + cg1_edu_grp +
  I_frisat + z_COI_nat + (1 | COMMAREA/ID)
```

```
Data: datlong
```

```
REML criterion at convergence: 27143
```

```
Scaled residuals:
```

Min	1Q	Median	3Q	Max
-3.3318	-0.5998	-0.0113	0.5645	4.2233

```
Random effects:
```

Groups	Name	Variance	Std.Dev.
ID:COMMAREA	(Intercept)	2254	47.48
COMMAREA	(Intercept)	0	0.00
Residual		8045	89.69

```
Number of obs: 2269, groups: ID:COMMAREA, 314; COMMAREA, 50
```

```
Fixed effects:
```

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	475.969	36.519	304.777	13.033	< 2e-16 ***
gender.v1Male	-32.434	7.064	302.846	-4.591	6.46e-06 ***
age.v1	-5.907	2.185	304.584	-2.703	0.00725 **
income.v1.10000	0.262	1.150	299.955	0.228	0.81987
cg1_edu_grp<=BS	-7.102	7.697	308.575	-0.923	0.35687
cg1_edu_grp>=MS	-3.796	12.857	307.151	-0.295	0.76799
I_frisat	35.286	4.240	1983.848	8.321	< 2e-16 ***
z_COI_nat	4.962	4.755	308.486	1.044	0.29743

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Correlation of Fixed Effects:
```

	(Intr)	gnd.1M	age.v1	i.1.10	c1__<=	c1__>=	I_first
gender.v1M1	-0.180						
age.v1	-0.975	0.130					
inc.1.10000	0.001	0.000	-0.111				
cg1_d_g<=BS	-0.084	-0.081	-0.044	-0.145			
cg1_d_g>=MS	-0.001	-0.153	-0.061	-0.254	0.468		

```

I_frisat      -0.019  0.002 -0.013  0.009 -0.003 -0.011
z_COI_nat      0.248 -0.039 -0.168 -0.110 -0.059 -0.117  0.010
optimizer (nloptwrap) convergence code: 0 (OK)
boundary (singular) fit: see help('isSingular')

```

```

mod1_outbed_commarea <- lmer(act_outbed.time_numeric ~ gender.v1 + age.v1 + income.v1.10000 +
summary(mod1_outbed_commarea)

```

```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: act_outbed.time_numeric ~ gender.v1 + age.v1 + income.v1.10000 +
      cg1_edu_grp + I_frisat + z_COI_nat + (1 | COMMAREA/ID)
Data: datlong

```

REML criterion at convergence: 9509.4

Scaled residuals:

Min	1Q	Median	3Q	Max
-7.2593	-0.4797	-0.0532	0.4536	8.8574

Random effects:

Groups	Name	Variance	Std.Dev.
ID:COMMAREA	(Intercept)	2.4334	1.5599
COMMAREA	(Intercept)	0.3151	0.5613
Residual		2.9202	1.7089

Number of obs: 2269, groups: ID:COMMAREA, 314; COMMAREA, 50

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	8.55200	1.08969	302.05527	7.848	7.39e-14 ***
gender.v1Male	0.31744	0.20975	297.24246	1.513	0.1312
age.v1	-0.02526	0.06476	295.88557	-0.390	0.6968
income.v1.10000	-0.01426	0.03429	295.75139	-0.416	0.6778
cg1_edu_grp<=BS	-0.14964	0.22634	291.84008	-0.661	0.5090
cg1_edu_grp>=MS	0.08334	0.37930	291.48623	0.220	0.8262
I_frisat	1.23710	0.08098	1966.14052	15.277	< 2e-16 ***
z_COI_nat	0.34264	0.16651	151.57889	2.058	0.0413 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

(Intr) gnd.1M age.v1 i.1.10 c1__<= c1__>= I_frst

```

gender.v1M1 -0.181
age.v1      -0.970  0.133
inc.1.10000 0.005 -0.012 -0.111
cg1_d_g<=BS -0.087 -0.091 -0.039 -0.150
cg1_d_g>=MS -0.006 -0.150 -0.059 -0.250  0.462
I_frisat    -0.011  0.001 -0.009  0.006 -0.003 -0.008
z_COI_nat   0.228 -0.022 -0.137 -0.073 -0.056 -0.130  0.007

```

```

mod1_onset_commarea <- lmer(act_onset.time_numeric ~ gender.v1 + age.v1 + income.v1.10000 +
summary(mod1_onset_commarea)

```

```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: act_onset.time_numeric ~ gender.v1 + age.v1 + income.v1.10000 +
      cg1_edu_grp + I_frisat + z_COI_nat + (1 | COMMAREA/ID)
Data: datlong

```

REML criterion at convergence: 9588.7

Scaled residuals:

	Min	1Q	Median	3Q	Max
	-5.4885	-0.5104	-0.0491	0.4569	7.1675

Random effects:

Groups	Name	Variance	Std.Dev.
ID:COMMAREA	(Intercept)	2.7502	1.6584
COMMAREA	(Intercept)	0.1358	0.3685
Residual		3.0054	1.7336

Number of obs: 2269, groups: ID:COMMAREA, 314; COMMAREA, 50

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	1.838e-02	1.135e+00	3.079e+02	0.016	0.987090
gender.v1Male	7.602e-01	2.188e-01	3.011e+02	3.474	0.000589 ***
age.v1	2.670e-02	6.763e-02	3.011e+02	0.395	0.693293
income.v1.10000	1.294e-02	3.576e-02	3.017e+02	0.362	0.717687
cg1_edu_grp<=BS	6.324e-02	2.369e-01	2.929e+02	0.267	0.789746
cg1_edu_grp>=MS	2.077e-01	3.969e-01	2.943e+02	0.523	0.601099
I_frisat	5.894e-01	8.216e-02	1.969e+03	7.173	1.03e-12 ***
z_COI_nat	1.815e-01	1.645e-01	5.676e+01	1.103	0.274531

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

```
(Intr) gnd.1M age.v1 i.1.10 c1__<= c1__>= I_frst
gender.v1Ml -0.181
age.v1      -0.972  0.132
inc.1.10000 0.003 -0.006 -0.109
cg1_d_g<=BS -0.085 -0.088 -0.042 -0.149
cg1_d_g>=MS -0.004 -0.152 -0.060 -0.253  0.464
I_friscat   -0.011  0.001 -0.009  0.006 -0.003 -0.007
z_COI_nat    0.241 -0.029 -0.149 -0.081 -0.057 -0.123  0.006
```

```
mod1_eff_commarea <- lmer(act_efficiency ~ gender.v1 + age.v1 + income.v1.10000 + cg1_edu_
summary(mod1_eff_commarea)
```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]

Formula: act_efficiency ~ gender.v1 + age.v1 + income.v1.10000 + cg1_edu_grp +
I_friscat + z_COI_nat + (1 | COMMAREA/ID)
Data: datlong

REML criterion at convergence: 15682.5

Scaled residuals:

Min	1Q	Median	3Q	Max
-8.3021	-0.4830	0.1165	0.5794	2.9021

Random effects:

Groups	Name	Variance	Std.Dev.
ID:COMMAREA	(Intercept)	38.729	6.223
COMMAREA	(Intercept)	1.004	1.002
Residual		44.892	6.700

Number of obs: 2269, groups: ID:COMMAREA, 314; COMMAREA, 50

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	96.51473	4.25004	306.79825	22.709	< 2e-16 ***
gender.v1Male	-2.26333	0.82050	303.68209	-2.758	0.00616 **
age.v1	-0.78438	0.25360	305.15531	-3.093	0.00217 **
income.v1.10000	0.39886	0.13400	302.92490	2.977	0.00315 **
cg1_edu_grp<=BS	-0.10266	0.88972	305.46701	-0.115	0.90821
cg1_edu_grp>=MS	-0.81390	1.48951	304.54547	-0.546	0.58518
I_friscat	0.54240	0.31751	1967.84113	1.708	0.08774 .

```
z_COI_nat          -0.09155    0.59738    84.81121   -0.153   0.87856
```

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Correlation of Fixed Effects:
```

```
      (Intr) gnd.1M age.v1 i.1.10 c1__<= c1__>= I_frst
gender.v1M1 -0.181
age.v1      -0.973  0.132
inc.1.10000 0.002 -0.004 -0.109
cg1_d_g<=BS -0.084 -0.086 -0.043 -0.148
cg1_d_g>=MS -0.003 -0.152 -0.060 -0.254  0.465
I_frisat    -0.011  0.001 -0.009  0.006 -0.003 -0.008
z_COI_nat   0.245 -0.033 -0.156 -0.087 -0.057 -0.120  0.007
```

```
mod1_awake_commarea <- lmer(act_awakening.min ~ gender.v1 + age.v1 + income.v1.10000 + cg1_
summary(mod1_awake_commarea)
```

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
```

```
Formula:
```

```
act_awakening.min ~ gender.v1 + age.v1 + income.v1.10000 + cg1_edu_grp +
  I_frisat + z_COI_nat + (1 | COMMAREA/ID)
```

```
Data: datlong
```

```
REML criterion at convergence: 23257.1
```

```
Scaled residuals:
```

```
      Min      1Q  Median      3Q      Max
-2.6376 -0.5937 -0.1529  0.4391  5.1297
```

```
Random effects:
```

```
Groups      Name      Variance Std.Dev.
ID:COMMAREA (Intercept) 820.02  28.636
COMMAREA    (Intercept)  23.02   4.798
Residual                    1329.11 36.457
```

```
Number of obs: 2269, groups: ID:COMMAREA, 314; COMMAREA, 50
```

```
Fixed effects:
```

```
              Estimate Std. Error      df t value Pr(>|t|)
(Intercept)    25.4401    20.1080  304.3166   1.265  0.20678
gender.v1Male     5.4273     3.8801  300.8676   1.399  0.16292
age.v1           2.9824     1.1997  302.5492   2.486  0.01346 *
```

```

income.v1.10000    -2.0729      0.6334  299.8664  -3.273  0.00119 **
cg1_edu_grp<=BS    -1.2713      4.2119  303.0500  -0.302  0.76298
cg1_edu_grp>=MS     1.7449      7.0479  302.1830   0.248  0.80463
I_frisat           4.5811      1.7267 1969.7096   2.653  0.00804 **
z_COI_nat          0.9219      2.8303   83.7996   0.326  0.74545

```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

```

(Intr) gnd.1M age.v1 i.1.10 c1__<= c1__>= I_frst
gender.v1M1 -0.181
age.v1      -0.973  0.132
inc.1.10000 0.003 -0.004 -0.110
cg1_d_g<=BS -0.085 -0.085 -0.043 -0.148
cg1_d_g>=MS -0.004 -0.152 -0.060 -0.254  0.466
I_frisat    -0.013  0.001 -0.010  0.007 -0.003 -0.009
z_COI_nat   0.245 -0.033 -0.155 -0.087 -0.058 -0.122  0.008

```

```

save.image("/Users/gregorymatthews/Dropbox/bohnert_sleep/bohnert_sleep_20250312.rda")

```